THE SEMESTER BOOKS

Book 3 · Design System & Building

Design system · Architecture · Engineering · Source code

One of three books that together cover everything about Semester: Book 1 Company, Brand & Operating · Book 2 Product · Book 3 Design System & Building. Compiled from *The Main Semester design system*.

Edition of 21 October 2026 · Pre-seed, founder-led · Large figures are labelled sample data

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BOOK 3

Design System & Building

The complete design system and how Semester is architected, built, tested and shipped.

Executive summary

Semester describes itself as the native Student Operating System for the whole educational journey: one governed, accessible platform for a student's academic, personal, campus, financial and career life, plus the institutional workspaces (faculty, advisor, registrar, finance, IT and security) that serve them. The design system in this book is the set of rules, tokens, components and patterns that make those surfaces look and behave like one product. Its character is calm, scholarly and source-aware: near-black chrome, warm off-white content, a restrained Semester blue reserved for action and focus, and a controlled vocabulary that keeps official, connected, personal and AI-generated information visibly distinct.

The company is early. The README gives its status as of October 2026: pre-seed, founder-led and bootstrapped, with a first paid pilot signed with Vanderbilt University (dates not set). HECVAT Full, a VPAT/ACR, SAML single sign-on and insurance are in progress, and large figures anywhere in the system are labelled sample data. Read the book with that in mind. Much of it is a design-level specification and hand-off package, not a record of shipped and tested software, and the sources say so repeatedly.

A few principles recur in every document. Name the source of every fact and rank sources honestly. Never let colour carry meaning alone; every status has a glyph and a word. Say what an action will do before it does it, with a way back. Default to private and ask for consent one decision at a time. Keep the institution as system of record and keep student records away from any AI model. Be respectful rather than celebratory: no streaks, ranks or confetti. These rules are written into tokens, components and checks rather than left to taste.

The system is concrete. One `tokens.json` holds 112 colour tokens plus spacing, radius, shadow, size, layer, leading and tracking tokens, with twelve type roles on three typefaces. It ships 56 documented components (57 exports counting Mark), a 108-glyph icon set, logo files, 33 specimen cards, eight written guidelines and 28 UI kits. The student app covers 64 destinations in seven navigation areas, with five canonical destinations: Today, My Path, Search, Plan and Me.

The design system is subordinate to the real codebase. The README names `harrisonjrubin7-cmyk/semester` (branch `main`) as ground truth, and the build documents repeat "repo beats design": extend what exists and record whether each item was found, extended or new. The app's token pipeline runs from `lib/look.ts` to `styles/tokens.css` to `lib/tokenexport.ts` to a generated `semester.tokens.json`, and a Claude Code toolkit checks the design project against it. The documents list the differences. The main one is palette: the app's rule is "one colour for attention, the rest in words and glyphs", while the founder's brief asked for Semester blue and distinct source hues. The system keeps the code's grounds, type, radii, motion and contrast floors, adopts the brief's palette, and treats the source hues as a proposal not yet in code.

Architecturally the product has three tiers: an experience platform, an education intelligence platform and an institutional infrastructure platform. Underneath is Postgres with row-level security and a tenant identifier on every row. The browser calls only server-side `api.*` functions, writes pass through audited commands and an outbox into read models stamped with version and freshness, and one AI gateway checks scope and policy, blocks student records, labels answers AI-assisted and logs evidence.

It is built through `handoff/BUILD.md`, a single entry point telling Claude Code or Codex to work one branch and one pull request per stream, from stream 00 (repository audit) to stream 13 (platform maturity). The package supplies nine pure-TypeScript decision modules, a database and API package with eight migrations, a Next.js reference app and a Figma

toolkit, plus a playbook, QA plan and accessibility plan. It is candid that the platform code has not been run, typechecked or tested. The closing chapter records the backlog and one open decision: whether the palette moves toward the indigo, emerald, amber and rose proposed in an outside brief.

The system in numbers

Fact	Value	Source
Colour tokens	112 total	tokens.json
Colour tokens by family	source semantics 12; status semantics 10; status primitives 14; blue ramp 9; neutral ramp 9; Ink 6; Parchment 6; Graphite 3; surfaces 6; text 6; chrome 6; brand 6; action 5; sterling 4; brass 4; borders 3; focus, accent, scrim 3	tokens.json
Themes	2 (Parchment light, Ink dark)	tokens.json
Grounds and accents in the app	13 grounds, 11 accents; this project ships Ink and Parchment	README; colors cards
Typefaces	Barlow, Barlow Condensed, Cinzel, system mono	tokens.json
Font files	6	tokens.json; fonts/
Type roles and scale	12 roles; 14-step scale from 28px to 10.5px	tokens.json; type-scale card
Line-height and tracking tokens	12 and 4	tokens.json
Layout spacing tokens	11 (8px grid plus padding)	tokens.json
Density steps	7 (2, 4, 6, 8, 10, 12, 16px) at 3 densities (1, 0.86, 0.74)	spacing card; guideline 06
Radius, shadow, size, layer tokens	4, 6, 11 and 8	tokens.json
Breakpoints	3 tiers: under 760px, 760 to 1099px, 1100px and up	guideline 07
Page padding	18, 20 and 30px by tier	guideline 07
Motion durations	130, 180, 240 and 280ms	motion card
Components	56 documented, 57 exports with Mark	components/; index.d.ts
Components by group	Actions 4; AI 2; Data 5; Forms 8; Glyph 3; Navigation 4; States 11; Surfaces 8; Trust 3; Governance 7; core primitives 2	README
Icons	108 glyphs, 1.5px stroke, 24px grid	README
StatusChip keys; SourceBadge sources	21; 10	component READMEs; index.d.ts
Provenance ladder	6 rungs plus 4 display-only kinds	guideline 01
Specimen cards; guideline documents	33; 8	guidelines/
UI kits; templates	28 folders; 21 folders	ui_kits/; templates/
Student app	64 screens, 26 fully recreated, 7 areas, 5 destinations	README
Master catalog	51 roles, 673 screens, 319 workflow steps (17 workflows), 88 systems, 82 documents	README
Architecture	3 tiers; 20 tables; 15 API domains; 10 API checks; 5 freshness states; 5 ADRs	architecture docs

Fact	Value	Source
Migrations; core modules	8 (010 to 080); 9	BUILD.md; semester-core README
Security findings fixed in 050	11	semester-platform README
Execution streams	14 (00 to 13)	execute/00-INDEX.md
CI jobs that block merge	7	Engineering Playbook
QA role journeys; severities	14; 4	QA Test Plan
Assistive technologies tested	5	Accessibility Test Plan
Figma mapping	7 collections, 48 semantic rows, 12 primitive rows, 11 components	FIGMA-MAPPING.md
Toolkit skills; router workflows	3; 12	toolkit and router READMEs

The sources disagree in a few places. Components number 56 in the README, 57 in NEXT-SESSION and 64 in BUILD.md; this book uses 56 documented and 57 exported. The Figma README says 49 semantic variables but the table has 48 rows. Role counts are 51 in the catalog, 49 in the role-pages note, and the Role lens covers 14 roles in the README but 22 in NEXT-SESSION. The execute index says "12 stream prompts" but lists 00 to 13. The front end is Vite and React in the playbook, baseline and explorer, but the Architecture Guide says Next.js; BUILD.md reconciles this with a separate `web/` Next.js project if `app/` is Vite. Environments number three in the architecture card and four in the guide.

Design principles

1. Name the source and rank it honestly. Facts show their origin in a controlled vocabulary. Stale ranks below "yours" on purpose, because an unchecked copy of something that may have moved is worth less than a guess that knows it is a guess. Example: a deadline shows a SourceBadge "Official", naming the Registrar and "Updated 2h ago".

2. Colour is never the only carrier. Every state is glyph, word and tone. Hues are desaturated stones used as ink, hairline and an 8 percent wash. Example: forced-colours mode thickens the solid provenance chip's border to 3px.

3. Say what will happen first. Consequences appear before confirm: what happens, exactly what, what stays, how to take it back. Example: "You leave CHEM 101 and a W is recorded on your transcript. It cannot be undone here."

4. The institution stays the system of record. Semester never calls itself the official gradebook, registrar or payment system. Example: "Semester does not process the payment; TouchNet does."

5. Student records never reach an AI model. Enforced in the gateway, not by convention. Example: the QA plan tests that a question containing a student ID is blocked, logged and answered with a friendly message.

6. Default private, one decision at a time. Decline has equal weight to accept. Example: "Only what you tick here can be shared."

7. Respectful, not celebratory. No streaks, ranks, confetti or personas. Example: "Registration submitted. Next: review your fall schedule."

8. Border first, shadow second. Hierarchy is hairline, then surface step, then shadow. Example: cards have a 1px border, 6px radius, no shadow, and are never nested.

9. Design all seven states. Loading, empty, partial or stale, offline, error, permission and success. Example: "Your draft is safe on this device."

10. One primary action per decision region. Example: PageHeader allows a kicker, one H1, one line of purpose, a source badge and at most one primary button.

11. Disclose in three layers. Immediate decision, explanation and action, then advanced detail. Example: "Your registration opens Tuesday at 8:00 AM. You have one hold." is layer 1.

12. Reuse; do not invent. Modules may not invent colours, spacing, status words or source labels. Example: states come from `SOURCES` and `STATUSES` in `components/trust/vocabulary.js`, never hand-typed.

13. Status, not claims. Never claim a certification not held. Example: "working toward WCAG 2.2 AA; ACR in preparation", and roadmap items carry "Planned", "Pilot" or "Preview" chips.

Foundations explained

Colour: grounds. Ink is the near-black family (`ink-bg` #090a0e, `ink-panel` #12141a, `ink-raise` #22262f, text #eceef2). Parchment is the warm off-white family (`parchment-bg` #f4f1ea, panel #fbf9f4, text #1b1a17). Content defaults to Parchment and flips to Ink under `[data-theme="dark"]` with tonal elevation, so surfaces lighten as they rise. Chrome (rail, tab bar, console bar) is always Ink. The production app offers 13 grounds and 11 accents (Sterling, Brass, Copper, Jade, Slate, Pewter, Industry, Oxblood, Moss, Indigo, Old gold); this project ships Ink and Parchment. Accents identify courses and action and never encode success or error.

Colour: roles and hues. Six surface, six text and three border tokens map onto the ramps; borders are hairlines at about 9, 16 and 34 percent ink on light. Action is Semester blue: blue 700 (#416180) on light and blue 400 (#94bce3) on dark, with hover and press one or two ramp steps away and a blue-600 focus ring on light. Four source states each have ink, line and wash tokens: official (#2c455d on light, #b5d9fd on dark), connected (#2d5c61, #9fcbcd), personal (#4a4b4f, #b9b9bd) and AI (#544a7f, #bcb2df). Status hues are pending (#7a5310, #e2c27a), danger (#a3301f, #f0907a) and success (#3e6141, #a9c9a3). Lines run about 40 to 50 percent alpha and washes 6 to 12. Sterling is for the mark and chrome highlights (a brushed gradient is used on wordmark text only); brass is an editorial accent, using its shade stop (#695a33) for text on light.

Colour: contrast. Contrast is measured against every surface of a ground. Body and UI text need 4.5:1, large text 3:1, and icons, borders, focus and chart marks 3:1. Tertiary text is about 3.4 to 3.7:1 and is for metadata only, never anything a student must act on.

Type. Barlow carries reading at 16px on a 1.55 line, with controls at 14px. Barlow Condensed 600 carries headings, labels and buttons at -0.015em. Cinzel is for rare editorial display of 42 to 56px and never appears in warnings, provenance, consent or accessibility text. System mono is for IDs and timestamps. The twelve roles run from editorial (42px) through display (26), page (24), section (20), card title (18), body (16), control (14), compact (12), label and caption (11), kicker (10.5, tracked 0.08em caps) and numeric (22, tabular). Measure is about 66 characters, sentence case is universal, and sizes scale with `--text-scale`, which reader settings move from 0.94 to 1.18.

Spacing and density. Layout uses an 8px grid (8, 16, 24, 32, 40, 48, 64, 80, 96px) with page padding of 18px widening at 760 and 1100px. Inside components, seven steps (2 to 16px) are multiplied by density 1, 0.86 or 0.74. Targets are 44px by default, 40px for icon buttons, 32px for compact rows and a 24px floor in crowded composites with keyboard parity.

Corners. "Drawn" corners are 3px for controls, 6px for cards and 10px for sheets, with a 999px pill for floating bars. Chips stay at 3px.

Elevation. Border first, shadow second. Light grounds use three ink-tinted shadows; dark grounds use three lift tokens. Shadows appear only on floating and modal layers. Eight stacking layers run from 0 to a 1500 curtain. Only the tab bar and focus bar are translucent (14px blur), with an opaque fallback under reduced transparency, increased contrast, Low stimulation and forced colours. Scrims are 50 percent Ink.

Motion. Hover and press take 130ms, state changes 180ms, panels 240ms and sheets 280ms, with progress at 180ms linear and the ease `cubic-bezier(.22,1,.36,1)`. Sheets rise 8px and fade. There is no bounce, parallax, shimmer, spinner, looping decoration or press scaling. All motion is zeroed by `prefers-reduced-motion` and `data-calm`, and no meaning depends on it.

Layout and breakpoints. Below 760px a labelled bottom TabBar carries the five destinations. From 760 to 1099px a 72px compact rail appears. From 1100px the SideRail is 248px (272px at very wide). A 44px SystemContextBar sits on top with location, term, Command-K search and sync state. Pages use a `minmax(0,1.6fr) minmax(300px,1fr)` grid that stacks below 1000px, with the next action leading the main column and "Needs you", context and policy in the aside. Reading measure is 66ch and operational pages cap at 1180px. The student app is mobile-first and offline-safe; staff and console are desktop-first; the public site goes single column under 860px; everything reflows at 320px and 200 percent zoom.

Accessibility floors. The target is WCAG 2.2 AA, a design requirement and explicitly not a conformance claim. Each page has one main landmark, one H1 and a skip link. Focus never lands under sticky chrome. Inputs are native with persistent labels, and errors preserve input. Overlays trap focus, close on Escape and return focus. Reader settings cover text size, density, line height, letter and word spacing, reading width (52 to 82 characters), an Atkinson Hyperlegible option, three motion modes, Increase contrast and 13 grounds. Status, provenance, consent text and accessibility controls are brand-exempt: body-grade type only and never restyled.

Component library guide

Each component has a README, type declarations and a live preview. All are bundled under `window.SemesterDesignSystem_e9f3ba` and typed together in `components/index.d.ts`.

Actions

- **Button.** One primary per decision region; 44px, or 32px with `size="sm"`. Variants: primary, secondary, ghost and danger (red ink and hairline, never a red fill). Props: `variant`, `size`, `icon`, `loading`, `disabled`.
- **IconButton.** Icon-only control; `label` (its accessible name and tooltip) is required.
- **QuickActions.** Two to five labelled starting actions for empty and onboarding states.
- **ActionPreview.** What happens, exactly what, what stays, taking it back. Props: `says`, `exactly`, `doesNotChange`, `recovery` (undo, request or none). Place it inside a Dialog for consequential actions.

AI

- **AIResponse.** A governed answer labelled AI-assisted with scope, numbered sources, confidence, cost and policy. Consequential `actions` must open a Dialog and ActionPreview.
- **ProvenanceChips.** Origin chip plus at most two cues and an age, spoken as one sentence.

Data

- **DataTable.** Compare columns; use a list when scanning or acting on rows. Props: `columns`, `rows`, `caption`.
- **Fields.** Label and value list for one record.
- **DecisionTrail.** Timeline of who did what, when and under which authority; states done, current, waiting, blocked.
- **MetricTile.** One figure with unit, source and age; never an outcome claim.
- **ProvenanceChart.** Bars encode provenance by pattern: solid, outline, hatched, dotted. The title states the finding.

Forms

- **TextField.** Visible label, hint and error; the placeholder is an example, never the label.
- **Select.** Native select in Semester's palette.
- **Segmented.** Two to four exclusive presentation choices, not navigation.
- **Choice.** Radio or checkbox row with a sentence on the consequence, for sharing decisions.
- **Switch.** On/off setting that shows the word On or Off.
- **Combobox.** Filtered listbox for courses, people and rooms.
- **DateField.** Native input that speaks the date back as "Thu 9 Oct 2026" via `said`.
- **VisibilityPicker.** The one privacy question with four answers: Only me, Course, Selected collaborators, Public portfolio. Offer only what the module can honour; it never grants access.

Glyph

- **Icon.** From the 108-glyph set; `label` makes it a named image.

- **Mark** and **Wordmark.** The three-slab mark and the SEMESTER lockup; `metal` paints sterling for dark chrome.
- **SectionLabel** and **Avatar** (core primitives). The tracked-caps kicker with optional count, and initials in a tinted square, never photos.

Navigation

- **SideRail.** Persistent 248px Ink rail; the current row has a blue wash.
- **TabBar.** Mobile bottom navigation, always labelled.
- **SystemContextBar.** Location, term, search, sync state.
- **CommandPalette.** Global search and commands with keyboard selection.

States

- **EmptyState.** Why it is empty plus one action.
- **LoadingState.** Structural placeholders and a spoken "Loading"; no spinner.
- **ErrorState.** Plain-language failure with a required recovery and optional reference; keeps the user's input.
- **SuccessState.** Acknowledgement pointing to what is next.
- **ProgressState.** Determinate wait with Cancel or Retry.
- **StepStatus.** Named steps (waiting, working, done, failed) so a system never fakes certainty.
- **PermissionNotice.** What changed about access, why, and where it is controlled.
- **OfflineStrip.** What still works and how many changes are queued.
- **SaveState.** Quiet "Saving, Saved" live region.
- **Notice.** One-line service notice.
- **UndoToast.** Confirmation with Undo for reversible work; never for consequential or financial actions.

Surfaces

- **ObjectCard.** The universal decision card: eyebrow, title, provenance, why, metadata, one primary and one secondary action.
- **ContextBar.** The object a screen is about, with states and actions.
- **NextSteps.** Ordered safe actions with one line of why.
- **Dialog.** Modal confirmation; focus lands on Cancel; `destructive` uses a red-ink button.
- **Sheet** (with **SourceDetails**). Bottom sheet or window; home of Source and details.
- **FocusBar.** Floating pill for focus sessions.
- **ScreenGuide.** "About this screen" disclosure at every screen's foot.
- **OpenIn.** Words-only links to every place an object matters.

Trust and provenance

- **SourceBadge.** Glyph, word and tone for origin; `onReport` adds "Report incorrect information".
- **SourceLine.** A thin brass rule from a statement to its source chip; used only where a real source exists.
- **StatusChip.** One of 21 shared status keys, always glyph and word.

Governance

- **PolicyBadge.** Policy name, version and owner; opens policy text in a Sheet.
- **ConsentBadge.** Private, shared (recipient, expiry), expired, revoked or required.
- **HealthBadge.** Healthy, watch, at risk or unknown, with driver and freshness.
- **ApprovalBanner.** Two-person approval state with buttons suited to the viewer.
- **ErrorSummary.** Form errors at the top (WCAG 3.3.1), each linking to its field.
- **SupportHandoff.** Hand-off to the owning office with channel, hours and status, without editing the official record.
- **PageHeader.** Kicker, one H1, purpose, source and at most one primary action.

The component library document marks FreshnessBadge, AuthorityBadge and RiskBanner as "Partial" and NotificationCenter as "Pattern only". The seven governance components closed gaps in October 2026.

Patterns and governance rules

Provenance ladder. Strongest first: official, connected, made (built by the app from the student's sources), yours, sample, stale. Display-only kinds are AI-assisted, estimated, needs review and external. Made and AI differ on purpose, since an extraction can misread and a draft can sound right while wrong. Estimates and AI are styled to be noticed, because students over-trust them. Freshness reads "Updated 3 days ago" or "Update time not recorded", never "just now" for an unknown time.

AI answer anatomy. Every answer shows the AI-assisted label, the allowed data scope (changeable before asking), numbered sources, confidence and limits, request cost, the governing policy, and actions that open a preview first. StepStatus shows the work. Never say "AI-powered", "intelligent", "knows you" or "accurate". When an instructor restricts AI, the course shows "Course AI policy, set by instructor" and disallowed actions are disabled with a reason, not hidden.

Consent. Private by default, never implying guardian access. Each ask states what, with whom, why, how long and how to undo. Before sharing, the student previews the exact payload and expiry. Shares are revocable beside where they are made, access history is visible, and protected categories (counselling, health, disability services) show a disabled switch with the reason.

Consequential actions.

Kind	Pattern
Reversible and personal	Do it, then UndoToast
Leaves Semester or visible to others	Dialog with ActionPreview
Academic record	In-place Review then Confirm, stating where the request lands
Financial	Dialog naming amount, method, processor and refund route
Institutional high-risk	Two-person duty, fresh MFA within 15 minutes, audit first

Dialog titles are questions naming the object, buttons name outcomes, focus lands on the safe choice, and recovery is required. A SuccessState receipt follows.

State matrix. Every data region designs loading, empty, partial or stale, offline, error, permission and success.

Data trust panel. Critical screens show Source, Updated, Authority, Status, Policy (versioned) and a path to official detail. Delayed data blocks destructive actions until confirmed at the source.

Content templates. The status sentence is the unit of voice: what is true, what it means for you, what you can do. Errors give what failed, what is safe, the way out and a reference such as SYNC-4F2A. Incident notices list what is affected, what is not (only if verified), what to do and the next update, and never say "minor" or "isolated". Onboarding starts from what the student has, asks consent one item at a time, uses no streaks, and always offers Skip. Dates read "Thu 9 Oct · 23:59". Emoji are never used.

Architecture

The architecture documents are labelled design-level. The Architecture Guide (v0.1, October 2026) says to "verify everything against the repo", so this chapter describes the intended architecture, not audited running software.

Three tiers. Tier 1, the experience platform ("Apple-like"), holds the Student OS, Course Studio, advisor, registrar, family, career, company site and operations surfaces. Tier 2, the education intelligence platform ("Google-like"), holds identity and education graphs, search, the AI gateway, policy and workflow engines, provenance, analytics and human escalation. Tier 3, the institutional infrastructure platform ("Microsoft-like"), holds the tenant control plane, RLS, SSO and SCIM, approvals, audit, the integration gateway (LTI, OneRoster, Edu-API), migration, feature flags, billing, trust, SLOs, backup and a developer ecosystem. The borrowing table says what not to copy: closed ecosystems, ad-tech incentives on student data, configuration overload and treating model output as official truth. A ten-step deployment ladder runs from development to multi-campus expansion, and each step needs a security review, accessibility review, rollback plan and release evidence. The integration doctrine is to connect first and replace a domain only where the institution chooses, in five phases from Connect to Education OS; phase one is "Now (paid pilot)".

Layers and data. The front end is the student app, the staff and console routes on the same shell, the design system and an offline queue. Middleware refreshes sessions, resolves the tenant, checks role by screen by grant, rate-limits (harder for AI), writes the audit event before the action, and applies per-tenant flags and kill switches. The server layer offers `api.*` read functions, `private.*` command functions requiring an idempotency key, edge functions, integrations and an outbox worker. The database is Supabase Postgres with RLS on every table. The design-level schema has 20 tables, including tenants, memberships, role_grants, enrollments, grades, holds, shares, approvals, audit_log, domain_outbox_events, ai_requests and integrations. Schemas are `core`, `edu` (student data), `community`, `market`, `ops` (company data, no student identifiers) and `audit`.

Read models, CQRS and freshness. A write succeeds against the authoritative table, an outbox event is written in the same transaction, a worker applies the projection idempotently, version and freshness are stamped, caches are invalidated, the browser gets a targeted refetch signal and the UI renders the read model. The `private`, `read_model` and `api` schemas keep that boundary visible. Freshness drives UI policy: live (up to 5 seconds) and recent (up to 60) behave normally; delayed (over 60) shows a badge and re-checks the source before consequential actions; failed or rebuilding shows the last good state and holds them. Rebuilds are blue-green behind a parity check. Realtime is limited to approvals, incidents, P0 and P1 support, rollout state and invalidation. The UI is optimistic: it shows "Saving" at once, confirms only when the projection version advances, and rolls back with a reason otherwise.

AI gateway. The gateway is the single egress to models (ADR-003). A request passes identity and scope, policy (course AI policy, institution switches, FERPA boundary), routing to an approved model, an answer labelled AI-assisted with sources and confidence, and an evidence log. Student records are blocked at the policy step. The allow-list admits course content, the student's own typed text and public catalog data, and blocks anything from `edu.*`. The starting code runs authentication, policy, gate, content scan, provider and audit, with a redaction module that blocks IDs, SSN, GPA, grades, transcripts, dates of birth and card numbers in prompts, a usage log that stores no prompt content, and a red-team suite that is a launch gate.

Environments. Local uses synthetic seed data. Preview deploys per pull request on synthetic data. Staging mirrors the institution's configuration with synthetic data and test IdP accounts. Production holds real records under the DPA with separate keys, named operators, MFA and logged access. Data is never copied between environments, and secrets live in

Vercel and Supabase settings.

API contract standard. APIs are organised by authority and risk. "High-impact writes never mutate raw tables from the browser; they go through capability-checked server commands." Fifteen domains are defined (identity, tenant, academic, learning, workspace, AI, student success, campus, career, family, integrations, commercial, trust, operations, platform), each with a write-control model. Every API answers ten questions: identity, scope, purpose, authority, policy, validation, idempotency, audit, freshness and error recovery. Examples include `POST /v1/academic/registration-requests` and `POST /v1/family/grants`.

Security model. Five principles: the institution is the system of record; student records never reach an AI model; company roles never see student data; consequential actions are previewed, attributed and logged; source and freshness show on every record. RLS checks the tenant against the JWT claim, service-role keys live only in workers, every query runs as the user, and every mutation writes an audit row. Marketplace payments go through a licensed processor, so Semester never holds funds. High-risk actions need a second approver and fresh MFA. A review of migrations 010 to 040 found 11 issues, including a critical one (staff self-inserting as CEO) and a high one (changing a profile's institution), fixed in migration 050, which must be applied before any data. Migrations 060 to 080 add audited command functions, the AI gateway tables, private storage and delegations. Five ADRs record the main choices: RLS as primary authorisation, read models over live SIS queries, the single AI egress, processor-based payments, and a separate `ops` schema.

Operations. The explorer records staged tenant rollouts, forward-only migrations with a parity check, monitoring of errors, projection lag, sync failures and AI spend, on-call response of 1 hour (P0) and 4 hours (P1), daily backups with point-in-time recovery and a quarterly restore test, quarterly access reviews, and postmortems within five days.

How it is built

Stack. The toolkit baseline (repository commit 5eba494, 5 October 2026) describes the real app as Vite, React and TypeScript under `app/`, with plain CSS, about 273 component files, about 95 lazy screens, 19 style contract tests, a 143-pairing contrast test and 12 CI workflows. Hosting is Vercel and Supabase; validation is Zod; tests are Vitest. The platform and router packages are Next.js App Router with Supabase.

Packages. `semester-core` is dependency-free, I/O-free TypeScript with nine tested modules: provenance, nextaction, access, freshness, consequence, aigateway, approvals, scoring and pilot. `semester-platform` holds database and API code for community, marketplace, ops and audit: migrations 010 core, 020 community, 030 market and 040 ops, then 050 to 080 for security, commands, AI and storage, plus API routes, validation and RLS, security, command and red-team tests. Its README says: "Not yet run, typechecked or tested." `workflow-router-app` is the AI Workflow Router and Benchmark Lab: Next.js, strict TypeScript, Supabase magic-link auth, Zod server actions, two migrations with organisation-scoped RLS, a seed of 12 workflows and Vitest suites; its README says checks "have not been run here". All code files carry a `.txt` suffix, which `install.sh` strips without overwriting.

Migrations and tests. Migrations run 010 to 080 in order, renumbered after the repository's latest, and applied ones are never edited. Each ships with RLS policies and tests in the same pull request (own tenant allowed, other tenant denied, company roles denied on student tables, expired grant denied), proved red then green. Test layers are unit, RLS, AI red-team, Playwright end-to-end and accessibility. `verify.sh` must end with "ALL GATES PASSED".

Execution plan. Each stream uses a branch `semester/<stream>`, ends with a pull request and a report, and stops for review.

Step	Stream	Gate
00	Repo audit and design-system sync	CI green; baseline approved
01	Database, tenancy, RLS, core logic	RLS red-then-green
02	Identity, SAML SSO, onboarding	Test login per pilot role
03	Student OS	Five states per screen; accessibility 95 or more
04	Staff and academic workspaces	Two-person approvals tested
05	Institution console, integrations	Zero unexplained mismatches
06	Governed AI gateway	Red-team suite passes
07	Community, career, family	Revocation immediate
08	Operations command center	Company-role isolation test
09	Security, privacy, accessibility	Restore drill timed
10	Pilot launch	Every go/no-go gate evidenced
11	Public site and grow	Every claim sourced
12	One place	No regulated step re-implemented
13	Platform maturity	Scorecard re-scored from evidence

The pilot-critical path is 00, 01, 02, 03, 06, 05, 09, 10. Humans alone bind insurance, sign the DPA, submit HECVAT Full, obtain attorney review, supply the Vanderbilt IdP details, approve the baseline and each pull request, approve cutover and choose the AI provider.

Engineering playbook. Three rules never bend: RLS on every table with a proving test; the browser calls `api.*` only; student records never reach an AI provider. `main` is always deployable, pull requests stay under about 400 lines, and the founder approves anything touching auth, RLS, the AI gateway or billing. Seven CI jobs block merge. Releases are tagged, smoke-tested in staging, backed up before migrations and watched for 30 minutes, and none ship in launch week, finals or registration opening except S1 and S2 fixes. Features ship behind per-tenant flags with kill switches. Dependency alerts are reviewed weekly. The product and engineering operations page admits gaps in a PRD template, ADR log and experiment registry.

QA and accessibility. The QA plan uses one synthetic account per role and 14 end-to-end journeys, such as a registrar override needing a second approver and a support user unable to read student tables. Defects run S1 (data exposure) to S4 (cosmetic). The accessibility plan targets WCAG 2.2 AA with an ACR on the VPAT 2.x template, runs axe in CI, and requires manual keyboard, screen-reader (NVDA, JAWS, VoiceOver, TalkBack), zoom, spacing, target and motion checks on pilot paths each release.

Claude Code and Figma toolkit. The toolkit adds three skills (`audit-semester-design-sync` , `build-semester-ui` , `create-semester-component`), an `.mcp.json` for the remote Figma MCP server with no credentials, a CLAUDE.md section, a zero-dependency audit script, a report script, Vitest proofs and `design-system:check` scripts, with no new dependency, Tailwind or Storybook. Authority runs from `look.ts` to `tokens.css` to `tokenexport.ts` to the generated JSON to `FIGMA-MAPPING.md` to Figma, and Figma is never written to. Figma names stay "proposed" until a file is connected.

What's next

The founder's top-priority ask is: "Every role in Semester. Finish, generate and produce everything I need to run, grow, build, operate, maintain, secure and launch Semester." NEXT-SESSION.md records launch, secure, build, operate, run and grow documents as done, plus one page for each of 49 catalogued roles.

Open work, in order: replace archetype screens with bespoke designs, highest traffic first; regenerate the master catalog documents; export the investor, institution and internal decks to PPTX; deepen registration readiness, support SLA and SIS reconciliation screens; build Course Studio and the moderator workspace; write the AI governance policy, data inventory, incident response and pilot delivery documents; and run the Claude Code toolkit in the real repository. An administrator manual and tutorials are requested twice and remain to do, as does a batch of device and media capabilities (camera, microphone, location, notifications, offline, multi-device sessions, personalisation) with consent, fallback and revoke paths.

One decision is open: indigo with emerald, amber and rose from an outside brief, or Semester blue with stone hues. The rules to keep are that nothing is verified without code, tests and release evidence; company roles never see student records; status replaces certifications not held; and external numbers are cited. The `github.md` note records that `docs/design-system/` and the `workflow-lab/` app have landed upstream, while the toolkit's `.mcp.json`, skills, scripts and baseline had not.

Reader's guide

- **Design system README.** What Semester is, voice, visual foundations, iconography, components and UI kits.
- **Tokens.** The reference generated from `tokens.json`.
- **Foundations.** Specimen cards for colour, type, spacing, density, layout, motion and data visualisation.
- **Brand specimens.** Logo system, mark, wordmark, iconography rules and voice.
- **Components, light.** Every component on the Parchment ground.
- **Components, light and dark.** The same components on Parchment and Ink.
- **Patterns.** Provenance ladder, AI anatomy, consent, consequential actions, state matrix, Source Line.
- **Guidelines.** Eight written deep dives, from provenance to content templates.
- **Design language.** The eight documents in `docs/system/design`.
- **Figma mapping.** Collections, variables and component mappings.
- **Architecture guide.** Principles, components, data model, flows, environments and ADRs.
- **Three-tier and API.** The three tiers, deployment ladder and API contract standard.
- **Architecture cards.** Layers, read-model pipeline, AI gateway, environments.
- **Architecture explorer.** OS map, 20-table schema, CQRS pipeline, lag simulator and phases.
- **Developer guide.** Setup, conventions, branching, testing and the new-feature checklist.
- **Engineering playbook.** Layout, CI, environments, releases and definition of done.
- **Product and engineering ops.** The company-OS page on owner, KPIs and approval thresholds.
- **QA plan.** Test levels, role journeys, negative tests and gates.
- **Accessibility plan.** WCAG 2.2 AA checks, assistive-technology matrix and ACR.
- **Build handoff.** `handoff/README.md` and `BUILD.md`.
- **Execution plan.** The master index and stream prompts 00 to 13.
- **Code packages.** READMEs for semester-core, semester-platform and the router app.
- **Toolkit.** The Claude Code and Figma toolkit documents.
- **Tooling cards.** Four summary cards for the tooling.
- **Workflow lab.** The AI router and benchmark prototype.
- **Prototype notes.** What is sample data and what is stored only on the device.
- **Prototype-to-codebase map.** The `github.md` screen map and sync history.
- **What's next.** The NEXT-SESSION backlog.
- **Source-code appendices.** `tokens.css` and `bundle.css`; component sources; semester-core; semester-platform (API routes, migrations, tests); the workflow-router app.

Glossary

Term	Meaning
Ask Semester	The AI assistant; no human name or persona, always labelled AI-assisted
AI gateway	Single egress to models; checks scope and policy, blocks records, logs evidence
Approval banner	Component showing two-person approval state
Audit before action	The audit event is written first, then the action applies
Authority	The office entitled to decide a fact, such as the Registrar
Batch N	The founder's plan to produce everything needed to run and launch Semester
Brand-exempt zones	Status, provenance, consent text and accessibility controls, never restyled
Break-glass	Emergency privileged access needing a reason and review
Calm mode	`data-calm`; zeroes motion and quiets decoration
Canonical destinations	Today, My Path, Search, Plan, Me
Chrome	The always-dark rail, tab bar and console bar
Connected	Provenance rung for a recently checked account the student connected
Consent badge	Private, shared, expired, revoked or required state
Consequence copy	Text stating what an action does, before confirm
Core	`semester-core`, nine tested decision modules
CQRS	Writes to authoritative tables, reads from projected models
Decision trail	Timeline of who did what under which authority
Density	Multiplier of 1, 0.86 or 0.74 on component spacing
DPA	Data processing agreement governing production data
Drawn corners	3px controls, 6px cards, 10px sheets
Freshness	Live, recent, delayed, failed or rebuilding
Ground	Base surface family: Ink or Parchment (13 in the app)
HECVAT	Vendor assessment toolkit; HECVAT Full is in progress
Idempotency key	Makes a retried command safe
Ink	The near-black ground, #090a0e
Kicker	10.5px tracked uppercase label
Made	Rung for content built by rule-based extraction from student sources
Master catalog	51 roles, 673 screens, 319 steps, 88 systems, 82 documents
Outbox	Events written in the same transaction as a change
Parchment	The warm off-white ground, #f4f1ea
Policy badge	Names governing policy, version, owner
Provenance	Origin and trust level of a fact
Read model	Projected table stamped with version and freshness

Term	Meaning
Repo beats design	Existing repository code wins over design proposals
RLS	Row-level security in Postgres
Semester blue	Action colour: #416180 light, #94bce3 dark
SIS	Student information system, the institution's record
Source badge	Glyph, word and tone showing origin
Source Line	Thin brass rule linking a statement to its source
Stale	Rung for data from a connection not checked lately
Status sentence	What is true, what it means, what you can do
Stream	One numbered unit of the execution plan
Tenant	An institution's isolated space; `tenant_id` on every row
Three-layer disclosure	Decision, explanation, advanced detail
Two-person duty	Requester and a different approver, fresh MFA, audit first
VPAT/ACR	Accessibility template and conformance report; in preparation
Visibility picker	The four-audience "Who can see this?" control
Wash	Low-alpha tint behind chips
WCAG 2.2 AA	Accessibility target, not a conformance claim
Yours	Rung for student-entered, unchecked information

PART 1

Design system

Tokens, foundations, every component in light and dark, patterns and the deep-dive guidelines.

Brand book

Semester is the native Student Operating System for the complete educational journey — one governed, accessible platform for a student's academic, personal, campus, financial, career and lifelong life, plus the institutional workspaces (faculty, advisor, registrar, finance, IT/security) that serve them. Surfaces include Student Today, Calendar, Tasks, Courses, Study, Write, Degree, Registration, Grades, Campus, Dining, Housing, Family, Career, Billing, Support, the AI Assistant, and the Institution Console.

The system is calm, scholarly and source-aware: near-black chrome, warm off-white content, restrained Semester blue for action and focus, and a controlled vocabulary that keeps **official, connected, personal and AI-generated** information visibly distinct.

Status (Oct 2026): pre-seed, founder-led, bootstrapped; first paid pilot signed with Vanderbilt University (dates not set). HECVAT Full, VPAT/ACR, SAML SSO and insurance are in progress. Large figures anywhere in this system are labelled sample data.

Sources

- **Codebase (ground truth):** https://github.com/harrisonjrubin7-cmyk/semester (branch `main`). Key files read: `DESIGN-SYSTEM-GUIDE.md`, `docs/COLOR-AND-DARK-MODE-SPEC.md`, `app/src/styles/{tokens,typefaces,industry,unity}.css`, `app/src/lib/{look,status,source}.ts`, `app/src/components/{Brand,Icons,icons.data,mark.data,SourceBadge,StatusNotice,StartToday}.tsx`, `app/src/components/unity/*`, `app/src/components/console/*`. Also read: `docs/BRAND-PLATFORM.md` (voice, naming, Source Line), `enrollment/StudentRegistration.tsx`, `gradebook/StudentGrades.tsx`, `AdvisorMeeting.tsx`. Explore the repo further — especially `docs/design/`, `company-site/` and `app/src/components/` — to build closer to the real product.
- **Uploads:** `uploads/download-1.png` (lit silver logo mark on Ink), `uploads/Semester build Architecture.pdf`, `uploads/audit [...].pdf/.md` (repo audit).
- **Brief:** the user's written design direction (source-status hues, Semester blue, warm off-white content). Where it diverges from the code it is noted below.

Reconciling brief and code

The app's own rule is "one colour for attention, the rest in words and glyphs", with a sterling/brass metal accent. The brief asks for a Semester blue action colour and distinct source hues. This system **keeps the code's grounds, type, radii, motion, glyph+word status vocabulary and contrast floors**, and **adopts the brief's palette** using hues pulled from the code's own measured ramps (Industry blue → Semester blue; Jade/Moss/Indigo/Old gold → teal/green/violet/amber). Every chip still carries a glyph and a word; colour is never the only carrier.

CONTENT FUNDAMENTALS

From `docs/BRAND-PLATFORM.md` (draft) and in-app copy. **Personality:** a good registrar's assistant crossed with a precision instrument — exact, calm, candid, warm sparingly, self-possessed. **Public descriptor:** "the student action platform" ("Student OS" is internal only — it implies replacement). **Assistant name:** *Ask Semester* — no human name, face or persona; always labelled AI-assisted.

The status sentence (the unit of Semester voice): *what is true → what that means for you → what you can do.* "Your Canvas calendar hasn't updated since Tuesday. Deadlines below may have moved. Reconnect it, or check the course page."

Prefer / avoid: "helps you organise" not "revolutionises"; "alongside your school's systems" not "replaces / all-in-one"; "AI-assisted, source-linked" not "AI-powered / intelligent / magic"; name the control, never "secure / private / compliant / certified"; "planned / in private beta / pilot" not "launching soon"; "can / is designed to" not "guarantees / always / 100%"; "student" not "user / learner". Destinations are always Today, My Path, Search, Plan, Me.

Consequence copy from the product: "Confirming puts you on the waitlist, number 3 in line. You are not enrolled until a seat comes to you." · "You leave CHEM 101 and a W is recorded on your transcript. It cannot be undone here." · "Worked out by Semester from released grades only; your instructor's final grade is the one that counts." · "Only what you tick here can be shared. Nothing else — not your study history, grades, notes or anything else — ever is."

- **Voice:** plain, precise, quietly warm. Speaks to the student as **"you"**; the product never says "we" about itself in-UI ("Semester has not checked it with your institution"). First person only in the assistant's own answer ("I'd place them…").
- **Casing:** Sentence case everywhere — buttons, titles, chips ("Needs confirmation", "Sync trouble"). UPPERCASE only for tracked kickers/eyebrows and the wordmark.
- **Say what, why, who decides.** Every state explains itself: *"You are offline. You can keep working — 2 changes will sync when you are back."* Errors say what is safe: *"Your draft is safe on this device."*
- **Name the source.** "Institution verified", "Imported", "Student entered", "Estimated", "AI-assisted", "Needs review". Never "verified" unless it was.
- **Consequences before confirm:** What happens / Exactly what / What stays the same / Taking it back. *"This can be reversed on request: ask the Registrar before Oct 20."*
- **Respectful, not celebratory:** no streaks, ranks, exclamation marks or confetti. *"Registration submitted. Next: review your fall schedule."*
- **Never overclaim:** don't call Semester the official gradebook, registrar or payment system; label demo data "Sample".
- **Numbers:** tabular figures; dates as "Thu 9 Oct · 23:59"; middle dot `·` separates metadata.
- **Emoji:** never. Unicode glyphs are used as status marks: ◆ official, ○ yours, ✦ AI, ⇄ connected/conflict, ? review, ! attention, ✓ done, ⊘ offline, … working, ◌ sample.

VISUAL FOUNDATIONS

- **Colour:** Ink (#090a0e family) for all chrome — rail, tab bar, console environment bar. Parchment (#f4f1ea / panel #fbf9f4) for content. Dark mode = Ink content via `[data-theme="dark"]` (tonal elevation: surfaces lighten as they rise). Semester blue 700 `#416180` is the one action/fill colour on light, 400 `#94bce3` on dark. Brass is an editorial

accent only. Source/status hues are desaturated stones, used as ink + hairline + 8% wash — never as large fills.

- **Type:** Barlow (body 16/1.55, controls 14), Barlow Condensed 600 (headings, buttons, labels, -0.015em), Cinzel (editorial display, 42–56px, rare), system mono (IDs, timestamps). Kickers: 10.5px, 0.08em tracked caps, secondary ink. Body measure ≈ 66ch.
- **Spacing:** density-aware 2px steps `--sp-1…7` (2–16px) inside components; 8px grid `--space-*` for layout. Page padding 18 / 20 / 30 by breakpoint. 44px default targets, 24px floor in crowded composites.
- **Backgrounds:** flat grounds only. No photography backgrounds, no gradients except the brushed sterling text gradient on the wordmark/chrome text. No patterns, textures or illustrations.
- **Borders & cards:** hierarchy by hairline first (9 / 16 / 34% ink), surface step second, shadow third. Cards: 1px border, 6px radius, 16px padding, no shadow, never nested. A card groups a *decision*, not a paragraph.
- **Corners:** "Drawn" 3 / 6 / 10px (control / card / sheet). Pills only for floating bars; chips are 3px.
- **Shadows:** light — ink-tinted `0 1px 2px` / `0 3px 10px` / `0 12px 32px`; dark — `--lift-1…3`. Used for floating and modal layers only.
- **Transparency & blur:** only on the tab bar and focus bar (14px blur), always with an opaque fallback under reduced transparency, increased contrast, Low stimulation and forced colours. Scrims are 50% Ink.
- **Motion:** 130ms hover/press, 180ms state change, 240ms panel, 280ms sheet; ease `cubic-bezier(.22,1,.36,1)`. Sheets rise 8px + fade. No bounce, parallax, shimmer, spinners or looping decoration. All zeroed by `prefers-reduced-motion` and `data-calm`.
- **Hover:** a blue 8% wash on buttons/rows, hairline darkens on inputs, text steps from secondary → primary. **Press:** 14% wash or one ramp step darker on primary. No scale/shrink.
- **Focus:** 2px blue-600 outline, 2px offset; selected list rows use a 3px leading edge + weight, never colour alone.
- **Imagery:** the only brand image is the lit silver mark on Ink (cool, metallic, no grain). No stock photography or AI imagery in-product.
- **Layout:** fixed dark side rail (248px) on desktop, labelled bottom tab bar on mobile, persistent system context bar with ⌘K search on top. Operational content ≤1180px; one H1 per screen.

ICONOGRAPHY

- The app ships its **own inline icon set** (`app/src/components/icons.data.ts`): Lucide-like, 24×24 grid, **1.5px stroke, round caps/joins, no fills** (one exception: the favourite star fills when on). 108 glyphs, copied programmatically to `assets/icons/*.svg` and to `components/icons/iconShapes.js` for the `<Icon>` component.
- Default size 19px in rows/rails, 21px in the tab bar, 15–17px inside buttons. Always `currentColor`.
- `assets/icons-sprite.svg` is the app's public sprite; logo files: `assets/logo-horizontal(-reversed).svg`, `logo-stacked(-reversed).svg`, `logo-brass.svg`, `logo-mono-black/white.svg`, `favicon.svg`, `logo-cobrand.svg` (see Brand → Logo system), `assets/logo-mark.svg` (flat, currentColor, generated from `mark.data.ts`), `assets/logo-mark-tile.svg` / `-maskable.svg` / `-512.png` (lit silver app icons), `assets/logo-mark-silver.png` (uploaded render).
- Unicode glyphs carry status (see Content). Emoji are never used. Icon-only controls always have an accessible name. If a glyph you need is missing, use the matching **Lucide** icon at stroke-width 1.5 (CDN) and flag it.

Components

Sources in `components/<group>/Name.jsx`; each component's guide and types in `components/<Name>/README.md` + `components/<Name>/<Name>.d.ts`, styled by `components/bundle.css`. Namespace: `window.SemesterDesignSystem_e9f3ba`.

- **core:** Button, IconButton, Icon, Wordmark (+ Mark), SectionLabel, Avatar
- **forms:** TextField, Select, Segmented, Choice, Switch, Combobox, DateField, VisibilityPicker
- **trust:** SourceBadge, SourceLine, StatusChip, ProvenanceChips, AIResponse (shared `vocabulary.js`)
- **feedback:** EmptyState, LoadingState, ErrorState, SuccessState, ProgressState, StepStatus, PermissionNotice, OfflineStrip, SaveState, Notice
- **surfaces:** ObjectCard, ContextBar, NextSteps, ActionPreview, Dialog, ScreenGuide, OpenIn, QuickActions
- **overlays:** Sheet (+ SourceDetails), UndoToast, FocusBar
- **navigation:** SideRail, TabBar, SystemContextBar, CommandPalette
- **data:** DataTable, Fields, DecisionTrail, MetricTile, ProvenanceChart
- **governance:** PolicyBadge, ConsentBadge, HealthBadge, ApprovalBanner, ErrorSummary, SupportHandoff, PageHeader

Intentional additions (no single source file): `AIResponse` (combines the app's AI disclosure rules into one frame), `Switch` / `Choice` (the app uses native inputs in `.visibility-choice`), `SideRail` (generalises `desk/Sidebar.tsx`), `CommandPalette` (from `unity/Combobox` styling + the brief's ⌘K), `DecisionTrail` (from `unity/DecisionTrail` + console approvals), `SourceLine` (the brand platform's proposed signature device, §2.6).

UI kits

All views follow the codebase's seven navigation areas (`app/src/lib/navareas.ts`): **Today** (what now?), **Plan** (what's coming?), **Learn** (how do I get the work done?), **Help** (who can help?), **Campus** (what's happening, where?), **Progress** (how am I doing?), **You** (settings). Every one of the 64 destinations in `app/src/screens.tsx` appears in the student App directory; 26 are fully recreated, the rest open an outline screen with their area and primary action.

- `ui_kits/student/` — Student (desktop). Today, Notifications · Calendar, Registration, Career, Billing · Courses, Grades, Study, Ask Semester, Write, Sheets, Decks · Support, Family · Campus, Maps, Mail, Community · Me, Degree, Advising · Settings (Appearance), Connect, Privacy & data · App directory.
- `ui_kits/staff/` — Faculty gradebook, TA grading queue, Advisor, Registrar desk, Finance student accounts, Student affairs support requests.
- `ui_kits/console/` — Institution console: Overview, Approvals, Support queue, Integrations, Security operations, Audit log.
- `ui_kits/external/` — Applicant, Guardian (family portal), Alumni, Employer, Marketplace partner.
- `ui_kits/mobile/` — Phone-width app on Parchment and Ink with the five-destination tab bar.

Audience coverage: students ✓ · applicants ✓ · faculty ✓ · TAs ✓ · advisors ✓ · registrars ✓ · finance ✓ · student affairs ✓ · IT/security ✓ · support/ops ✓ · guardians ✓ · alumni ✓ · employers ✓ · marketplace partners ✓.

AI Workflow Router & Benchmark Lab

- `ui_kits/workflow-lab/` — interactive prototype (Internal tools card): 6-input router with forced-v0 rules, confidence, reasons, limits, escalation, prompts, GitHub plan, Supabase prompt, MD/JSON export; sortable 16-column feature matrix; 12 canonical workflows; benchmark grid with persistence tests, persistence ceiling, 0–5 grading, weighted /5 and /100 scores, leaderboard, CSV/Markdown export. Runs are saved on this device only and labelled prototype mode.
- `handoff/workflow-router-app/` — the production app source: Next.js App Router + TypeScript strict + Tailwind + Supabase (@supabase/ssr, magic link, middleware refresh), Zod, Server Actions, export route handlers, migrations `001_core_workflow_router.sql` / `002_benchmark_grading.sql` with org-scoped RLS (owner/admin/member/viewer), `seed.sql` with the 12 workflows, honest manual adapters, persistence + source-structure probes, vitest suites and an RLS test plan. Code files carry `.txt`; run `install.sh`. Not yet typechecked, linted, tested or built.

Master control & company site

- `ui_kits/master-control/` — the hub: product front end, console & back end, business OS, company site & marketing, design system & engineering — every kit, deck, document and hand-off one click away.
- `ui_kits/company-site/` — homepage following the brand platform: hero, "status is part of the product" map (use now / pilot / later), value props, institutions, trust (status, not certifications), footer.

Role lens

- `ui_kits/role-lens/` — every one of the 64 screens through 14 roles (student, applicant, guardian, alumni, faculty, TA, advisor, registrar, finance, student affairs, IT/security, support, employer, partner): access level (full · scoped · shared · read · audited · none), how the screen renders for that role, and its backend (sources, tables, RLS, API). Design-level model in `roles.js`; verify against migrations.

Business workspace

- `ui_kits/business/` — gap analysis, sortable launch tracker (56 items, CSV), 90-day roadmap, pilot/contract checklist, risk register, hiring plan, fundraising readiness; `business-data.js` is the single record. `Operating Pack.html` is the printable document.
- Decks as templates: `templates/investor-deck`, `templates/institution-deck`, `templates/internal-plan-deck`.
- Inputs: `uploads/Business Master Checklist…pdf`, `uploads/now what else…pdf`, founder answers (LLC, solo, bootstrapping, signed paid pilot with DPA/FERPA, HECVAT Full, SAML SSO, VPAT, insurance; Vercel + Supabase; AI never sees student records).

Claude Code toolkit (`handoff/claude-code-toolkit/`)

Repo-ready files that wire this system into Claude Code + Figma MCP without changing the app's architecture: two skills (`audit-semester-design-sync` , `build-semester-ui`), `.mcp.json` , `docs/FIGMA-MAPPING.md` (collections, 49 semantic + 12 primitive variables, 11 components, and this project's proposal tokens marked as *not in code*), a zero-dependency audit + report script, focused vitest proofs, and the three `design-system:*` package scripts. `PROMPTS.md` has the exact Claude Code commands. Every mapped path was checked against `app/design-tokens/semester.tokens.json` .

Design System tab, group by group

Each group in the tab lists what it holds, where it lives, and the rule to keep.

Foundations

- **Colors**: `guidelines/colors-*.html` , `dataviz.html` . Ink chrome, Parchment content, Semester blue 700/400, metals, source semantics on both grounds, provenance-by-pattern data viz. *Rule:* colour is never the only carrier.
- **Type**: `guidelines/type-*.html` . Barlow body, Barlow Condensed headings, Cinzel display (rare), mono for IDs. *Rule:* sentence case, 66ch measure, tabular figures.
- **Spacing**: `guidelines/spacing.html` , `archived/guidelines/radii.html` , `layout.html` , `states-interaction.html` . 2px density steps, 8px grid, corners 3/6/10, breakpoints, hover/press/focus. *Rule:* no scale or bounce on press.
- **Brand**: `components/LogoSystem/preview.html` , `components/LogoMark/preview.html` , `components/WordmarkLockup/preview.html` , `components/Iconography/preview.html` . Mark, wordmark, **logo system** (horizontal, stacked, reversed, brass, mono, favicon, co-brand lockup in `assets/logo-*.svg`), icon set. *Rule:* Semester leads in co-brand lockups, partner logo at equal cap height, never recolour the mark.

Components (`components/<Name>/` , namespace `window.SemesterDesignSystem_e9f3ba`)

- **Actions**: Button, IconButton, QuickActions, ActionPreview. Primary = one per view, consequence copy before confirm.
- **AI**: AIResponse, ProvenanceChips. Always labelled AI-assisted, sources shown, human confirms the action.
- **Data**: DataTable, Fields, DecisionTrail, MetricTile, ProvenanceChart.
- **Forms**: TextField, Select, Segmented, Choice, Switch, Combobox, DateField, VisibilityPicker.
- **Glyph**: Icon, Mark, Wordmark. 1.5px stroke, `currentColor` , accessible names on icon-only controls.
- **Navigation**: SideRail, TabBar, SystemContextBar, CommandPalette (⌘K).
- **States**: Empty, Loading, Error, Success, Progress, StepStatus, Permission, Offline, SaveState, Notice, UndoToast.
- **Surfaces**: ObjectCard, ContextBar, NextSteps, Dialog, Sheet, FocusBar, ScreenGuide, OpenIn.
- **Trust & provenance**: SourceBadge, SourceLine, StatusChip. The official / connected / personal / AI distinction lives here.
- **Patterns** (`guidelines/pattern-*.html`): AI answer anatomy, provenance ladder, consent & sharing, consequential actions.

Product

- **Student app**: `ui_kits/student/`. All 64 destinations across Today, Plan, Learn, Help, Campus, Progress and You, with Tweaks and live Ask Semester.
- **Mobile**: `ui_kits/mobile/`. Five-destination tab bar on Parchment and Ink.
- **Product**: `ui_kits/community/`, the community and live marketplace: feed, groups, events, Q&A, clubs and leaders, mentorship, campus marketplace with live drops, messages, privacy and safety. Campus-verified only; never touches student records.
- **Staff workspaces**: `ui_kits/staff/`. Faculty, TA, advisor, registrar, finance, student affairs.
- **Institution console**: `ui_kits/console/`. Overview, approvals, support, integrations, security, audit.
- **External audiences**: `ui_kits/external/`. Applicant, guardian, alumni, employer, marketplace partner.
- **Forms & interfaces**: `ui_kits/forms/`. 12 working forms, role capabilities, API, command and event contracts.
- **Wireframes**: `ui_kits/wireframes/`. Every screen × role in low-fi.

Maps of the whole

- **Master control**: `ui_kits/master-control/`. The hub; every kit, deck, document and hand-off is one click away. *Rule:* every new file gets linked here.
- **Master catalog**: `ui_kits/master-catalog/`. 51 roles, 673 screens, 319 workflow steps (17 workflows), 88 systems and 82 docs, each with a status.
- **Atlas**: `ui_kits/atlas/`. Each role's features, screens and functions.
- **Maps card**: `guidelines/maps-of-the-whole.html`. **Documents card**: `guidelines/company-documents.html` (Business group). **Iconography rules**: `guidelines/iconography-rules.html` (Brand group).
- **Role lens**: `ui_kits/role-lens/`. Access level, rendering, tables, RLS and API per role × screen.

Architecture

- **Architecture / Education OS architecture**: `ui_kits/architecture/`. OS map against legacy systems, full stack and data (20 tables), CQRS read-model pipeline, lag simulator, AI gateway, migration phases. *Rule:* the institution stays the system of record; AI never receives student records.

Company

- **Business**: `ui_kits/business/`. Workspace (gap analysis, launch tracker, roadmap, pilot checklist, risks, hiring, fundraising), Operating Manual, Operating Pack, Market Research, Vanderbilt one-pager, invite email, campus map.
- **Operations**: `ui_kits/ops/`. 12 consoles: social media control, finance, marketing, sales and RevOps, customer success, support, trust & safety plus community and marketplace ops, legal and compliance, people, IT and security, data, partnerships. Role-gated. *Rule:* everything is labelled sample data and contains no student PII.
- **Investor**: `ui_kits/investor/`. Investor and board portal.
- **Company site**: `ui_kits/company-site/`. Home, institutions, pilot, pricing, trust, company.
- **Compliance**: `ui_kits/compliance/`. WCAG 2.2, FTC proof rules, release gate. *Rule:* state your status; never claim a certification you don't hold.

- **Product brief package**: `docs/brief/`. Index, one-pager, FAQ. Overview deck and role sheets are next.

- **Architecture cards**: `guidelines/arch-*.html` — stack layers, read-model pipeline & freshness, AI gateway, environments.

Tooling

- **Tooling cards**: `guidelines/tooling-*.html` — Claude Code toolkit, semester-core, semester-platform, AI workflow router & benchmark lab.
- **Internal tools**: `ui_kits/workflow-lab/`, the AI workflow router and benchmark lab.
- **Tooling**: `handoff/claude-code-toolkit/`, `handoff/semester-core/`, `handoff/workflow-router-app/`. See `handoff/README.md`.

Templates (`templates/`, entry `<Slug>.dc.html` + `ds-base.js`)

- Decks: company-deck (14 slides), investor-deck, institution-deck, internal-plan-deck.
- Screens: student-screen (rail + context bar + main/aside), console-screen (environment bar + DataTable + DecisionTrail).
- Reports: pilot results report.

What's next

See `NEXT-SESSION.md` for the ordered backlog: render every missing screen, capability map, workflow runner, site operations, the full public site, personalization capabilities, formal documents, the rest of the brief package, and the PPTX exports.

Index

- `styles.css` — entry; `@import`s only.
- `tokens.json` (compiled to `tokens.css`) and `components/bundle.css`; the original `tokens/` stylesheets are in `archived/tokens/`.
- `fonts/` — Barlow 400/500/700, Barlow Condensed 400/600, Cinzel (OFL).
- `assets/` — logos, app icons, `icons/` (108 SVGs), sprite.
- `guidelines/` — specimen cards (Colors, Type, Spacing, Brand, Patterns) and deep-dive docs:
- `01-provenance-and-trust.md` — the ladder, made vs AI, where cues go
- `02-ai-governance.md` — Ask Semester, AIResponse anatomy, course policy
- `03-consequential-actions.md` — undo vs preview vs two-person duty
- `04-consent-and-family.md` — default private, one decision at a time, access history
- `05-institutional-surfaces.md` — context bars, tables, authority scope
- `06-accessibility.md` — contrast, structure, reader settings, brand-exempt zones
- `07-layout-and-responsive.md` — tiers, shell, page grid, destinations
- `08-content-templates.md` — status sentence, empty/error/success, incidents, onboarding
- `components/`, `ui_kits/` — see above.

- `assets/notes/SKILL.from-standalone.md`, `github.md`, `thumbnail.html`.

Caveats: Atkinson Hyperlegible (the app's legibility body option) and Latin-Extended font subsets were not copied. The production app offers 13 grounds × 11 accents; only Ink and Parchment ship here.

Where the system lives

- **Tokens:** `tokens.json` is the one source for colours, type, spacing, radii, shadows, sizes, layers, leading and tracking; `tokens.css` is generated from it. Values that are not tokens (the sterling gradient, calc type scales, density steps, motion) and the base and component styles are in `components/bundle.css`.
- **Components:** the bundle is `components/bundle.js` (namespace `window.SemesterDesignSystem_e9f3ba`); each component has `components/<Name>/README.md`, `<Name>.d.ts` and a live `preview.html`; all types together are in `components/index.d.ts`.
- **Moved aside:** the old stylesheets, the original `_ds_bundle.js`, group cards, `_ds_manifest.json` and specimen pages now covered by tokens are in `archived/`.

Design tokens

Content defaults to Parchment; [data-theme="dark"] flips content to Ink. Chrome (rail, tab bar, console header) is always Ink. Source and status hues always travel with a word and a glyph.

Colour — primitives

	TOKEN	VALUE	USAGE
	`ink-void`	`#040507`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`ink-bg`	`#090a0e`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`ink-panel`	`#12141a`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`ink-hero`	`#191c23`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`ink-raise`	`#22262f`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`ink-fg`	`#eceef2`	Ink ground ramp: near-black, cooled. Default dark ground and always-dark chrome.
	`parchment-void`	`#e8e4dc`	Parchment ground ramp: warm off-white. Default light content ground.
	`parchment-bg`	`#f4f1ea`	Parchment ground ramp: warm off-white. Default light content ground.
	`parchment-panel`	`#fbf9f4`	Parchment ground ramp: warm off-white. Default light content ground.
	`parchment-hero`	`#ffffff`	Parchment ground ramp: warm off-white. Default light content ground.
	`parchment-raise`	`#ffffff`	Parchment ground ramp: warm off-white. Default light content ground.
	`parchment-fg`	`#1b1a17`	Parchment ground ramp: warm off-white. Default light content ground.
	`graphite-bg`	`#16181c`	Graphite: alternate dark ground.
	`graphite-panel`	`#1f2229`	Graphite: alternate dark ground.
	`graphite-raise`	`#333843`	Graphite: alternate dark ground.
	`sterling`	`#d4d9e2`	Semester color.
	`sterling-bright`	`#f6f8fb`	Sterling metal, for the mark and chrome highlights.
	`sterling-deep`	`#9ca3b2`	Sterling metal, for the mark and chrome highlights.
	`sterling-shade`	`#4c5561`	Sterling metal, for the mark and chrome highlights.
	`brass`	`#d8c79a`	Semester color.

	TOKEN	VALUE	USAGE
	`brass-bright`	`#f2e7c8`	Brass metal, for editorial accents.
	`brass-deep`	`#b09f76`	Brass metal, for editorial accents.
	`brass-shade`	`#695a33`	Brass metal, for editorial accents.
	`blue-100`	`#eef6ff`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-200`	`#d6ebff`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-300`	`#b5d9fd`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-400`	`#94bce3`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-500`	`#749dc4`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-600`	`#597ea3`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-700`	`#416180`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-800`	`#2c455d`	Semester blue (the Industry ramp): interactive focus and trust.
	`blue-900`	`#1d2d3d`	Semester blue (the Industry ramp): interactive focus and trust.
	`neutral-100`	`#f5f5f8`	Industry neutral ramp.
	`neutral-200`	`#e7e7ea`	Industry neutral ramp.
	`neutral-300`	`#d4d4d7`	Industry neutral ramp.
	`neutral-400`	`#b7b7ba`	Industry neutral ramp.
	`neutral-500`	`#98989b`	Industry neutral ramp.
	`neutral-600`	`#7a7a7d`	Industry neutral ramp.
	`neutral-700`	`#5d5d60`	Industry neutral ramp.
	`neutral-800`	`#424244`	Industry neutral ramp.
	`neutral-900`	`#2b2b2d`	Industry neutral ramp.
	`official-ink`	`#2c455d`	Official status primitive: text on light grounds.
	`official-tint`	`#b5d9fd`	Official status primitive: text on dark grounds.
	`connected-ink`	`#2d5c61`	Connected status primitive: text on light grounds.
	`connected-tint`	`#9fcbcd`	Connected status primitive: text on dark grounds.
	`personal-ink`	`#4a4b4f`	Personal status primitive: text on light grounds.
	`personal-tint`	`#b9b9bd`	Personal status primitive: text on dark grounds.
	`ai-ink`	`#544a7f`	Ai status primitive: text on light grounds.
	`ai-tint`	`#bcb2df`	Ai status primitive: text on dark grounds.

	TOKEN	VALUE	USAGE
	`pending-ink`	`#7a5310`	Pending status primitive: text on light grounds.
	`pending-tint`	`#e2c27a`	Pending status primitive: text on dark grounds.
	`risk-ink`	`#a3301f`	Risk status primitive: text on light grounds.
	`risk-tint`	`#f0907a`	Risk status primitive: text on dark grounds.
	`success-ink`	`#3e6141`	Success status primitive: text on light grounds.
	`success-tint`	`#a9c9a3`	Success status primitive: text on dark grounds.
	`chrome-text-secondary`	`rgba(236,238,242,.64)`	Chrome (rail, tab bar, console bar): always dark.
	`chrome-line`	`rgba(236,238,242,.10)`	Chrome (rail, tab bar, console bar): always dark.
	`chrome-active`	`rgba(148,188,227,.14)`	Chrome (rail, tab bar, console bar): always dark.
	`scrim`	`rgba(4,5,7,.5)`	Backdrop behind sheets and dialogs.

Colour — semantic (light · dark)

LIGHT · DARK	TOKEN	LIGHT	DARK	USAGE
	`surface-sunken`	`{parchment-void}`	`{ink-void}`	Wells and recessed areas below the canvas.
	`surface-canvas`	`{parchment-bg}`	`{ink-bg}`	Page background.
	`surface-base`	`{parchment-panel}`	`{ink-panel}`	Panels and cards on the canvas.
	`surface-quiet`	`#f1ede4`	`#161820`	Quiet fills: hover rows, secondary panels.
	`surface-raised`	`{parchment-hero}`	`{ink-hero}`	Raised surfaces: inputs, popovers, hero cards.
	`surface-overlay`	`{parchment-raise}`	`{ink-raise}`	Overlays: menus, sheets, dialogs.
	`text-primary`	`{parchment-fg}`	`{ink-fg}`	Body and heading text on every surface-* token.
	`text-secondary`	`rgba(27,26,23,.68)`	`rgba(236,238,242,.64)`	Supporting text on surface-canvas, surface-base and surface-raised.
	`text-tertiary`	`rgba(27,26,23,.52)`	`rgba(236,238,242,.42)`	Placeholders and least-important meta text. Contrast note: about 3.4–3.7:1 on the surfaces in both themes, below 4.5:1 for body text; keep it to large text (24px+) or non-essential hints, and use text-secondary for anything a person must read.
	`text-disabled`	`rgba(27,26,23,.52)`	`rgba(236,238,242,.42)`	Disabled control text.
	`text-on-action`	`#ffffff`	`#08090c`	Text on action-primary fills.
	`text-link`	`{blue-700}`	`{blue-300}`	Inline links on surface-* grounds.

LIGHT · DARK	TOKEN	LIGHT	DARK	USAGE
	`border-subtle`	`rgba(27,26,23,.09)`	`rgba(236,238,242,.08)`	Hairline dividers.
	`border-default`	`rgba(27,26,23,.16)`	`rgba(236,238,242,.14)`	Default control and card borders.
	`border-strong`	`rgba(27,26,23,.34)`	`rgba(236,238,242,.30)`	Hover borders and emphasized edges.
	`action-primary`	`{blue-700}`	`{blue-400}`	Primary button fill; one per decision region.
	`action-primary-hover`	`{blue-800}`	`{blue-300}`	Primary button hover.
	`action-primary-press`	`{blue-900}`	`{blue-500}`	Primary button pressed.
	`action-secondary`	`rgba(65,97,128,.08)`	`rgba(148,188,227,.10)`	Secondary button hover wash.
	`action-secondary-hover`	`rgba(65,97,128,.14)`	`rgba(148,188,227,.18)`	Secondary button pressed wash.
	`focus-color`	`{blue-600}`	`{blue-400}`	Focus ring on every interactive element.
	`accent-editorial`	`{brass-shade}`	`{brass}`	Editorial accent (kickers, rules) in brand moments.
	`source-official`	`{official-ink}`	`{official-tint}`	Text and glyph color for the official source state. Every state also carries a word and a glyph.
	`source-official-line`	`rgba(44,69,93,.45)`	`rgba(181,217,253,.40)`	Border for the official source state. Every state also carries a word and a glyph.
	`source-official-wash`	`rgba(44,69,93,.08)`	`rgba(181,217,253,.08)`	Background wash for the official source state. Every state also carries a word and a glyph.
	`source-connected`	`{connected-ink}`	`{connected-tint}`	Text and glyph color for the connected source state. Every state also carries a word and a glyph.
	`source-connected-line`	`rgba(45,92,97,.45)`	`rgba(159,203,205,.40)`	Border for the connected source state. Every state also carries a word and a glyph.
	`source-connected-wash`	`rgba(45,92,97,.08)`	`rgba(159,203,205,.08)`	Background wash for the connected source state. Every state also carries a word and a glyph.
	`source-personal`	`{personal-ink}`	`{personal-tint}`	Text and glyph color for the personal source state. Every state also carries a word and a glyph.
	`source-personal-line`	`rgba(74,75,79,.40)`	`rgba(185,185,189,.34)`	Border for the personal source state. Every state also carries a word and a glyph.
	`source-personal-wash`	`rgba(74,75,79,.06)`	`rgba(185,185,189,.06)`	Background wash for the personal source state. Every state also carries a word and a glyph.
	`source-ai`	`{ai-ink}`	`{ai-tint}`	Text and glyph color for the ai source state. Every state also carries a word and a glyph.
	`source-ai-line`	`rgba(84,74,127,.45)`	`rgba(188,178,223,.42)`	Border for the ai source state. Every state also carries a word and a glyph.

LIGHT · DARK	TOKEN	LIGHT	DARK	USAGE
	`source-ai-wash`	`rgba(84,74,127,.08)`	`rgba(188,178,223,.09)`	Background wash for the ai source state. Every state also carries a word and a glyph.
	`status-pending`	`{pending-ink}`	`{pending-tint}`	Text and glyph color for the pending status state. Every state also carries a word and a glyph.
	`status-pending-line`	`rgba(160,112,24,.50)`	`rgba(226,194,122,.45)`	Border for the pending status state. Every state also carries a word and a glyph.
	`status-pending-wash`	`rgba(196,146,48,.12)`	`rgba(226,194,122,.09)`	Background wash for the pending status state. Every state also carries a word and a glyph.
	`status-danger`	`{risk-ink}`	`{risk-tint}`	Text and glyph color for the danger status state. Every state also carries a word and a glyph.
	`status-danger-line`	`rgba(163,48,31,.45)`	`rgba(240,144,122,.45)`	Border for the danger status state. Every state also carries a word and a glyph.
	`status-danger-wash`	`rgba(163,48,31,.08)`	`rgba(240,144,122,.09)`	Background wash for the danger status state. Every state also carries a word and a glyph.
	`status-success`	`{success-ink}`	`{success-tint}`	Text and glyph color for the success status state. Every state also carries a word and a glyph.
	`status-success-line`	`rgba(62,97,65,.45)`	`rgba(169,201,163,.42)`	Border for the success status state. Every state also carries a word and a glyph.
	`status-success-wash`	`rgba(62,97,65,.08)`	`rgba(169,201,163,.08)`	Background wash for the success status state. Every state also carries a word and a glyph.
	`status-neutral`	`{text-secondary}`	`{text-secondary}`	Text and glyph color for the neutral status state. Every state also carries a word and a glyph.
	`chrome-surface`	`{ink-bg}`	`{ink-bg}`	Chrome (rail, tab bar, console bar): always dark.
	`chrome-surface-raised`	`{ink-panel}`	`{ink-panel}`	Chrome (rail, tab bar, console bar): always dark.
	`chrome-text`	`{ink-fg}`	`{ink-fg}`	Chrome (rail, tab bar, console bar): always dark.
	`brand-canvas`	`{surface-canvas}`	`{surface-canvas}`	Brand role shared with the company site.
	`brand-surface`	`{surface-base}`	`{surface-base}`	Brand role shared with the company site.
	`brand-ink`	`{text-primary}`	`{text-primary}`	Brand role shared with the company site.
	`brand-muted`	`{text-secondary}`	`{text-secondary}`	Brand role shared with the company site.
	`brand-accent`	`{action-primary}`	`{action-primary}`	Brand role shared with the company site.
	`brand-focus`	`{focus-color}`	`{focus-color}`	Brand role shared with the company site.

Typography

Barlow for product reading, Barlow Condensed for headings, labels and controls, Cinzel for editorial display moments only, system mono for identifiers and timestamps. Sizes scale with —text-scale (see components/bundle.css).

FAMILY	STACK
`display`	'Cinzel','Barlow Condensed',Georgia,serif
`heading`	'Barlow Condensed',system-ui,sans-serif
`body`	'Barlow',system-ui,sans-serif
`mono`	`ui-monospace,'SF Mono',Menlo,Consolas,monospace`

FONT FILE	FAMILY	WEIGHT
`fonts/Barlow-400-latin.woff2`	Barlow	400
`fonts/Barlow-500-latin.woff2`	Barlow	500
`fonts/Barlow-700-latin.woff2`	Barlow	700
`fonts/BarlowCondensed-400-latin.woff2`	Barlow Condensed	400
`fonts/BarlowCondensed-600-latin.woff2`	Barlow Condensed	600
`fonts/Cinzel-latin.woff2`	Cinzel	400 600

Roles

`role-editorial` · 42px / 1.1 · 500 · display
Editorial headlines on the company site; Cinzel moments only.

Your lab report is due Thursday

`role-display` · 26px / 1.15 · 600 · heading
Display numbers and hero headings in product.

Your lab report is due Thursday

`role-page` · 24px / 1.2 · 600 · heading
Page titles (h1).

Your lab report is due Thursday

`role-section` · 20px / 1.2 · 600 · heading
Section headings (h2).

Your lab report is due Thursday

`role-card-title` · 18px / 1.25 · 600 · heading
Card titles.

Your lab report is due Thursday

`role-body` · 16px / 1.55 · 400 · body
Body copy.

Your lab report is due Thursday

`role-control` · 14px / 1.2 · 600 · heading
Button and control labels.

Your lab report is due Thursday

`role-compact` · 12px / 1.45 · 400 · body
Dense rows and compact UI.

Your lab report is due Thursday

`role-label` · 11px / 1.3 · 500 · body
Field labels and meta.

Your lab report is due Thursday

`role-caption` · 11px / 1.45 · 400 · body
Captions.

Your lab report is due Thursday

`role-kicker` · 10.5px / 1.3 · 500 · body
Uppercase kickers above headings.

YOUR LAB REPORT IS DUE THURSDAY

`role-numeric` · 22px / 1.15 · 600 · heading
Metric values (tabular numerals).

Your lab report is due Thursday

Spacing

8px macro grid. The density-aware —sp-1…7 steps stay in components/bundle.css (they multiply by —density).

TOKEN	VALUE		USAGE
`space-1`	8px		8px macro grid for layout gaps and section spacing.
`space-2`	16px		8px macro grid for layout gaps and section spacing.
`space-3`	24px		8px macro grid for layout gaps and section spacing.
`space-4`	32px		8px macro grid for layout gaps and section spacing.
`space-5`	40px		8px macro grid for layout gaps and section spacing.
`space-6`	48px		8px macro grid for layout gaps and section spacing.
`space-8`	64px		8px macro grid for layout gaps and section spacing.

TOKEN	VALUE		USAGE
`space-10`	80px		8px macro grid for layout gaps and section spacing.
`space-12`	96px		8px macro grid for layout gaps and section spacing.
`page-pad`	18px		Page and chrome padding at narrow widths; widens at 760px and 1100px.
`chrome-pad`	18px		Page and chrome padding at narrow widths; widens at 760px and 1100px.

Radius

"Drawn" corners, the app default.

TOKEN	VALUE		USAGE
`r-sm`	3px		Drawn corners: controls.
`r-md`	6px		Drawn corners: cards.
`r-lg`	10px		Drawn corners: surfaces and sheets.
`shape-pill`	999px		Pills and chips.

Shadow

Border first, shadow second.

TOKEN	VALUE		USAGE
`lift-1`	`0 1px 2px rgba(0,0,0,.4)`		Lift on dark grounds. Border first, shadow second.
`lift-2`	`0 1px 2px rgba(0,0,0,.45),0 6px 18px rgba(0,0,0,.35)`		Lift on dark grounds. Border first, shadow second.
`lift-3`	`0 1px 3px rgba(0,0,0,.5),0 12px 36px rgba(0,0,0,.5)`		Lift on dark grounds. Border first, shadow second.
`shadow-sm`	`0 1px 2px rgba(43,43,45,.14)`		Ink-tinted shadow on light grounds. Border first, shadow second.
`shadow-md`	`0 3px 10px rgba(43,43,45,.16)`		Ink-tinted shadow on light grounds. Border first, shadow second.

TOKEN	VALUE	USAGE
`shadow-lg`	`0 12px 32px rgba(43,43,45,.22)`	Ink-tinted shadow on light grounds. Border first, shadow second.

Sizes and targets

TOKEN	VALUE	USAGE
`rail-w`	`248px`	Side rail width.
`rail-w-compact`	`72px`	Collapsed side rail width.
`layout-measure`	`100%`	Default content measure.
`layout-operational`	`1180px`	Max width of operational (console) pages.
`target-min`	`24px`	Smallest hit target.
`target-compact`	`32px`	Compact control height (dense rows).
`target-icon`	`40px`	Icon button size.
`target-primary`	`44px`	Standard control height.
`focus-ring-width`	`2px`	Focus outline width.
`focus-ring-offset`	`2px`	Focus outline offset.
`blur-chrome`	`14px`	Backdrop blur behind translucent chrome.

Layers (z-index)

TOKEN	VALUE	USAGE
`layer-base`	`0`	Stacking order: base.
`layer-raised`	`1`	Stacking order: raised.
`layer-sticky`	`20`	Stacking order: sticky.
`layer-chrome`	`21`	Stacking order: chrome.
`layer-overlay`	`80`	Stacking order: overlay.
`layer-menu`	`90`	Stacking order: menu.
`layer-skip`	`100`	Stacking order: skip.
`layer-curtain`	`1500`	Stacking order: curtain.

Leading (line height)

TOKEN	VALUE	USAGE
`leading-none`	`1`	Line height: none.
`leading-display-xl`	`1.08`	Line height: display-xl.
`leading-display-lg`	`1.1`	Line height: display-lg.
`leading-display`	`1.15`	Line height: display.
`leading-display-sm`	`1.2`	Line height: display-sm.
`leading-display-xs`	`1.25`	Line height: display-xs.
`leading-tight`	`1.3`	Line height: tight.
`leading-tight-plus`	`1.35`	Line height: tight-plus.
`leading-normal`	`1.45`	Line height: normal.
`leading-relaxed`	`1.5`	Line height: relaxed.
`leading-relaxed-plus`	`1.55`	Line height: relaxed-plus.
`leading-loose`	`1.6`	Line height: loose.

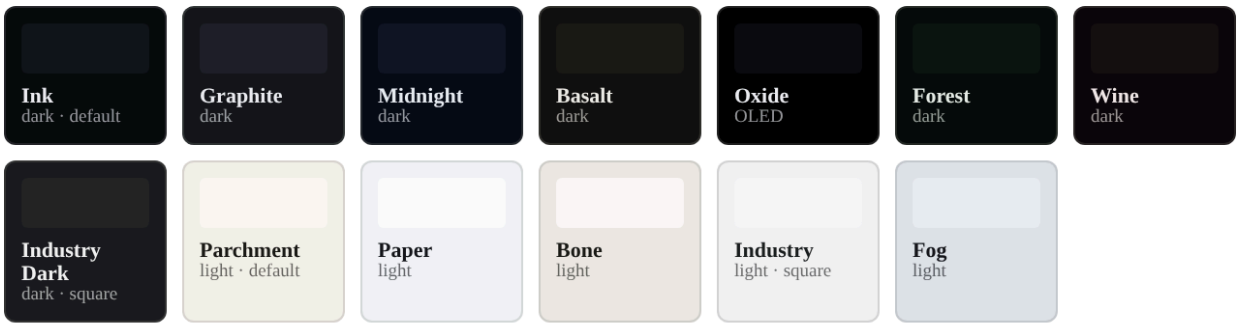
Tracking (letter spacing)

TOKEN	VALUE	USAGE
`tracking-heading`	`-0.015em`	Heading letter spacing.
`tracking-kicker`	`0.08em`	Kicker (uppercase label) letter spacing.
`tracking-caps`	`0.1em`	All-caps letter spacing.
`tracking-wordmark`	`0.22em`	Wordmark letter spacing.

Foundations

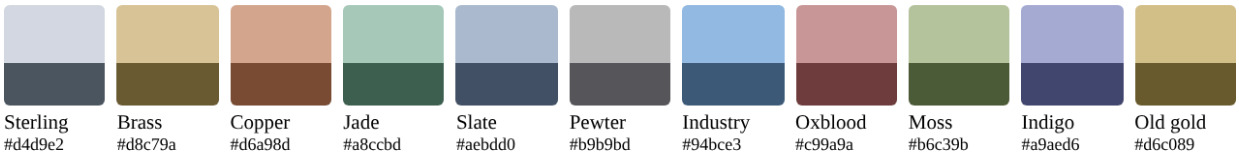
All 13 grounds (reader setting)

The app lets each reader choose a ground; this system ships Ink + Parchment



The 11 metal accents

base (on dark) and shade (text on light) — desaturated metals and stones



Industry → Semester blue in this system. Accent colours identify courses and action; they never encode success or error.

Metals

Sterling (wordmark/chrome), Brass (editorial accent)



Components on Ink

The same components with [data-theme=

!Assignment · Action required

Lab report 3: Enzyme kinetics

◆Official · BIO 210Institution verified by BIO 210

Due Thu 9 Oct · 23:59

Open draft View rubric

Submit Save draft

✦AI-assistedWritten with Semester's assistant. It is not an official answer — check anything you act on.

⇄ConnectedSynced from a connected account or feed. Semester has not checked it with your institution.

!Hold on account

⊘You are offline. You can keep working — 2 changes will sync when you are

Data visualisation encoding

Provenance by pattern: solid, outline, hatched, dotted — survives greyscale

Official / connected	Yours	Estimated / AI-derived	Out of date
data-enc="solid"	data-enc="outline"	data-enc="hatched"	data-enc="dotted"

Full type scale

Every —type-* step with its job; all × —text-scale

--type-2xl	28px	The figure a screen is about
--type-xl	26px	The one heading on a screen
--type-display-lg	24px	Screen title
--type-display	22px	Figure read at a glance
--type-display-sm	20px	Section heading
--type-display-xs	19px	Card or item title
--type-lg	18px	A card’s own title
--type-md	17px	List rows, the thing you tap
--type-base	16px	Body
--type-control	14px	Inputs and buttons
--type-sm-plus	12.5px	A caption with something to say
--type-sm	12px	Captions, second-rank prose
--type-xs	11px	Labels, kickers, meta
--type-2xs-plus	10.5px	Caps kicker over a section

Editorial display — Cinzel

Major statements and editorial moments only

The whole of the semester.

Fall Term · Northfield University

--font-display · Cinzel 400–600 · 42–56px · leading 1.08

Headings — Barlow Condensed

Page, section, card, and control labels

Page title · 24	**Today, Tuesday 7 October**
Display · 26	**Degree progress**
Section · 20	**This week in BIO 210**
Card · 18	**Registration opens tomorrow**

Body — Barlow

Product reading, forms, operational text

Body 16 / 1.55. Your lab report is due Thursday. The methods section is drafted; results and discussion remain. Start with the data table — it is the part the rubric weighs most.

Row 17 — the thing you tap

Control 14 — inputs & buttons

Caption 12 — second-rank prose

Meta 11 — labels and timestamps

Kickers, numerals & mono

Caps kickers, tabular figures, identifiers

Official · Updated today
Begin today

3.62 GPA · 94 / 120 cr

req_7Hk2Q9
2026-10-03 14:02 UTC

Spacing scales

Density-aware 2px steps for UI · 8px grid for layout

sp-1	2px
sp-2	4px
sp-3	6px
sp-4	8px
sp-5	10px
sp-6	12px
sp-7	16px

space-1	8px
space-2	16px
space-3	24px
space-4	32px
space-6	48px
space-8	64px

Density settings

Comfortable 1 · Snug .86 · Tight .74 — scales the 2px steps, never targets

COMFORTABLE · 1

Assignment
Lab report 3

Due Thu · 23:59

SNUG · 0.86

Assignment
Lab report 3

Due Thu · 23:59

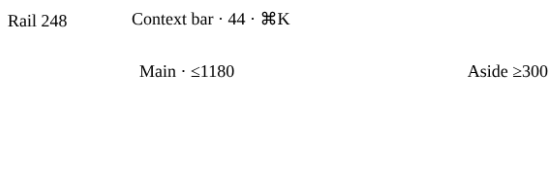
TIGHT · 0.74

Assignment
Lab report 3

Due Thu · 23:59

Layout & breakpoints

Rail + context bar + content; page padding by tier



<760	Tab bar · page-pad 18 · rail 0
760–1099	Compact rail 72 · pad 20
≥1100	Rail 248 · pad 30
READING	66ch measure
OPS MAX	1180px

Motion

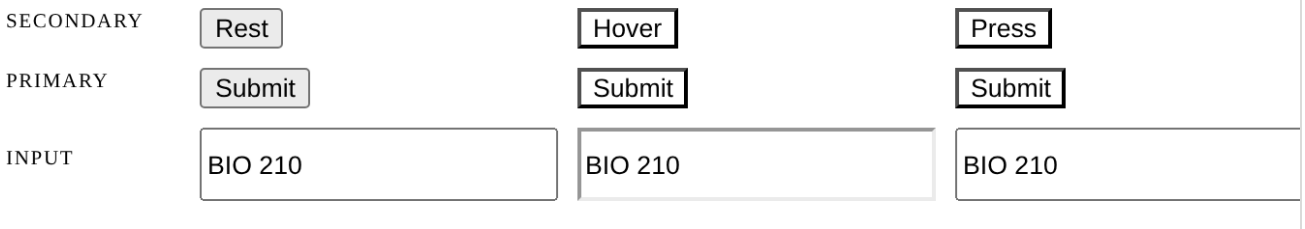
Role durations · zeroed by reduced motion and Less motion

fast	standard	slow	sheet	progress
130ms	180ms	240ms	280ms	180ms linear
hover, press	save, insert	panel	sheet, dialog	progress

ease cubic-bezier(.22,1,.36,1) · emphasized (.2,.8,.2,1) · no bounces, no parallax, no looping decoration

Interaction states

Rest, hover, press, focus, disabled — no scale, no bounce

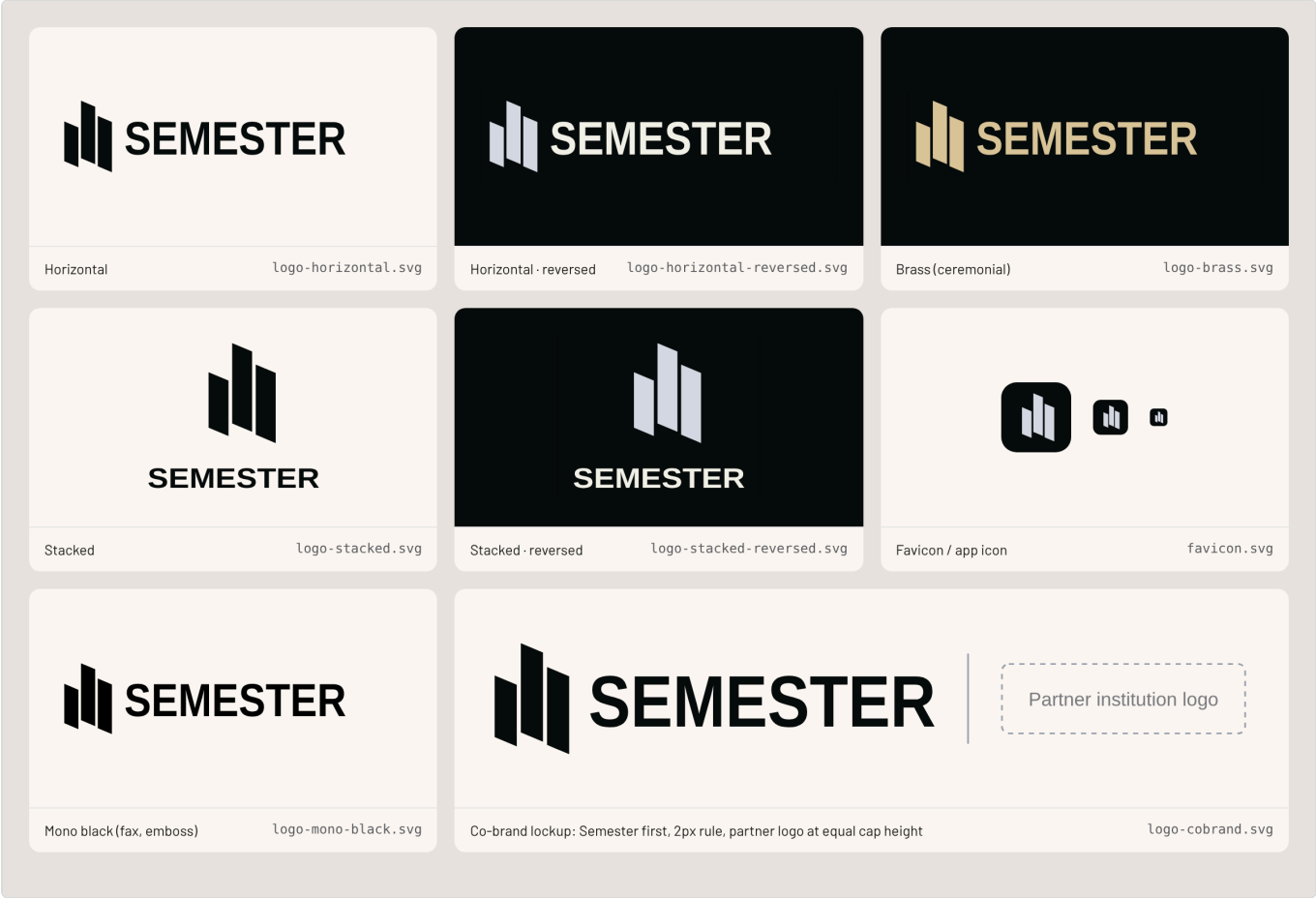


Hover = 8% blue wash · Press = 14% / one ramp step darker · Focus = 2px blue-600 ring, 2px offset · Disabled = 45% opacity + not-allow

Brand

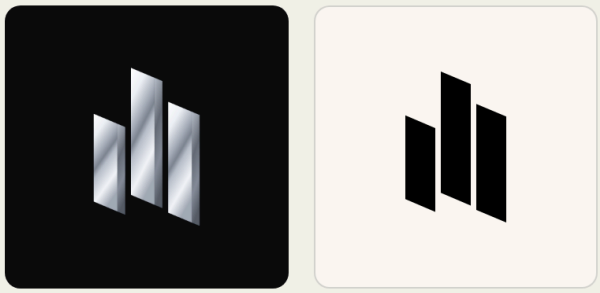
Logo system

Horizontal, stacked, mono, reversed, favicon, co-brand



Logo mark

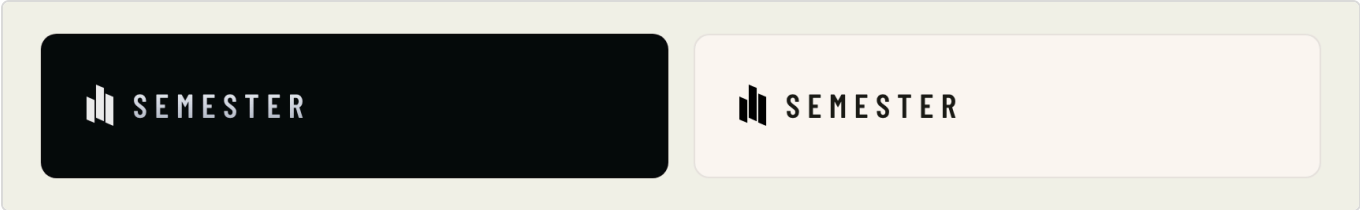
Three slabs — a term rising, cresting, running out



Lit silver on Ink for app icons and splash; flat currentColor everywhere in-app so it follows the ground. Never recolour into the accent, add glow, or place on photography.

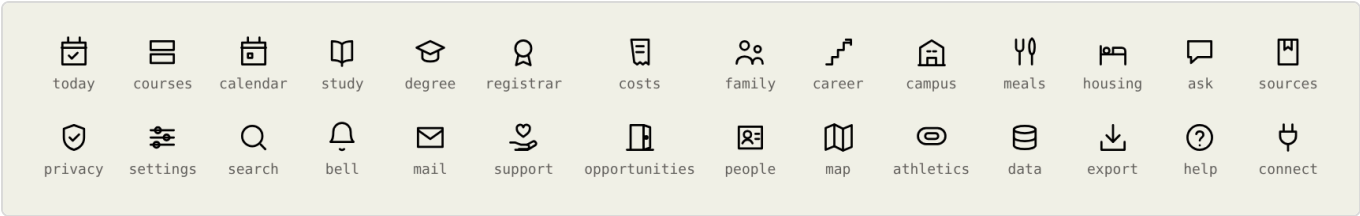
Wordmark lockup

Mark + SEMESTER in Barlow Condensed, tracked



Iconography

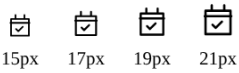
Lucide-like, 24 grid, 1.5 stroke, round caps — the app's own set



Iconography rules

Sizes, stroke, colour, accessible names, status glyphs, files

Sizes



15–17 in buttons · 19 in rows and rails · 21 in the tab bar

Drawing

24×24 grid, 1.5px stroke, round caps and joins, no fills (favourite star fills when on). Always currentColor. 108 glyphs.

Missing glyph
Matching Lucide icon at stroke 1.5, flagged

Status glyphs

◆ official ○ yours ✦ AI

⇄ connected ? review ! attention

✓ done ⊘ offline … working ◌

sample

No emoji. Icon-only controls always have an accessible name.

Voice — prefer / avoid

From docs/BRAND-PLATFORM.md §1.7

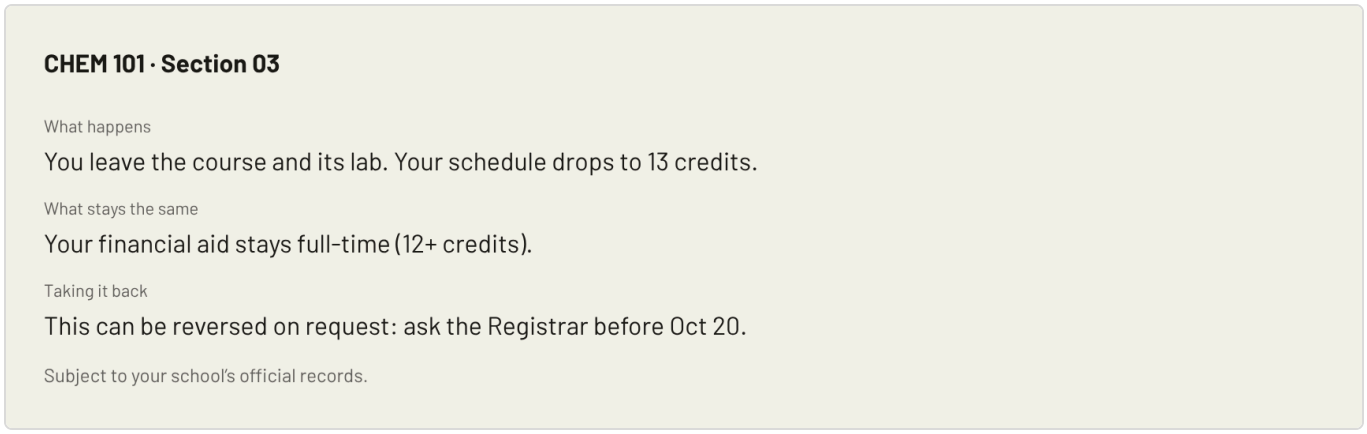
✓ helps you organise · turns … into	✕ revolutionises · transforms · supercharge
✓ alongside your school's systems	✕ replaces · all-in-one · everything school
✓ AI-assisted, source-linked · Ask Semester	✕ AI-powered · intelligent · knows you · magic
✓ built with controls for [named control]	✕ secure · private · compliant · certified
✓ planned · in private beta · in a pilot	✕ launching soon · coming Spring
✓ can · is designed to	✕ guarantees · always · never fails · 100%
✓ student	✕ user · learner
✓ Your Canvas calendar hasn't updated since Tuesday. Deadlines may have moved. Reconnect it.	✕ Oops! Something went wrong 😕

Components

Every component in `components/bundle.js`, rendered from its preview. Each entry gives its guide (README.md) and its props (from `components/index.d.ts`).

Actions

ActionPreview



What is about to happen, said before it happens — What happens / Exactly what / What stays / Taking it back.

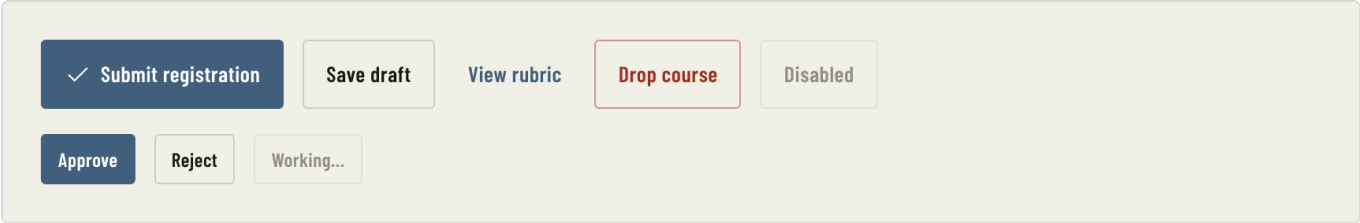
```
<ActionPreview subject="Drop CHEM 101" says="You leave the course and its waitlist." recovery=
{{kind:"request",how:"ask the Registrar before Oct 20."}} />
```

Put inside Dialog for consequential, irreversible or financial actions.

PROPS

```
import * as React from 'react';
/**
 * What is about to happen, said before it happens — What happens / Exactly what / What stays / Taking it back.
 */
export interface ActionPreviewProps { subject?: string; says: React.ReactNode; exactly?: React.ReactNode;
doesNotChange?: React.ReactNode; recovery: { kind: "undo" | "request" | "none"; how?: string }; subjectTo?:
React.ReactNode; provenance?: React.ReactNode; }
export declare function ActionPreview(props: ActionPreviewProps): JSX.Element;
```

Button



Action button — one primary per decision region; 44px tall by default, condensed-caps-free label in Barlow Condensed 600.

```
<Button variant="primary" icon="check">Submit registration</Button>
```

primary (Semester blue fill), secondary (hairline), ghost (text), danger (red ink, hairline — never a red fill). size="sm" = 32px for dense rows.

PROPS

```
import * as React from 'react';
/**
 * Action button — one primary per decision region; 44px tall by default, condensed-caps-free label in Barlow
Condensed 600.
 * @startingPoint section="Actions" subtitle="Action button — one primary per decision region; 44px tall by
default, condensed" viewport="700x400"
 */
export interface ButtonProps { variant?: "primary" | "secondary" | "ghost" | "danger"; size?: "md" | "sm";
icon?: string; iconAfter?: string; block?: boolean; loading?: boolean; disabled?: boolean; children?:
React.ReactNode; onClick?: (e: any) => void; type?: "button" | "submit"; }
export declare function Button(props: ButtonProps): JSX.Element;
```

IconButton



Icon-only control with a required accessible name (also its tooltip).

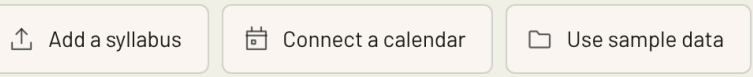
```
<IconButton icon="bell" label="Notifications" />
```

40px (md) / 32px (sm). Never ship an icon-only control without `label`.

PROPS

```
import * as React from 'react';
/**
 * Icon-only control with a required accessible name (also its tooltip).
 */
export interface IconButtonProps { icon: string; label: string; size?: "md" | "sm"; pressed?: boolean; onClick?:
(e: any) => void; }
export declare function IconButton(props: IconButtonProps): JSX.Element;
```

QuickActions



Row of 2–5 labelled starting actions for empty and onboarding states.

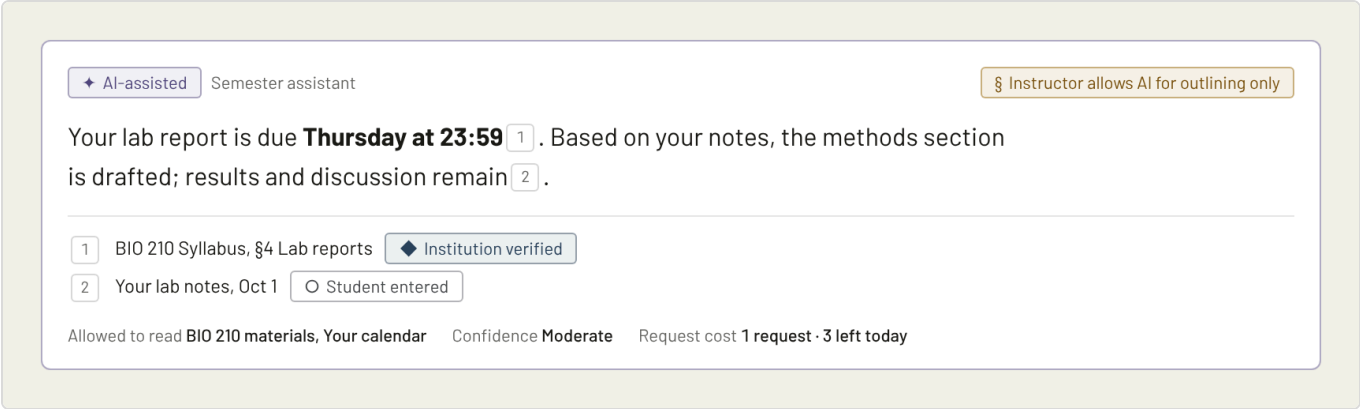
```
<QuickActions actions={[{label:"Add a syllabus",icon:"import"}]} />
```

PROPS

```
import * as React from 'react';
/** Row of 2–5 labelled starting actions for empty and onboarding states. */
export interface QuickActionsProps { actions: { label: string; icon?: string; onClick?: () => void }[]; }
export declare function QuickActions(props: QuickActionsProps): JSX.Element;
```

AI

AIResponse



A governed assistant answer — labelled AI-assisted with data scope, numbered sources, confidence, request cost and policy.

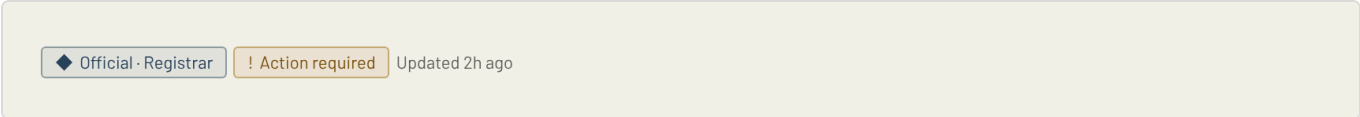
```
<AIResponse scope={["BIO 210 materials"]} sources={[{title:"Syllabus",kind:"official"}]}
confidence="Moderate">…</AIResponse>
```

Use `.ai-cite` spans for inline [n] citations. Consequential actions go in `actions` and must open a Dialog/ActionPreview before running.

PROPS

```
import * as React from 'react';
/**
 * A governed assistant answer — labelled AI-assisted with data scope, numbered sources, confidence, request
cost and policy.
 * @startingPoint section="AI" subtitle="A governed assistant answer — labelled AI-assisted with data scope,
numbered sou" viewport="700x400"
 */
export interface AIResponseProps { children: React.ReactNode; scope?: string[]; sources?: { title: string;
kind?: string }[]; confidence?: string; cost?: string; policy?: string; actions?: React.ReactNode; model?:
string; }
export declare function AIResponse(props: AIResponseProps): JSX.Element;
```

ProvenanceChips



A fact's origin chip plus at most two cues and its age, spoken as one sentence.

```
<ProvenanceChips origin="official" by="Registrar" cues={["action-required"]} age="Updated today"
/>
```

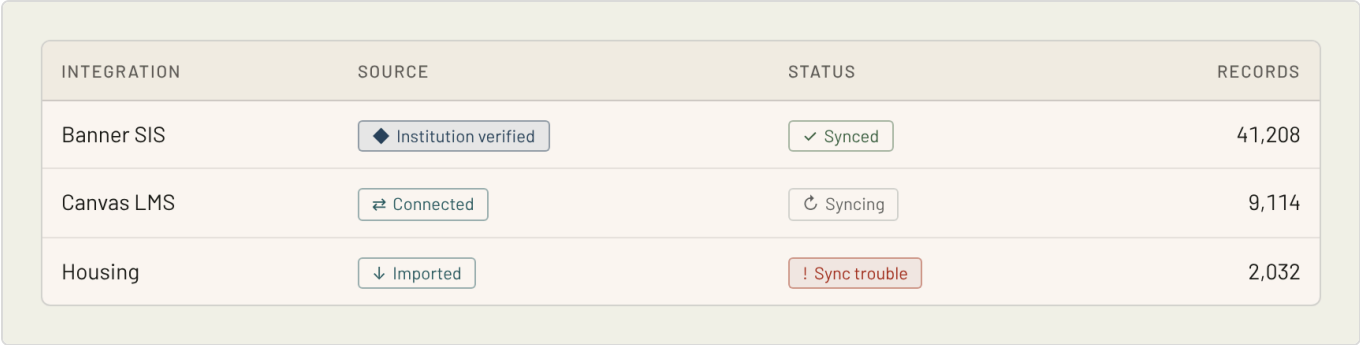
Origin always shows; only two cues render — the rest belong in Source & details.

PROPS

```
import * as React from 'react';
/**
 * A fact's origin chip plus at most two cues and its age, spoken as one sentence.
 */
export interface ProvenanceChipsProps { origin: string; by?: string; cues?: string[]; age?: string; }
export declare function ProvenanceChips(props: ProvenanceChipsProps): JSX.Element;
```

Data

DataTable



Crisp comparison table with caps headers, hairline rows, tabular numerals.

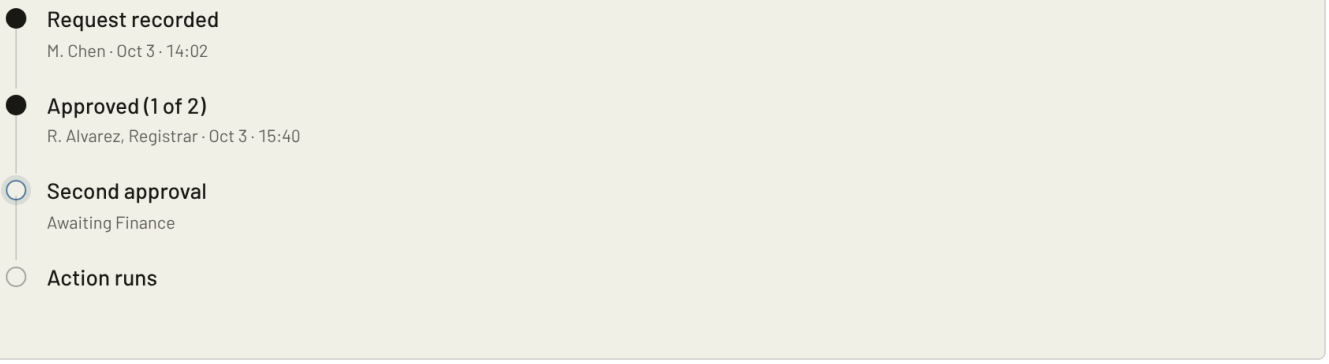
```
<DataTable caption="Holds" columns={[{key:"name",label:"Hold"}]} rows={[{name:"Immunization"}]}
/>
```

Use a list when scanning/acting on rows; a table only to compare columns.

PROPS

```
import * as React from 'react';
/**
 * Crisp comparison table with caps headers, hairline rows, tabular numerals.
 * @startingPoint section="Data" subtitle="Crisp comparison table with caps headers, hairline rows, tabular
numerals." viewport="700x400"
 */
export interface DataTableProps { columns: { key: string; label: string; numeric?: boolean; render?: (row: any)
=> React.ReactNode }[]; rows: any[]; caption?: string; }
export declare function DataTable(props: DataTableProps): JSX.Element;
```

DecisionTrail



Auditable timeline of who did what, when, and under which authority.

```
<DecisionTrail items={[{title:"Request recorded",who:"M. Chen",when:"Oct 3 · 14:02"}]} />
```

PROPS

```
import * as React from 'react';
/**
 * Auditable timeline of who did what, when, and under which authority.
 */
export interface DecisionTrailProps { items: { title: string; who?: string; when?: string; detail?: string;
state?: "done" | "current" | "waiting" | "blocked" }[]; label?: string; }
export declare function DecisionTrail(props: DecisionTrailProps): JSX.Element;
```

Fields



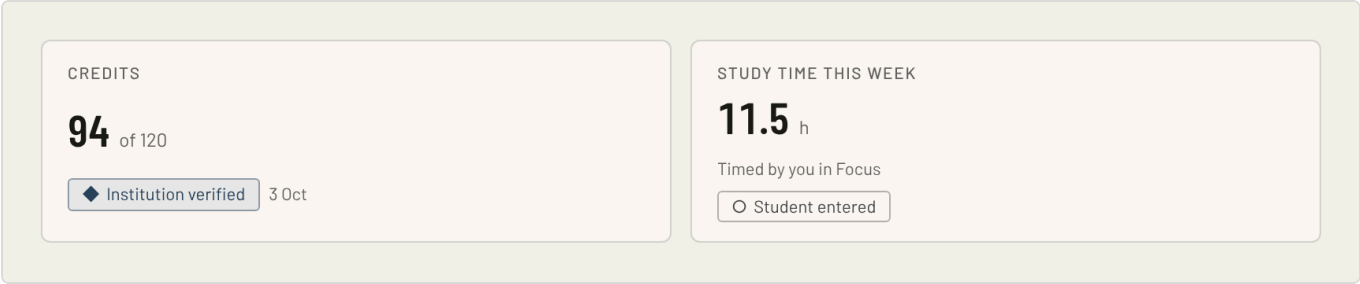
Label/value definition list for records and context.

```
<Fields items={[{field:"Environment",value:"Production"}]} />
```

PROPS

```
import * as React from 'react';
/**
 * Label/value definition list for records and context.
 */
export interface FieldsProps { items: { field: string; value: React.ReactNode; mono?: boolean }[]; label?:
string; }
export declare function Fields(props: FieldsProps): JSX.Element;
```

MetricTile



One figure worth reading first, with unit, source and age.

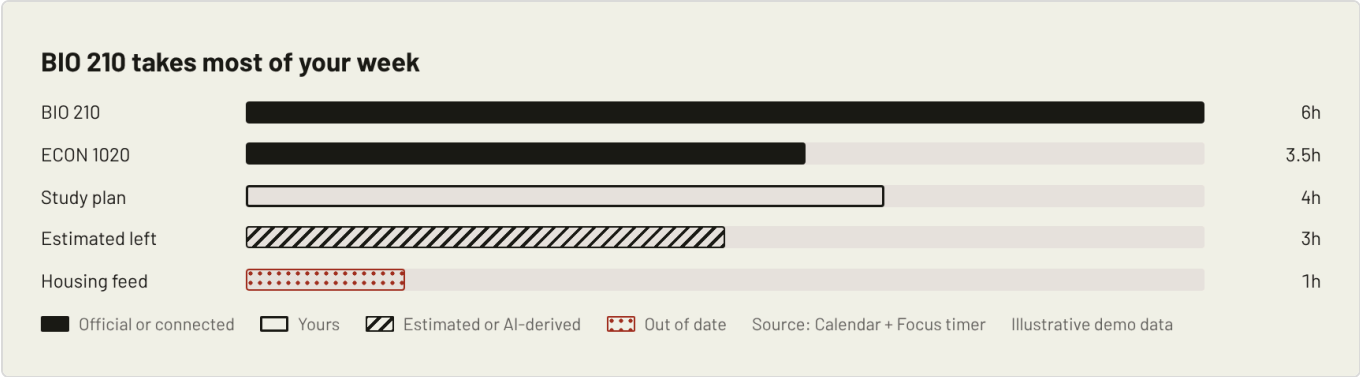
```
<MetricTile label="Credits" value="94" unit="of 120" source="official" updated="3 Oct" />
```

Never an outcome claim (GPA lift, time saved).

PROPS

```
import * as React from 'react';
/** One figure worth reading first, with unit, source and age. */
export interface MetricTileProps { label: string; value: string; unit?: string; source?: string; updated?:
string; delta?: string; }
export declare function MetricTile(props: MetricTileProps): JSX.Element;
```

ProvenanceChart



Bar chart that encodes provenance: solid official/connected, outline yours, hatched estimated/AI, dotted stale.

```
<ProvenanceChart title="Hours this week, by course" rows={[{label:"BIO
210",value:6,source:"personal"}]} unit="h" />
```

Title states the finding; every bar labelled directly; "Illustrative demo data" when `demo`.

PROPS

```
import * as React from 'react';
/** Bar chart that encodes provenance: solid official/connected, outline yours, hatched estimated/AI, dotted
stale. */
export interface ProvenanceChartProps { title: string; rows: { label: string; value: number; source?: string }
[]; unit?: string; max?: number; source?: string; demo?: boolean; }
export declare function ProvenanceChart(props: ProvenanceChartProps): JSX.Element;
```

Forms

Choice



Radio/checkbox row with a sentence explaining the consequence — used for visibility and sharing choices.

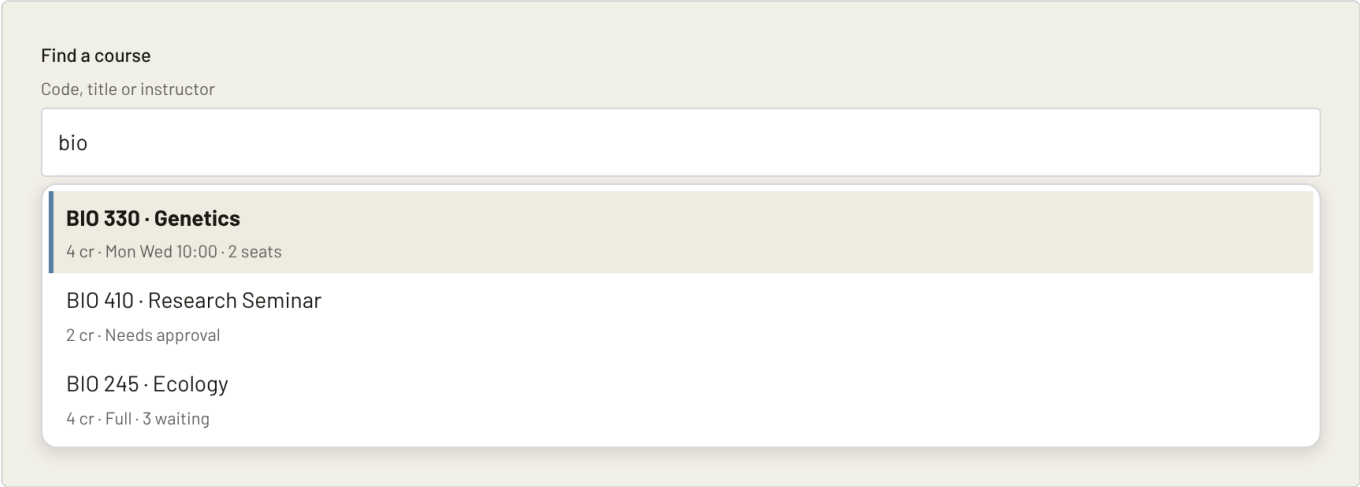
```
<Choice name="vis" label="Only me" about="Nobody else can see this." checked />
```

24px+ target, 44px row. Use for consent/sharing where the consequence must be read.

PROPS

```
import * as React from 'react';
/**
 * Radio/checkbox row with a sentence explaining the consequence — used for visibility and sharing choices.
 */
export interface ChoiceProps { type?: "radio" | "checkbox"; label: string; about?: string; checked?: boolean;
onChange?: (e: any) => void; name?: string; value?: string; disabled?: boolean; }
export declare function Choice(props: ChoiceProps): JSX.Element;
```

Combobox



Text box with a filtered listbox beneath — course search, people, rooms.

```
<Combobox label="Find a course" options={[{label:"BIO 330 · Genetics",detail:"4 cr · MW 10:00"}]}
/>
```

Selected row = 3px leading edge + weight. Escape closes; Enter picks.

PROPS

```
import * as React from 'react';
/** Text box with a filtered listbox beneath — course search, people, rooms. */
export interface ComboboxProps { label: string; hint?: string; options: { label: string; detail?: string }[];
value?: string; onChange?: (q: string) => void; onSelect?: (o: any) => void; placeholder?: string; defaultOpen?:
boolean; }
export declare function Combobox(props: ComboboxProps): JSX.Element;
```

DateField



Native date/time input with the chosen date spoken back unambiguously.

```
<DateField label="Due" value="2026-10-09" said="Thu 9 Oct 2026" />
```

Always show `said` in "Thu 9 Oct 2026" form so locale order can't mislead.

PROPS

```
import * as React from 'react';
/** Native date/time input with the chosen date spoken back unambiguously. */
export interface DateFieldProps { label: string; hint?: string; value?: string; onChange?: (e: any) => void;
withTime?: boolean; said?: string; }
export declare function DateField(props: DateFieldProps): JSX.Element;
```

Segmented



2–4 mutually exclusive view choices (e.g. Cards / Rows / Timeline).

```
<Segmented label="Feed" options={["Cards","Rows","Timeline"]} value="Cards" />
```

Selected = inverted ink. 32px tall; reserved for presentation choices, not navigation.

PROPS

```
import * as React from 'react';
/**
 * 2–4 mutually exclusive view choices (e.g. Cards / Rows / Timeline).
 */
export interface SegmentedProps { options: (string | { value: string; label: string })[]; value: string;
onChange?: (v: string) => void; label: string; }
export declare function Segmented(props: SegmentedProps): JSX.Element;
```

Select



Native select drawn in Semester's palette.

```
<Select label="Term" options={["Fall 2026","Spring 2027"]} />
```

Use native semantics; don't imitate a select with a menu.

PROPS

```
import * as React from 'react';
/**
 * Native select drawn in Semester's palette.
 */
export interface SelectProps { label: string; hint?: string; options: (string | { value: string; label: string
})[]; value?: string; onChange?: (e: any) => void; id?: string; }
export declare function Select(props: SelectProps): JSX.Element;
```

Switch



On/off setting that speaks its state as a word beside the track.

```
<Switch label="Share grades with Alex" checked />
```

role="switch"; On/Off text is always visible.

PROPS

```
import * as React from 'react';
/**
 * On/off setting that speaks its state as a word beside the track.
 */
export interface SwitchProps { checked?: boolean; onChange?: (v: boolean) => void; label: string; disabled?:
boolean; }
export declare function Switch(props: SwitchProps): JSX.Element;
```

TextField



Labelled input/textarea with associated hint and error — the label is always visible.

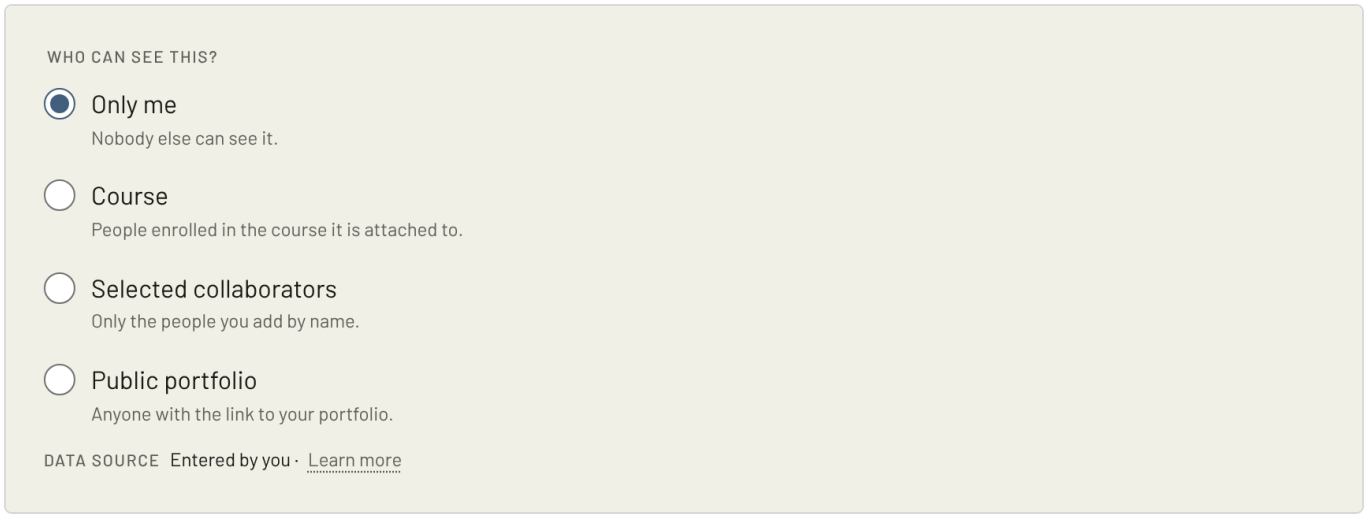
```
<TextField label="Preferred name" hint="Shown to classmates and instructors." />
```

`error` sets aria-invalid and a "!"-prefixed message. Placeholder is an example, never the label.

PROPS

```
import * as React from 'react';
/**
 * Labelled input/textarea with associated hint and error — the label is always visible.
 * @startingPoint section="Forms" subtitle="Labelled input/textarea with associated hint and error — the label
is always vis" viewport="700x400"
 */
export interface TextFieldProps { label: string; hint?: string; error?: string; multiline?: boolean; id?:
string; value?: string; placeholder?: string; onChange?: (e: any) => void; disabled?: boolean; }
export declare function TextField(props: TextFieldProps): JSX.Element;
```

VisibilityPicker



"Who can see this?" — the one privacy pattern, with the same four answers in every module.

```
<VisibilityPicker value="only-me" allowed={["only-me","course"]} />
```

Only offer audiences the module can actually honour. `locked` draws the statement alone. Presentation only — never grants access itself.

PROPS

```
import * as React from 'react';
/** "Who can see this?" — the one privacy pattern, with the same four answers in every module. */
export interface VisibilityPickerProps { value?: "only-me" | "course" | "collaborators" | "portfolio";
onChange?: (v: string) => void; allowed?: string[]; locked?: boolean; origin?: "student-entered" | "institution-
provided" | "connected-system"; onLearnMore?: () => void; }
export declare function VisibilityPicker(props: VisibilityPickerProps): JSX.Element;
```

Glyph

Avatar



Initials in a tinted square that follows the corner setting; profile glyph when no name.

```
<Avatar name="Maya Lindqvist" size={40} />
```

No photos, never derived from an email address.

PROPS

```
import * as React from 'react';
/** Initials in a tinted square that follows the corner setting; profile glyph when no name. */
export interface AvatarProps { name?: string; size?: number; }
export declare function Avatar(props: AvatarProps): JSX.Element;
```

Icon



Semester line glyph (Lucide-like, 24 grid, 1.5 stroke) from the app's own icon set — 108 names, see assets/icons/.

```
<Icon name="calendar" size={19} />
```

`label` makes it a named image; otherwise decorative. Names: today, courses, calendar, study, degree, registrar, costs, family, career, campus, meals, housing, ask, sources, privacy, settings, search, bell…

PROPS

```
import * as React from 'react';
/**
 * Semester line glyph (Lucide-like, 24 grid, 1.5 stroke) from the app's own icon set — 108 names, see
assets/icons/.
 * @startingPoint section="Glyph" subtitle="Semester line glyph (Lucide-like, 24 grid, 1.5 stroke) from the
app's own icon s" viewport="700x400"
 */
export interface IconProps { name: string; size?: number; label?: string; strokeWidth?: number; style?:
React.CSSProperties; className?: string; }
export declare function Icon(props: IconProps): JSX.Element;
export declare const ICON_NAMES: string[];
```

Wordmark



The Semester lockup — three-slab mark + SEMESTER in Barlow Condensed caps; also exports `Mark`.

```
<Wordmark size={20} metal />
```

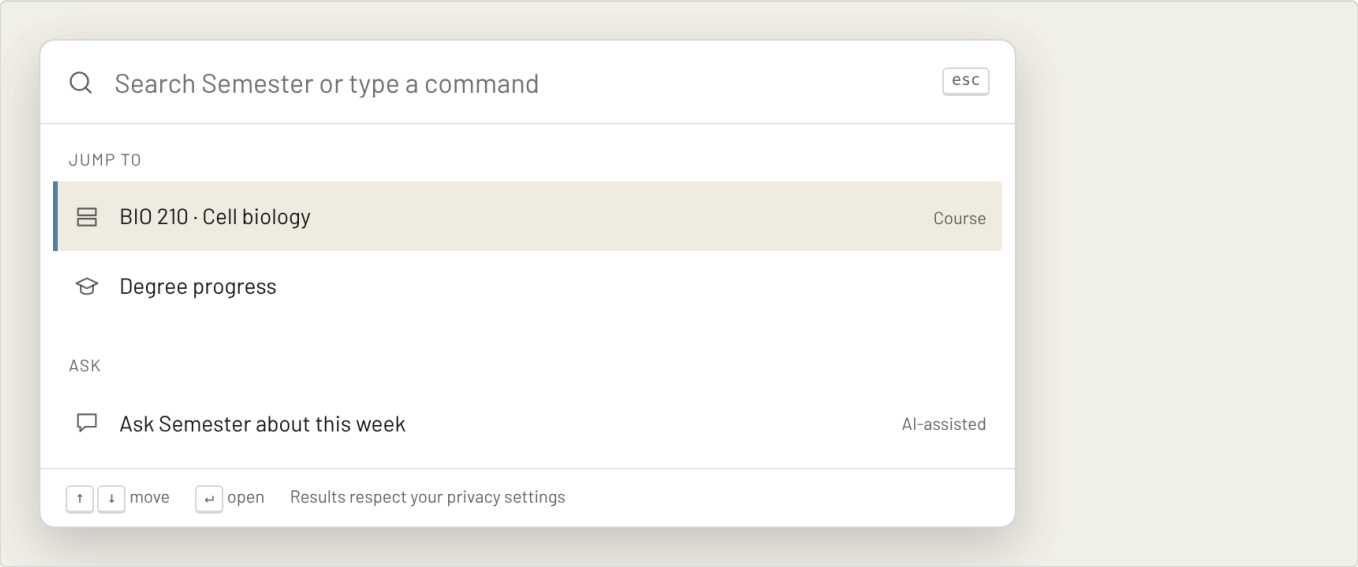
`metal` paints brushed sterling for dark chrome. `<Mark size={24}/>` alone for tiles/favicons.

PROPS

```
import * as React from 'react';
/**
 * The Semester lockup — three-slab mark + SEMESTER in Barlow Condensed caps; also exports `Mark`.
 */
export interface WordmarkProps { size?: number; metal?: boolean; showWord?: boolean; style?:
React.CSSProperties; }
export declare function Wordmark(props: WordmarkProps): JSX.Element;
export declare function Mark(props: { size?: number; style?: React.CSSProperties; className?: string }):
JSX.Element;
```

Navigation

CommandPalette



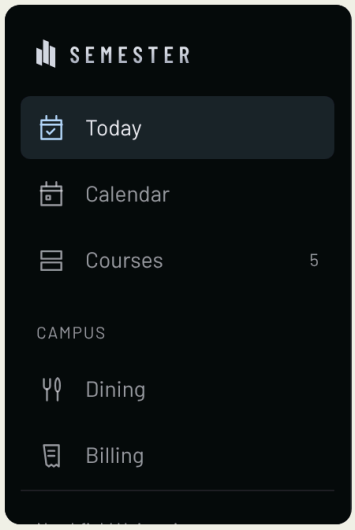
Global search and command palette (⌘K) with keyboard selection.

```
<CommandPalette groups={[{label:"Jump to",items:[{label:"BIO 210",icon:"courses"}]}]} />
```

PROPS

```
import * as React from 'react';
/**
 * Global search and command palette (⌘K) with keyboard selection.
 */
export interface CommandPaletteProps { groups: { label: string; items: { label: string; icon?: string; detail?:
string }[] }[]; placeholder?: string; onSelect?: (item: any) => void; }
export declare function CommandPalette(props: CommandPaletteProps): JSX.Element;
```

SideRail



Desktop side rail on dark Ink chrome — persistent, quiet, labelled.

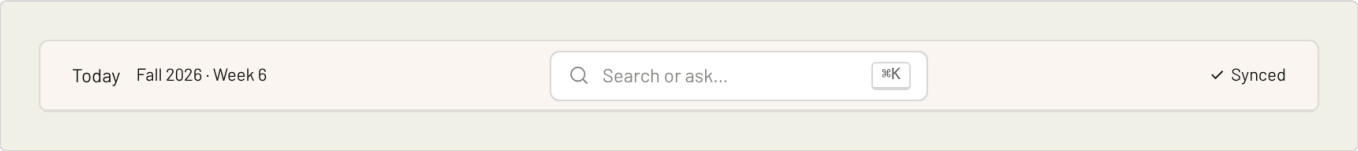
```
<SideRail current="today" groups={[{items:[{id:"today",label:"Today",icon:"today"}]}]} />
```

248px wide. Current row = blue-tinted wash + blue glyph.

PROPS

```
import * as React from 'react';
/**
 * Desktop side rail on dark Ink chrome — persistent, quiet, labelled.
 * @startingPoint section="Navigation" subtitle="Desktop side rail on dark Ink chrome — persistent, quiet,
labelled." viewport="700x400"
 */
export interface SideRailProps { groups: { label?: string; items: { id: string; label: string; icon: string;
count?: number }[] }[]; current?: string; onNavigate?: (id: string) => void; foot?: React.ReactNode; brand?:
boolean; }
export declare function SideRail(props: SideRailProps): JSX.Element;
```

SystemContextBar



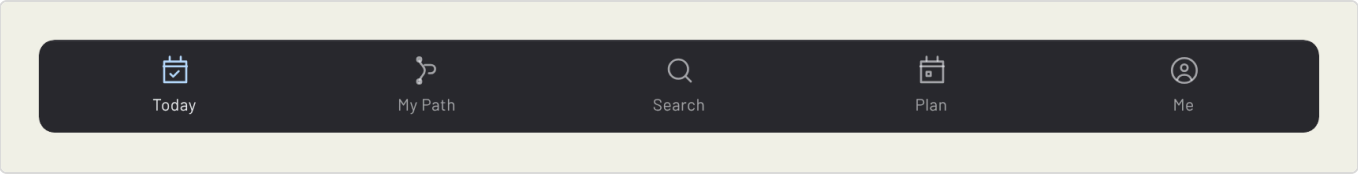
Persistent top bar: location, term, ⌘K search, sync state, quiet tools.

```
<SystemContextBar location="Today" term="Fall 2026 · Week 6" sync="synced" />
```

PROPS

```
import * as React from 'react';
/**
 * Persistent top bar: location, term, ⌘K search, sync state, quiet tools.
 */
export interface SystemContextBarProps { location?: string; term?: string; sync?: string; onSearch?: () => void;
actions?: React.ReactNode; }
export declare function SystemContextBar(props: SystemContextBarProps): JSX.Element;
```

TabBar



Mobile bottom navigation for the five canonical destinations, always labelled.

```
<TabBar current="today" items={[{id:"today",label:"Today",icon:"today"}]} />
```

Frosted Ink only where blur is supported; opaque otherwise.

PROPS

```
import * as React from 'react';
/**
 * Mobile bottom navigation for the five canonical destinations, always labelled.
 */
export interface TabBarProps { items: { id: string; label: string; icon: string }[]; current?: string;
onNavigate?: (id: string) => void; }
export declare function TabBar(props: TabBarProps): JSX.Element;
```

States

EmptyState



Explains why a list is empty and offers one relevant next action.

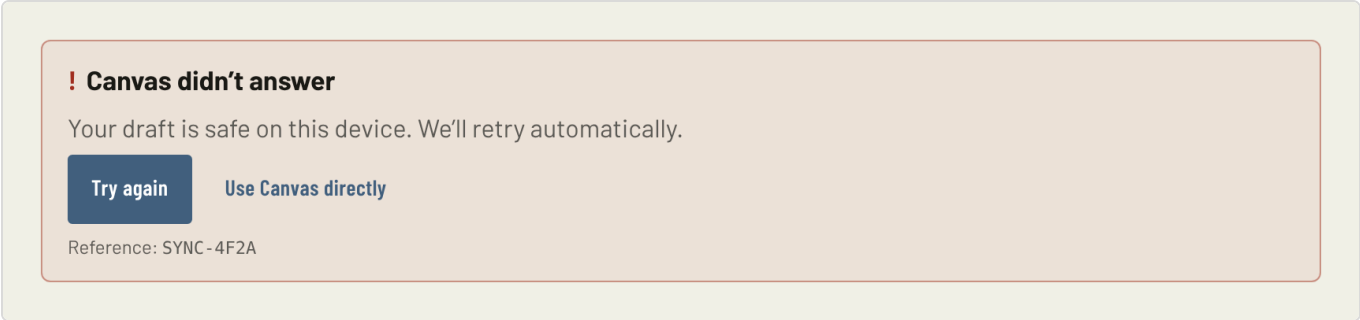
```
<EmptyState title="No tasks due this week" body="Tasks you add or that courses publish appear
here." action={<Button>Add a task</Button>} />
```

Dashed hairline frame; never decorative illustration.

PROPS

```
import * as React from 'react';
/**
 * Explains why a list is empty and offers one relevant next action.
 * @startingPoint section="States" subtitle="Explains why a list is empty and offers one relevant next action."
viewport="700x400"
 */
export interface EmptyStateProps { title: string; body?: string; action?: React.ReactNode; }
export declare function EmptyState(props: EmptyStateProps): JSX.Element;
```

ErrorState



Plain-language failure with a required recovery action and an optional support reference.

```
<ErrorState title="Canvas didn’t answer" body="Your draft is safe on this device."
reference="SYNC-4F2A" />
```

role="alert"; preserve the user's input, separate retryable failure from permission.

PROPS

```
import * as React from 'react';
/**
 * Plain-language failure with a required recovery action and an optional support reference.
 */
export interface ErrorStateProps { title: string; body: string; recoverLabel?: string; onRecover?: () => void;
secondaryLabel?: string; onSecondary?: () => void; reference?: string; }
export declare function ErrorState(props: ErrorStateProps): JSX.Element;
```

LoadingState

Still structural placeholders plus a spoken "Loading …" — no spinner or shimmer.

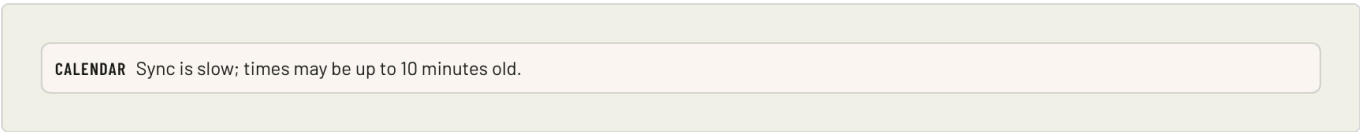
```
<LoadingState what="your courses" />
```

Only when the final shape is known; otherwise say what is happening in words.

PROPS

```
import * as React from 'react';
/**
 * Still structural placeholders plus a spoken "Loading …" — no spinner or shimmer.
 */
export interface LoadingStateProps { what?: string; bars?: number[]; }
export declare function LoadingState(props: LoadingStateProps): JSX.Element;
```

Notice



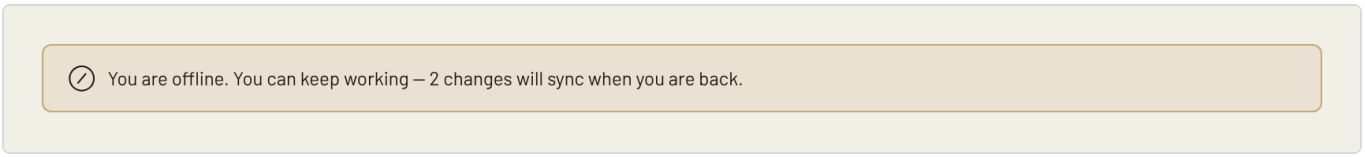
One-line service or context notice (role=status).

```
<Notice head="Calendar">Sync is slow right now; times may be up to 10 minutes old.</Notice>
```

PROPS

```
import * as React from 'react';
/**
 * One-line service or context notice (role=status).
 */
export interface NoticeProps { head?: string; children: React.ReactNode; }
export declare function Notice(props: NoticeProps): JSX.Element;
```

OfflineStrip



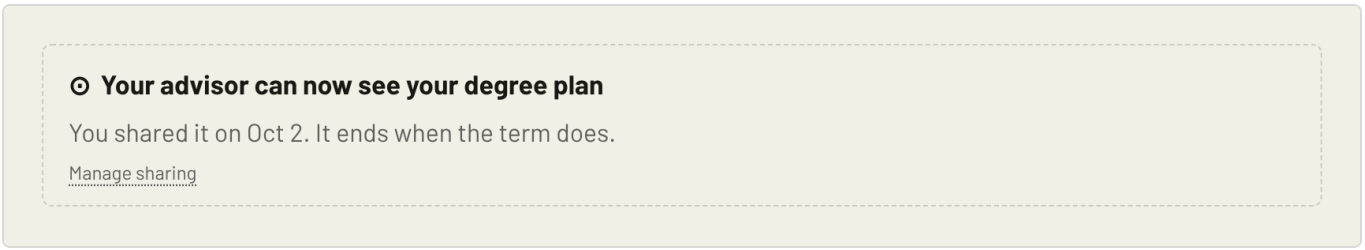
Offline strip that says what still works and that changes are queued.

```
<OfflineStrip queued={2} />
```

PROPS

```
import * as React from 'react';
/**
 * Offline strip that says what still works and that changes are queued.
 */
export interface OfflineStripProps { queued?: number; syncs?: boolean; }
export declare function OfflineStrip(props: OfflineStripProps): JSX.Element;
```

PermissionNotice



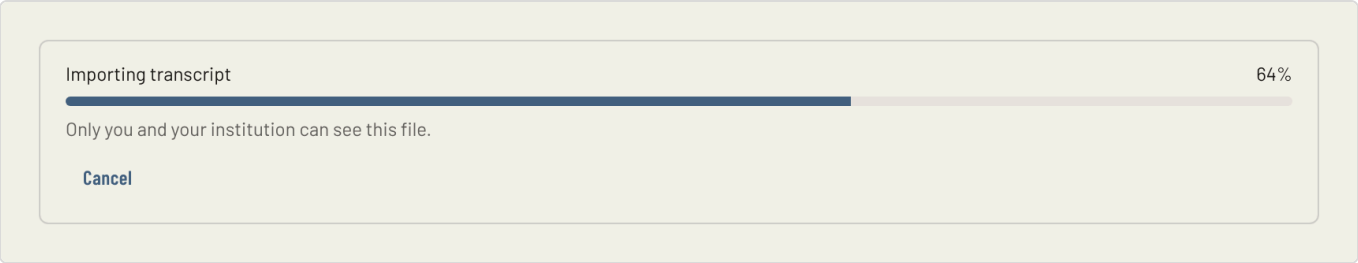
States what changed about access, why, and where it is controlled.

```
<PermissionNotice changed="Your advisor can now see your degree plan" why="You shared it on Oct
2." controlLabel="Manage sharing" />
```

PROPS

```
import * as React from 'react';
/**
 * States what changed about access, why, and where it is controlled.
 */
export interface PermissionNoticeProps { changed: string; why: string; controlLabel?: string; onControl?: () =>
void; }
export declare function PermissionNotice(props: PermissionNoticeProps): JSX.Element;
```

ProgressState



A determinate wait (upload, import) with real

```
<ProgressState label="Importing transcript" done={64} total={100} note="Only your institution and you can see this file." />
```

`failed` swaps the bar for the reason + Retry.

PROPS

```
import * as React from 'react';
/**
 * A determinate wait (upload, import) with real <progress>, percentage in words, Cancel/Retry.
 */
export interface ProgressStateProps { label: string; done?: number; total?: number; note?: string; failed?: string; onCancel?: () => void; onRetry?: () => void; }
export declare function ProgressState(props: ProgressStateProps): JSX.Element;
```

SaveState



The quiet "Saving… → Saved" line near the work, in a polite live region.

```
<SaveState status="saved" />
```

offline/conflict/sync-error announce assertively.

PROPS

```
import * as React from 'react';
/**
 * The quiet "Saving… → Saved" line near the work, in a polite live region.
 */
export interface SaveStateProps { status?: string; }
export declare function SaveState(props: SaveStateProps): JSX.Element;
```

StepStatus



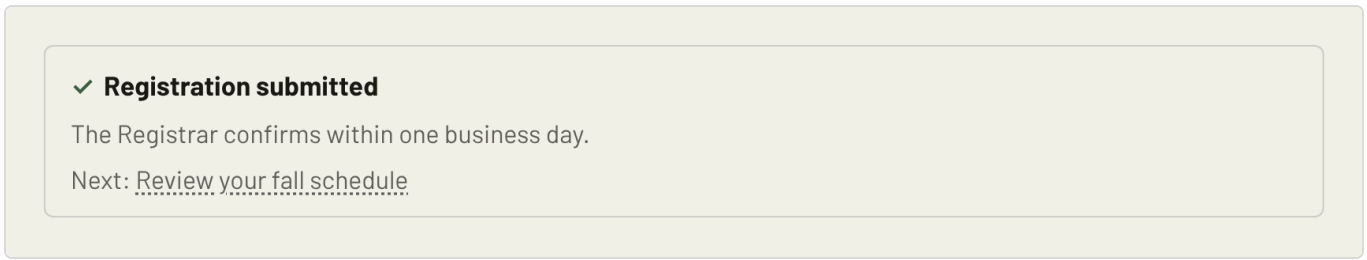
A process in named steps — shows the AI or system working instead of faking certainty.

```
<StepStatus steps={[{label:"Reading sources",state:"done"},{label:"Drafting",state:"working"}]}
/>
```

PROPS

```
import * as React from 'react';
/**
 * A process in named steps — shows the AI or system working instead of faking certainty.
 */
export interface StepStatusProps { steps: { label: string; state: "waiting" | "working" | "done" | "failed" }[];
label?: string; }
export declare function StepStatus(props: StepStatusProps): JSX.Element;
```

SuccessState



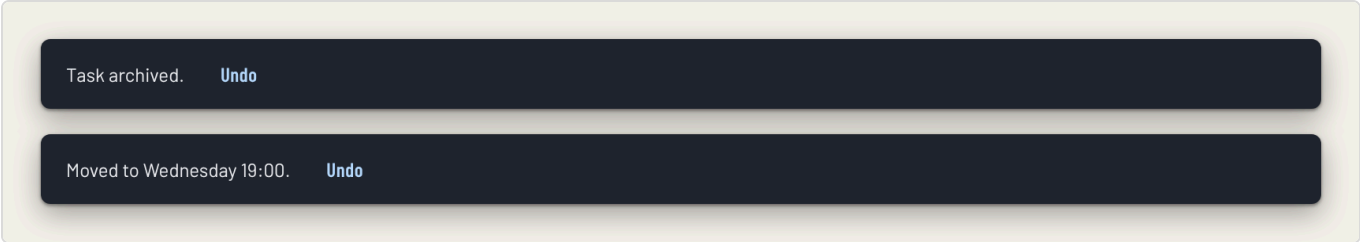
Respectful acknowledgement that points at what is next — no confetti, streaks or ranks.

```
<SuccessState title="Registration submitted" nextLabel="Review your fall schedule" />
```

PROPS

```
import * as React from 'react';
/**
 * Respectful acknowledgement that points at what is next — no confetti, streaks or ranks.
 */
export interface SuccessStateProps { title: string; body?: string; nextLabel?: string; onNext?: () => void; }
export declare function SuccessState(props: SuccessStateProps): JSX.Element;
```

UndoToast



One-sentence confirmation with Undo for reversible work — instead of an "are you sure?" dialog.

```
<UndoToast message="Task archived." onUndo={() => {}} />
```

Ink surface on every ground. Never for consequential or financial actions — those use Dialog + ActionPreview.

PROPS

```
import * as React from 'react';
/** One-sentence confirmation with Undo for reversible work — instead of an "are you sure?" dialog. */
export interface UndoToastProps { message: string; onUndo?: () => void; undoLabel?: string; }
export declare function UndoToast(props: UndoToastProps): JSX.Element;
```

Surfaces

ContextBar



The object a screen is about, with its states and actions.

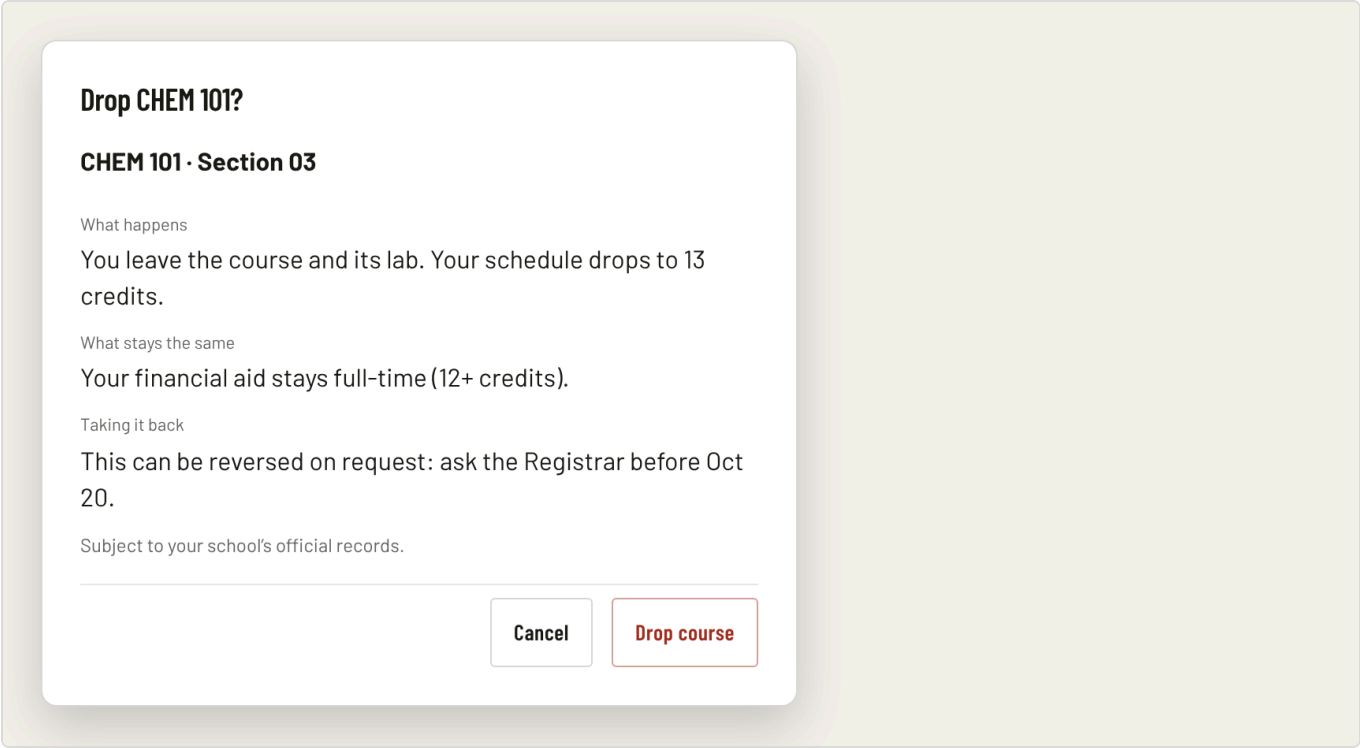
```
<ContextBar kicker="BIO 210 · Assignment" title="Lab report 3" states={<StatusChip
status="draft"/>} />
```

Inside an existing panel it drops its own frame.

PROPS

```
import * as React from 'react';
/**
 * The object a screen is about, with its states and actions.
 */
export interface ContextBarProps { kicker?: string; title: string; states?: React.ReactNode; actions?:
React.ReactNode; }
export declare function ContextBar(props: ContextBarProps): JSX.Element;
```

Dialog



Modal confirmation — focus lands on Cancel, Escape closes, names object + consequence + recovery.

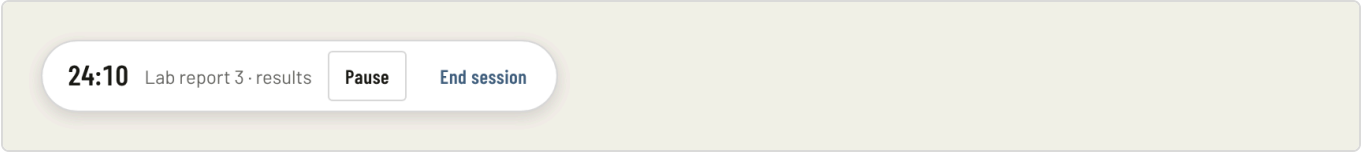
```
<Dialog title="Submit registration?" confirmLabel="Submit"><ActionPreview … /></Dialog>
```

`destructive` uses red-ink button. `inline` renders without scrim (for docs).

PROPS

```
import * as React from 'react';
/**
 * Modal confirmation — focus lands on Cancel, Escape closes, names object + consequence + recovery.
 */
export interface DialogProps { open?: boolean; title: string; children?: React.ReactNode; confirmLabel?: string;
cancelLabel?: string; onConfirm?: () => void; onCancel?: () => void; destructive?: boolean; inline?: boolean; }
export declare function Dialog(props: DialogProps): JSX.Element;
```

FocusBar



Floating pill during a focus session: time left, task, Pause, End.

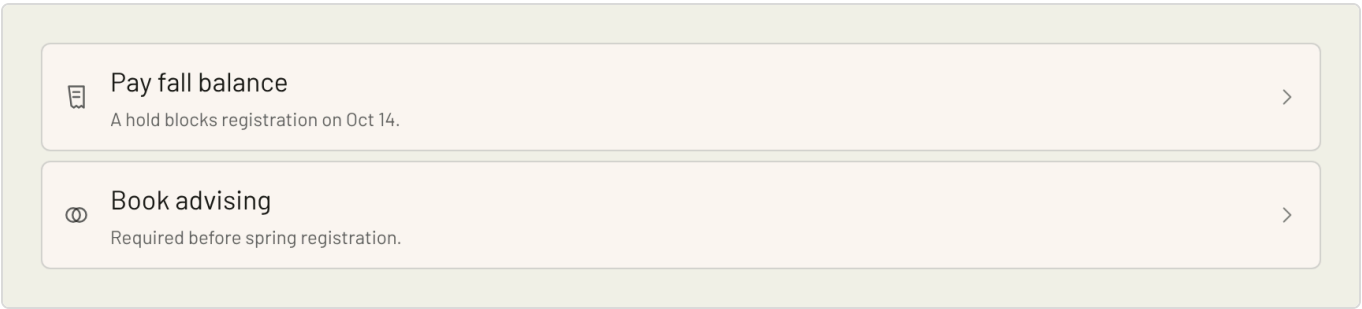
```
<FocusBar task="Lab report 3 · results" remaining="24:10" />
```

The only frosted overlay besides the tab bar; opaque under reduced transparency.

PROPS

```
import * as React from 'react';
/** Floating pill during a focus session: time left, task, Pause, End. */
export interface FocusBarProps { task: string; remaining?: string; onPause?: () => void; onEnd?: () => void;
breakDue?: string; }
export declare function FocusBar(props: FocusBarProps): JSX.Element;
```

NextSteps



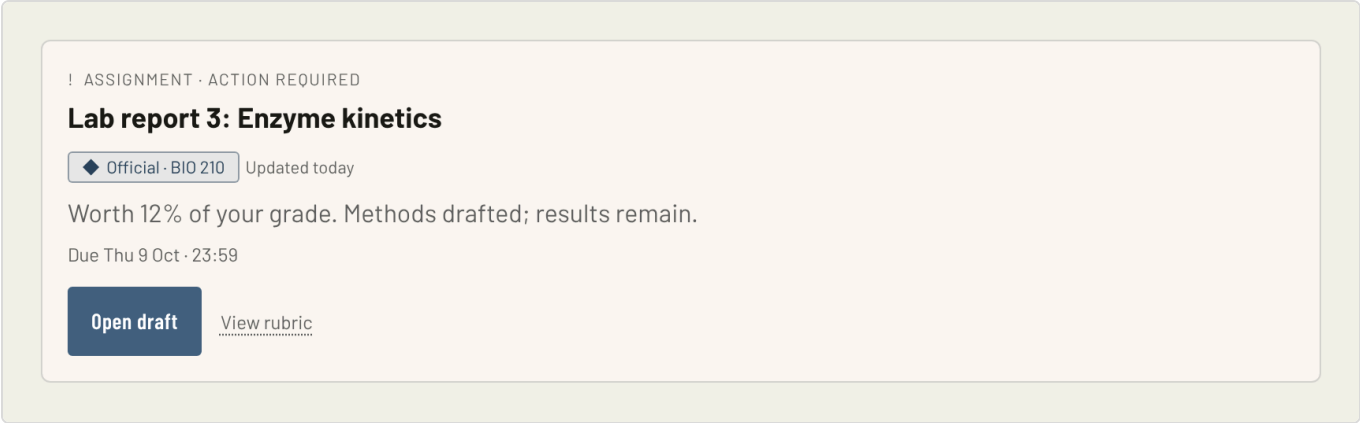
Ordered list of next safe actions, each with one line of why.

```
<NextSteps steps={[{label:"Pay fall balance",why:"A hold blocks registration on Oct
14.",icon:"costs"}]} />
```

PROPS

```
import * as React from 'react';
/**
 * Ordered list of next safe actions, each with one line of why.
 */
export interface NextStepsProps { steps: { label: string; why?: string; icon?: string; aside?: React.ReactNode;
onClick?: () => void }[]; }
export declare function NextSteps(props: NextStepsProps): JSX.Element;
```

ObjectCard



The universal decision card: eyebrow → title → provenance → why → metadata → one primary + one quiet secondary.

```
<ObjectCard kind="Assignment" title="Lab report 3" provenance={{origin:"official",by:"BIO 210"}}
explanation="Worth 12% of your grade." metadata="Due Thu 9 Oct · 23:59" primary={{label:"Open
draft"}} secondary={{label:"View rubric"}} />
```

A card groups a decision, not every paragraph. Never nest cards.

PROPS

```
import * as React from 'react';
/**
 * The universal decision card: eyebrow → title → provenance → why → metadata → one primary + one quiet
secondary.
 * @startingPoint section="Surfaces" subtitle="The universal decision card: eyebrow → title → provenance → why →
metadata → one" viewport="700x400"
 */
export interface ObjectCardProps { kind?: string; statuses?: string[]; title: string; provenance?: { origin:
string; by?: string; cues?: string[]; age?: string }; explanation?: string; metadata?: string; primary?: {
label: string; onClick?: () => void }; secondary?: { label: string; onClick?: () => void }; tone?: "pending" |
"danger"; level?: 2 | 3 | 4; children?: React.ReactNode; }
export declare function ObjectCard(props: ObjectCardProps): JSX.Element;
```

OpenIn



"Open in" row — the way from one object to every place it matters (Plan, Study, Calendar).

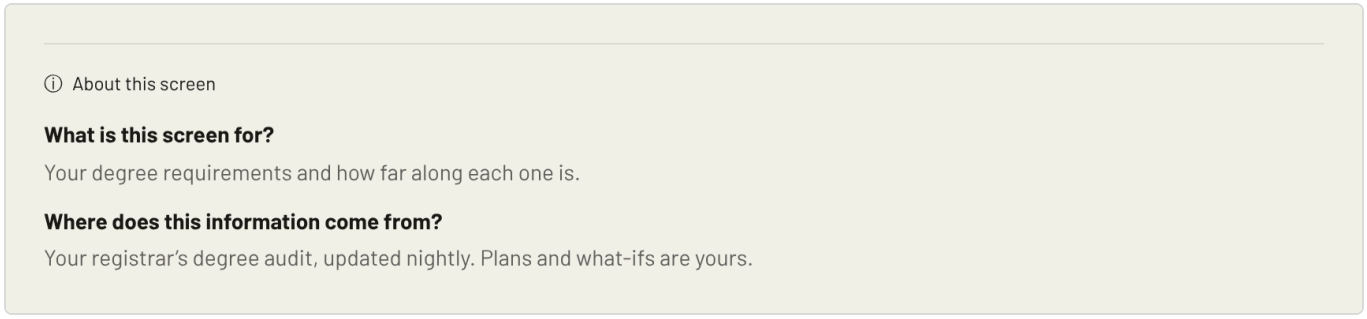
```
<OpenIn about="BIO 210" targets={["Plan","Study","Calendar"]} />
```

Words, never icons alone.

PROPS

```
import * as React from 'react';
/** "Open in" row — the way from one object to every place it matters (Plan, Study, Calendar). */
export interface OpenInProps { targets: string[]; about: string; onOpen?: (t: string) => void; }
export declare function OpenIn(props: OpenInProps): JSX.Element;
```

ScreenGuide



"About this screen" disclosure at the foot of every screen — what it is for, where its data comes from, who decides.

```
<ScreenGuide items={[{q:"Where does this information come from?",a:"Your registrar’s degree
audit, updated nightly."}]} />
```

PROPS

```
import * as React from 'react';
/** "About this screen" disclosure at the foot of every screen — what it is for, where its data comes from, who
decides. */
export interface ScreenGuideProps { items: { q: string; a: string }[]; open?: boolean; }
export declare function ScreenGuide(props: ScreenGuideProps): JSX.Element;
```

SectionLabel

The small uppercase kicker that heads a section, with an optional right-aligned count.

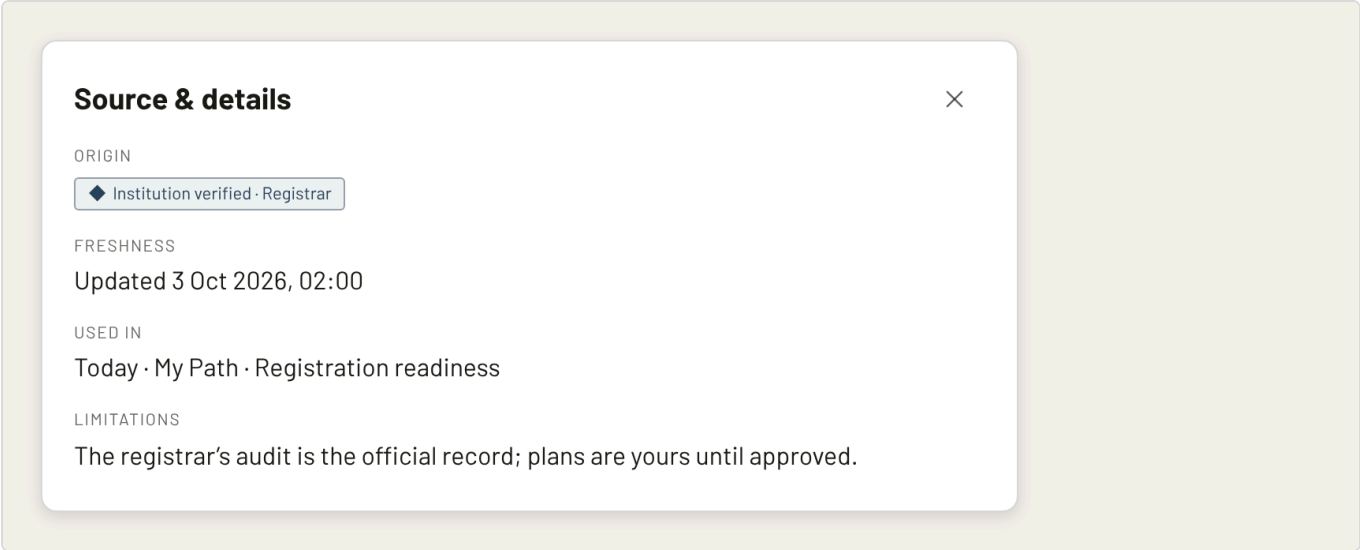
```
<SectionLabel aside="3">Begin today</SectionLabel>
```

Tracked 0.08em caps at 10.5px, secondary ink.

PROPS

```
import * as React from 'react';
/**
 * The small uppercase kicker that heads a section, with an optional right-aligned count.
 */
export interface SectionLabelProps { children: React.ReactNode; aside?: React.ReactNode; as?: "h2" | "h3" |
"h4"; style?: React.CSSProperties; }
export declare function SectionLabel(props: SectionLabelProps): JSX.Element;
```

Sheet



Bottom sheet on mobile, window on desktop; the home of "Source & details". Also exports `SourceDetails`.

```
<Sheet title="Source & details"><SourceDetails items={[{label:"Origin",value:"Official · Registrar"}]} /></Sheet>
```

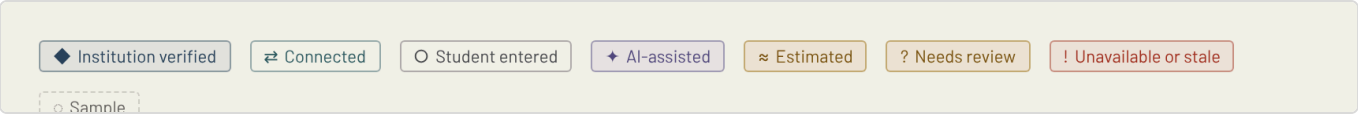
Rows: Origin, Freshness, Used in, Who can see this, Limitations, Report.

PROPS

```
import * as React from 'react';
/** Bottom sheet on mobile, window on desktop; the home of "Source & details". Also exports `SourceDetails`. */
export interface SheetProps { title: string; children?: React.ReactNode; onClose?: () => void; inline?: boolean; actions?: React.ReactNode; }
export declare function Sheet(props: SheetProps): JSX.Element;
export declare function SourceDetails(props: { items: { label: string; value: React.ReactNode }[] }): JSX.Element;
```

Trust & provenance

SourceBadge



Where a fact came from, beside the fact — controlled vocabulary (glyph + word + tone).

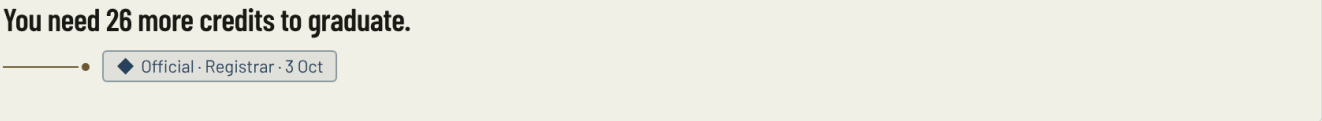
```
<SourceBadge source="official" by="Registrar" updated="Updated 2h ago" />
```

official navy (solid), connected teal, personal graphite, ai violet (always labelled), estimated/review amber, stale red, sample dashed. `onReport` adds "Report incorrect information".

PROPS

```
import * as React from 'react';
/**
 * Where a fact came from, beside the fact — controlled vocabulary (glyph + word + tone).
 * @startingPoint section="Trust & provenance" subtitle="Where a fact came from, beside the fact — controlled
vocabulary (glyph + word + " viewport="700x400"
 */
export interface SourceBadgeProps { source: "official" | "connected" | "imported" | "personal" | "ai" |
"estimated" | "review" | "stale" | "external" | "sample"; updated?: string; by?: string; onReport?: () => void;
}
export declare function SourceBadge(props: SourceBadgeProps): JSX.Element;
```

SourceLine



The Source Line — a thin brass/editorial rule from a statement to its source chip; Semester's signature device (docs/BRAND-PLATFORM.md §2.6).

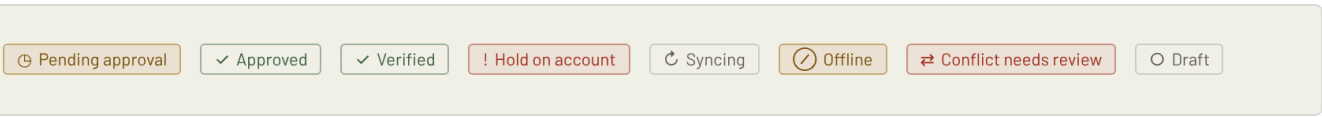
```
<SourceLine source="official" by="Registrar" when="3 Oct"><h2>You need 26 more credits to graduate.</h2></SourceLine>
```

Never decorative: use only where a real source label exists. The chip carries the word; the line is reinforcement.

PROPS

```
import * as React from 'react';
/** The Source Line — Semester's signature device: a thin editorial rule from a statement to its source chip. */
export interface SourceLineProps { children: React.ReactNode; source?: string; by?: string; when?: string }
export declare function SourceLine(props: SourceLineProps): JSX.Element;
```

StatusChip



One status chip — glyph + word + tone from the shared status table.

```
<StatusChip status="pending" />
```

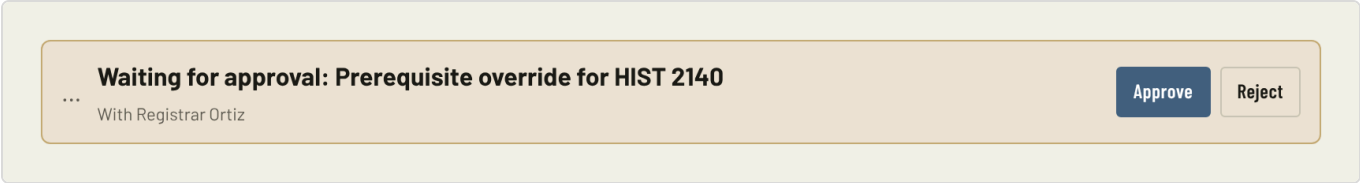
Keys: saving, saved, syncing, synced, offline, queued, conflict, sync-error, needs-confirmation, action-required, pending, approved, rejected, verified, hold, faculty-approved, course-provided, updated-today, draft, live, beta.

PROPS

```
import * as React from 'react';
/**
 * One status chip — glyph + word + tone from the shared status table.
 */
export interface StatusChipProps { status: string; label?: string; }
export declare function StatusChip(props: StatusChipProps): JSX.Element;
```

Governance

ApprovalBanner



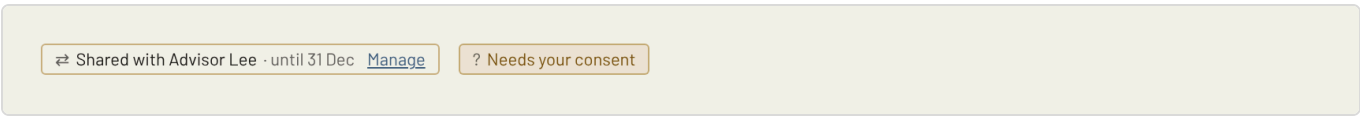
Approval state for a consequential action (two-person rule): required → pending → approved/rejected, with the right buttons for the viewer. Pair with ActionPreview and DecisionTrail.

```
<ApprovalBanner state="pending" what="Prerequisite override for HIST 2140" approver="Registrar Ortiz" onApprove={a} onReject={r} />
```

PROPS

```
import * as React from 'react';
export interface ApprovalBannerProps { state?: "required" | "pending" | "approved" | "rejected"; what: string; approver?: string; requestedBy?: string; onRequest?: () => void; onApprove?: () => void; onReject?: () => void; }
export declare function ApprovalBanner(props: ApprovalBannerProps): JSX.Element;
```

ConsentBadge



Consent state for a student-owned item: private, shared (with recipient and expiry), expired, revoked, or consent required. Always glyph + words.

```
<ConsentBadge state="shared" recipient="Advisor Lee" expires="31 Dec" onManage={open} />
```

PROPS

```
import * as React from 'react';
export interface ConsentBadgeProps { state?: "private" | "shared" | "expired" | "revoked" | "required"; recipient?: string; expires?: string; onManage?: () => void; }
export declare function ConsentBadge(props: ConsentBadgeProps): JSX.Element;
```

ErrorSummary



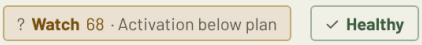
Summary of form errors at the top of a form (WCAG 3.3.1). Each item links to and focuses its field by id. Move focus to the summary on failed submit.

```
<ErrorSummary ref={r} errors={[{ field: "email", message: "Enter a full email address" }]} />
```

PROPS

```
import * as React from 'react';
export interface ErrorSummaryProps { errors: { field: string; message: string }[]; title?: string; }
export declare const ErrorSummary: (props: ErrorSummaryProps & { ref?: any }) => JSX.Element | null;
```

HealthBadge



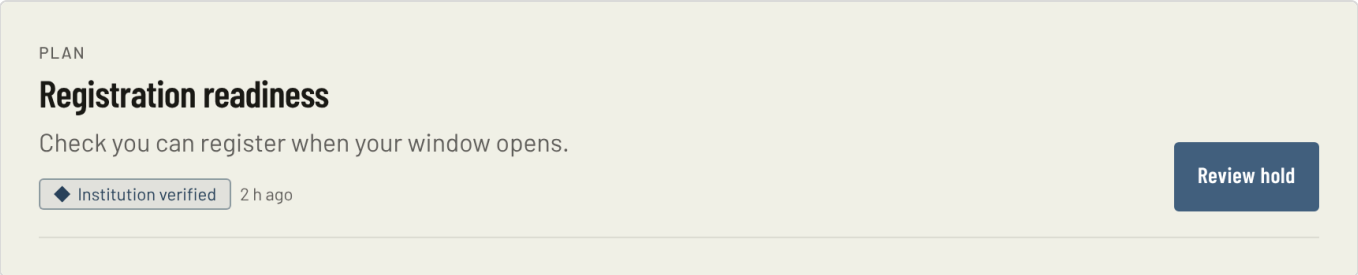
Health band for a tenant, integration or customer: band + optional score + top driver + freshness. Explains, never just colours.

```
<HealthBadge band="watch" score={68} driver="Activation below plan" updated="2 h ago" />
```

PROPS

```
import * as React from 'react';
export interface HealthBadgeProps { band?: "healthy" | "watch" | "atrisk" | "unknown"; score?: number; driver?:
string; updated?: string; }
export declare function HealthBadge(props: HealthBadgeProps): JSX.Element;
```

PageHeader



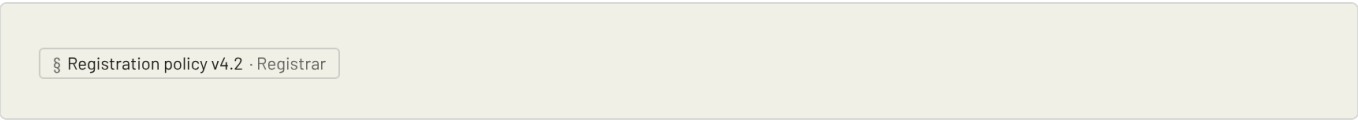
The universal page head from the UI grammar: kicker, one H1, one line on what the page helps you do, source/freshness, and at most one primary action.

```
<PageHeader kicker="Plan" title="Registration readiness" purpose="Check you can register when your window opens." source={<SourceBadge source="official" updated="2 h ago" />} actions={<Button variant="primary">Review hold</Button>} />
```

PROPS

```
import * as React from 'react';
export interface PageHeaderProps { kicker?: string; title: string; purpose?: string; source?: React.ReactNode; actions?: React.ReactNode; }
export declare function PageHeader(props: PageHeaderProps): JSX.Element;
```

PolicyBadge



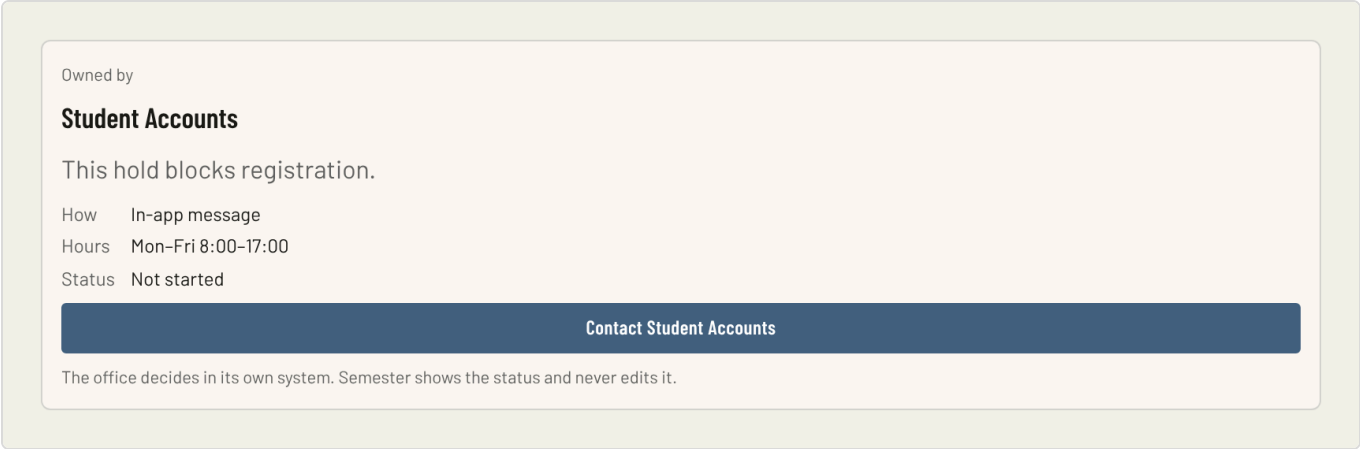
Names the policy that governs a fact, action or AI answer: name, version, owner. Clickable when onOpen is set (opens the policy text in a Sheet).

```
<PolicyBadge name="Registration policy" version="4.2" owner="Registrar" />
```

PROPS

```
import * as React from 'react';
export interface PolicyBadgeProps { name: string; version?: string; owner?: string; onOpen?: () => void; }
export declare function PolicyBadge(props: PolicyBadgeProps): JSX.Element;
```

SupportHandoff



Official hand-off to the owning office (holds, aid, accommodations, counseling). Names the owner, the channel and hours, and tracks status without editing the official record.

```
<SupportHandoff office="Student Accounts" reason="This hold blocks registration." channel="In-app message" hours="Mon–Fri 8–5" onStart={go} />
```

PROPS

```
import * as React from 'react';
export interface SupportHandoffProps { office: string; reason?: string; channel?: string; hours?: string; status?: "ready" | "sent" | "received" | "resolved"; reference?: string; onStart?: () => void; }
export declare function SupportHandoff(props: SupportHandoffProps): JSX.Element;
```

Patterns

AI answer anatomy

Label · scope · sources · uncertainty · cost · policy · human-confirmed action

✦AI-assistedWritten with Semester's assistant. It is not an official answer — check anything you act on.Semester assistant§Outlining allowed in BIO 210

About 3 hours remain on Lab report 31. Results is the largest gap2.

1. 1BIO 210 Syllabus §4◆Institution verifiedConfirmed by your institution's own system.
2. 2Your lab notes, Oct 1○Student enteredYou typed this in. Semester has not checked it with your institution.

1. Always labelled AI-assisted — never a persona
2. Allowed data scope, named
3. Numbered sources with their own provenance
4. Confidence stated, and what it can't know
5. Request cost and remaining allowance
6. Course/institution policy that applies
7. Consequential actions open a preview first

Provenance ladder

Strongest first; stale ranks below yours on purpose (lib/where.ts)

0	◆Institution verifiedConfirmed by your institution's own system.	Your institution's own record, read from its system.
1	⇄ConnectedSynced from a connected account or feed. Semester has not checked it with your institution.	From a calendar or account you connected, checked recently.
2	○Made hereSemester built this from sources you gave it — a rule-based extraction, not an AI draft.	The app built this from sources you gave it.
3	○Student enteredYou typed this in. Semester has not checked it with your institution.	You entered this yourself. Nothing has checked it.
4	○SampleIllustrative data. Not live.	Shipped as a demonstration. Not your semester.
5	!Unavailable or staleSemester cannot confirm that this information is current.	From a connection not checked lately. It may have moved.

Source Line

Signature device — a statement, a rule, its source. Only where a real source exists

You need 26 more credits to graduate.

The chip carries the word; the brass/editorial line is reinforcement only. Never decorative — absent source, absent line. Works on screens, decks, social cards and reports.

Consent & sharing

One decision at a time · default private · visible history · revoke anywhere

ASK

Share final grades with Jonas?

Course letter grades once posted by the Registrar. Not assignment scores, feedback or anything else.

Share final grades | Not now

OnFinal grades · shared since Aug 28

SHOW WHAT HAPPENED

1. Jonas viewed final grades
   Guardian · Sep 30 · 08:02
2. You turned on Final grades
   You · Aug 28 · 16:40
3. Invitation accepted
   Jonas Lindqvist · Aug 28 · 16:31

Consequential action

Review → preview → confirm → receipt. Reversible work gets Undo instead

Withdraw from CHEM 101?

CHEM 101 · Section 03

What happens
: You leave CHEM 101 and a W is recorded on your transcript.

What stays the same
: Your other courses and aid status (13 credits, full-time) stay as they are.

Taking it back
: This can’t be undone. Ask the Registrar if this was a mistake.

Subject to your school’s official records.

RULES

· Name the object in the title and both buttons.

· Focus lands on the safe choice.

· Destructive = red ink + hairline, never a red fill.

· State recovery: undo · on request · can’t be undone.

· Financial and record changes always preview.

AFTER

✓Withdrawn from CHEM 101

A W is recorded. Your schedule is now 13 credits.

State matrix

Every data region designs all seven states

LOADING

Loading this screen…

EMPTY

No tasks this week

Add a task

PARTIAL / STALE

HousingRoom data may be 6 hours old.

OFFLINE

⊘You are offline. You can keep working — 2 changes will sync when you are back.

ERROR

!Canvas didn’t answer

Your draft is safe.

Retry

PERMISSION

⊙Only your advisor can edit this

Set by the Registrar.

SUCCESS

✓Registration submitted

Next: Review your spring week

Guidelines

Provenance and trust

Source: `docs/TRUST-CUES-AND-SOURCE-PRESENTATION.md`, `app/src/lib/where.ts`, `lib/status.ts`, `lib/source.ts`.

The ladder (strongest first)

Rank	Key	Word	Sentence
0	official	Official / Institution verified	Your institution's own record, read from its system.
1	connected	Connected	From a calendar or account you connected, checked recently.
2	made	Made here	The app built this from sources you gave it (rule-based extraction).
3	personal (yours)	Yours / Student entered	You entered this yourself. Nothing has checked it.
4	sample	Sample	Shipped as a demonstration. Not your semester.
5	stale	Out of date	From a connection not checked lately. It may have moved.

Display-only kinds: **AI-assisted, source-linked** (a model's draft), **Estimated** (Semester's arithmetic), **Needs review**, **External**.

- `stale` ranks below `yours` on purpose: an unchecked copy of something that may have moved is worth less than a guess that knows it is a guess.
- `made` ≠ `ai`. An extraction can misread; a draft can be wrong while sounding right. Different glyphs (○ vs ✦), different sentences.
- Estimates and AI are styled to be **noticed** (amber/violet wash) — the opposite of "official looks louder", because they are what students over-trust.
- Never claim `official` for anything the institution did not send.

Where cues go

1. The eyebrow or state row of an ObjectCard / ContextBar — never louder than the primary action.
2. A SectionLabel aside (`<SourceBadge>` beside "Released grades").
3. The **Source & details** sheet: Origin · Freshness · Used in · Who can see this · Limitations · Report incorrect information.
4. Charts: by pattern (solid / outline / hatched / dotted), plus a source line and "Illustrative demo data" when applicable.

Rules for new work

1. Say a state through `SOURCES` / `STATUSES` in `components/trust/vocabulary.js`; never hand-type "Official" or "Synced".
2. Anything with a source gets a Source & details route.
3. Anything a model drafted is `ai`, lists its sources and limitations, and is editable before it is kept.
4. Freshness is written as "Updated 3 days ago", or "Update time not recorded" — never "just now" for an unknown time.

AI governance in the interface

Source: `docs/BRAND-PLATFORM.md` §6.1, `TRUST-CUES-AND-SOURCE-PRESENTATION.md` (AI presentation), `components/StudyStudio.tsx`.

Stance: assistive, source-linked, reviewable, interruptible. A tool, not a person. Name: **Ask Semester**. No human name, avatar face, sparkle orb or "I" in marketing.

Every AI answer shows (AIResponse)

1. The label **AI-assisted** (violet chip, ✦) — always, wherever it speaks.
2. **Allowed data scope** ("Allowed to read BIO 210 materials, Your calendar"). The student can change scope before asking.
3. **Numbered sources** with their own provenance chips; inline `.ai-cite` markers.
4. **Confidence** and what it can't know ("time estimates are mine, not your instructor's").
5. **Request cost** and remaining allowance.
6. **Policy** that applies (course AI policy set by the instructor; institution switches).
7. **Actions** — anything that writes, sends, shares, pays or registers opens Dialog + ActionPreview first.

Show the work

Use StepStatus ("Reading sources → Drafting → Matching quotations → Ready"), marking the step that failed. Never fake an instant, certain answer. No shimmer, no typing animation.

Words

Say: "Ask Semester drafts a plan from the sources you gave it. Review it before you use it." · "It can be wrong. Check the source linked beside each answer." Never: "AI-powered", "intelligent", "knows you", "accurate", "no hallucinations", "handles it for you", "Ask anything".

Course policy

When an instructor restricts AI (e.g. outlining only), show it on the course workspace (object card "Course AI policy · set by instructor") and as a policy chip on every answer in that course; disable the disallowed actions with a reason rather than hiding them.

Consequential actions

Source: `components/unity/ActionPreview.tsx`, `enrollment/StudentRegistration.tsx`, `console/Approvals.tsx`.

Kind of action	Pattern
Reversible, personal (archive, move, rename)	Do it, then **UndoToast** ("Task archived. Undo"). No dialog.
Leaves Semester or someone else sees it (share, send, submit)	**Dialog + ActionPreview**: What happens / Exactly what / What stays the same / Taking it back
Academic record (enroll, drop, withdraw, regrade)	Two-step **Review → Confirm** in place; state where the request will land ("number 3 in line"; "a W is recorded")
Financial (pay, refund, waive)	Dialog + ActionPreview naming amount, method, processor ("Semester does not process the payment; TouchNet does"), refund route
Institutional high-risk (release hold, change role, break-glass)	**Two-person duty**: request → approve (≠ requester) → act; MFA fresh within 15 min; audit event written first

Writing the dialog

- Title is a question naming the object: "Drop CHEM 101?"
- Buttons name the outcome: "Keep CHEM 101" / "Drop — confirm". Focus lands on the safe one.
- Destructive uses red ink + red hairline (`variant="danger"`), never a solid red fill.
- Recovery is required: `undo` · `request` (with how) · `none` (with what to do if it was a mistake).
- Afterwards: a SuccessState receipt with what changed and "Next:" — never "All set!" while anything is pending.

Consent, sharing and family access

Source: `components/unity/Visibility.tsx`, `AdvisorMeeting.tsx`, brief §5 (Family and Guardian Portal).

- **Default private.** Guardians, advisors and classmates see nothing by default. Never imply default guardian access.
- **One decision at a time.** Each ask states what, with whom, why, for how long, and how to undo. Decline is equal weight.
- **The same question everywhere:** VisibilityPicker — Only me · Course · Selected collaborators · Public portfolio. Offer only the audiences the module can honour.
- **Show exactly what leaves.** Before a share, preview the payload ("will see exactly this, as 'Maya Lindqvist', until 5 Nov 2026"). Private notes are never shared or exported.

- **Expiry and revoke.** Shares end when the student says; revoke and delete live next to every share.
- **Access history.** DecisionTrail of who viewed what, when. Students see when a share was opened.
- **Protected categories** (counseling, health, disability services) are never shareable automatically; show the switch disabled with the reason.
- **Guardian register:** the consent boundary is the hero of the screen — what's shared, since when, by whose choice.

Institutional surfaces

Source: `components/console/*`, `institutional/*`, brief §4.

- **Context first.** Every console page opens with Environment (word + shape: ◆ Production), Scope, Operator, Role (grants), MFA freshness, Session, Support access. Staff workspaces carry a scope bar: Role · Scope · Signed in · "Every record view is logged".
- **Tables over cards** for anything compared across rows (integrations, audit, requests). Caps headers, hairline rows, tabular numerals, mono for IDs and correlation ids.
- **Fields** (label/value dl) for a single record.
- **DecisionTrail** for anything with authority: who, when, under which grant.
- **Errors separate cause classes:** retryable failure vs permission vs configuration. Name the system and the consequence for students ("Students see a stale label on Housing").
- **Implementation honesty.** Use "Planned", "Pilot", "Preview" chips beside the capability — never a roadmap presented as live.
- **Authority scope is always visible** for faculty and advisors (Can see / Can approve / Cannot). No copy suggests the tool judges staff or ranks students.

Accessibility contract

Target WCAG 2.2 AA (a design requirement, not a conformance claim).

Contrast — measured against every surface of a ground, not the flattering one

Element	Requirement
Body / UI text	≥ 4.5:1
Large text (≥24px or ≥18.66px bold)	≥ 3:1
Icons, borders, focus, chart marks	≥ 3:1 vs neighbour
Tertiary ("faint") ink	metadata only, never information a student must act on
Brass	base stop on dark only; shade stop (#695a33) for text on light

Structure & input

- One `main`, one visible H1, skip link; heading levels describe structure.
- Keyboard order follows reading order; focus never lands under sticky chrome (scroll margins 96 / 84px).
- 44px ordinary targets; 24px floor only in crowded composites with keyboard parity. Icon-only controls always named.
- Native inputs; persistent labels; hints and errors associated via aria-describedby; errors preserve input.
- Overlays: focus trap, Escape, outside dismissal, focus return; triggers report aria-expanded.

Never colour alone

Every state: glyph + word + tone. Provenance origin has a second carrier in fill (solid / outline / dashed); forced colours thicken the solid chip's border to 3px.

Reader settings the tokens respond to

Text size 0.94–1.18× (`--text-scale`) · Density 1 / .86 / .74 · Line height 1.35–1.95 · Letter/word spacing · Reading width 52–82ch · Body face incl. Atkinson Hyperlegible · Motion: Follow device / Less motion / Low stimulation (also removes gradients, shadows, quieter badges) · Increase contrast raises faint/dim/hairline alphas · 13 grounds + Match my device.

Motion

All roles zero under reduced motion and `data-calm`. No essential meaning depends on motion. No flashing, no parallax, no looping decoration, no countdown theatre.

Brand-exempt zones

Status/warnings, provenance, consent text and accessibility controls: body-grade type only (never Cinzel), never restyled by tenants, campaigns or themes, never animated away or hidden behind hover.

Layout and responsive behaviour

Tier	Width	Navigation	Page pad
Compact	< 760px	Labelled bottom TabBar: Today · My Path · Search · Plan · Me	18px
Medium	760–1099px	Compact rail (72px, icons + tooltip names)	20px
Wide	≥ 1100px	SideRail 248px (272 at very wide) with group labels	30px

- **Shell:** SideRail (Ink) │ SystemContextBar (44px: location · term · ⌘K · sync · tools) over a scrolling page.

- **Page:** pagehead (kicker · H1 · one-line purpose · ≤2 actions) → `cols` grid `minmax(0,1.6fr) minmax(300px,1fr)` → main column leads with the next action; aside holds "Needs you", context, policy. Stacks to one column under 1000px.
- **Measures:** reading 66ch; operational content ≤ 1180px.
- **Tables** keep row identity and the first action on compact screens; the table region scrolls, not the document.
- **Five canonical destinations** are the spine: Today, My Path, Search, Plan, Me. Specialist tools open in context (OpenIn), never as a sixth global product.
- **Workspace modes** (presentation only): Guided · Focused (navigation steps back, FocusBar is the way out) · Detailed (source sentences up front) · Accessibility.

Content templates

From `docs/BRAND-PLATFORM.md` §7.

Status sentence — what is true → what it means for you → what you can do.

> "Your Canvas calendar hasn't updated since Tuesday. Deadlines below may have moved. Reconnect it, or check the course page."

Empty state — why it's empty + one action. "No grades released yet. Your instructor has not released any BIO 210 grades for Fall 2026. Grades appear here once they do." [Work out what you need]

Error — what failed, what's safe, the way out, a reference. "Canvas didn't answer. Your draft is safe on this device." [Try again] [Use Canvas directly] Reference: SYNC-4F2A

Success — acknowledge, then Next. "Registration submitted. Next: review your spring week."

Permission change — what changed, why, who controls it, where to change it.

Incident — Status · What is affected · What is NOT affected (only if verified) · What it means for you · What you can do now · What we know / don't know · Next update (only if operations can keep it). Never "minor", "isolated", "out of an abundance of caution".

Onboarding — "Semester starts with what you have. Pick one: add a syllabus, connect a calendar, or use sample data." → first real next step with its source label → consent one at a time → no streaks, no points, Skip always available.

Dates — "Thu 9 Oct · 23:59" in UI; "3 Oct 2026" in documents. Always unambiguous; time zone when it crosses one.

Architecture and tooling

Maps of the whole

Master control, catalog, role lens — how the pieces point at each other

Master control

ui_kits/master-control/ — product, console & back end, business OS, company site, design system & engineering

Rule
Every new file is linked here

Master catalog

51 roles · 673 screens · 319 workflow steps (17 workflows) · 88 systems · 82 docs

Rule
Status from evidence: code, tests, release

Role lens

64 screens × 14 roles: rendering, sources, tables, RLS, API

Access levels
full · scoped · shared · read · audited · none

Rule: company roles (investor, CS, RevOps) never see student records. Role model in roles.js is design-level; verify against migrations.

Stack layers

Front end → middleware → API → database, with the rule for each

Front end

Student app
Five destinations, 64 screens

Staff + console
Same shell, role-scoped routes

Design system
Tokens → semantic CSS → components

Offline
Queue mutations, replay on reconnect

Middleware

Auth proxy
Refresh session; reject expired

Tenant resolver
Host/SSO → tenant_id on every request

Capability check
role × screen × grant first

Audit writer
Event written before the action runs

Server / API

Query RPCs
api.* reads of read models

Commands
private.* fns; idempotency key

Integrations
LTI 1.3, OneRoster, SIS read, SAML

AI gateway
Scope, policy, citations, spend log

Database

Supabase Postgres
RLS on every table

tenant_id
On every row

private.*
Authoritative, never exposed

read_model.*
Projections + version + freshness

Read-model pipeline & freshness

Write → outbox → projection → refetch; lag decides what the UI allows

WRITE → READ

1 write		2 outbox		3 project		4 stamp		5 invalidate		6 signal		7 render
Authoritative write	→	Event, same transaction	→	Worker applies, idempotent	→	Version + freshness	→	Cache namespaces	→	Targeted refetch	→	Server read model

FRESHNESS → UI POLICY (FRESHNESS.TS)

State	Lag	Display	Consequential actions
Live	≤ 5 s	Normal	Allowed if approvals satisfied
Recent	≤ 60 s	Normal	Normal guardrails
Delayed	> 60 s	Delay badge + age	Re-check source first
Failed / Rebuilding	—	Last good + notice	Held

AI gateway

Six checks between a question and an answer — student records never leave

Request		Identity + scope		Policy		Routing		Answer		Evidence
Student, staff or institution asks	→	Who; which data they may read	→	Course AI policy, institution switches, FERPA boundary	→	Approved model per task; search, calendar, materials	→	Labelled AI-assisted, sources, confidence	→	Citations, cost, outcome logged

Environments

Local, staging, production on Vercel + Supabase — data never crosses

Environment	Hosting	Data	Access
Local	Supabase CLI + Next.js dev	Synthetic seed	Engineers
Staging	Vercel preview + Supabase staging	Synthetic + institution test IdP accounts	Engineers, institution IT testers
Production	Vercel production + Supabase production (separate keys)	Real student records under the DPA	Named operators, MFA, access logged

Rule: no copying data between environments; secrets set per environment. Cutover steps: docs/launch/Cutover Plan.html

Document library by stream

docs/ — what exists per stream and what is next

Stream	Built	Next
Brief	One-pager, FAQ, role sheets, overview deck	Real screenshot on slide 4
Guides	Architecture, developer, user	Admin manual, tutorials
Launch	Runbook, cutover plan, day-1/week-1 support	—
Secure	10 policies; HECVAT, SOC 2, pen test, insurance readiness	Stream index
Legal (drafts)	Terms, privacy, community, subprocessors	MSA, order form, DPA, cookie, accessibility
Build	—	Engineering playbook, QA and accessibility test plans

Rule: formal documents are doc_page files under docs/<stream>/, linked from docs/index.html. Legal stays draft until attorney review.

semester-core

Workflow decision logic as pure TypeScript — handoff/semester-core/

Module	Decides	Rendered by
provenance.ts	Source rank, conflicts, freshness label	SourceBadge, ProvenanceChips
nextaction.ts	Order of a student's next actions	Today, NextSteps
access.ts	Role × screen, share expiry, break-glass	Role lens, route guards
freshness.ts	Lag → display / safe / destructive	Read models, lag simulator
consequence.ts	Undo / preview / two-person	Dialog, ActionPreview
aigateway.ts	Scope + policy before any model call	AIResponse
approvals.ts	Two-person duty state machine	Console approvals
scoring.ts	Weighted 0–5 grading	Workflow lab
pilot.ts	Launch readiness	Business gap analysis

No dependencies, no I/O, vitest proof per module. Files end in .txt; strip on copy to app/src/lib/core/.

semester-platform

Migrations 010–040 with RLS, API routes and tests — handoff/semester-platform/

010 Core	020 Community	030 Market	040 Ops
core.institutions, core.profiles, audit.events	groups, members, posts, reports, events, rsvps	listings, offers, drops, bids	staff, accounts, invoices, social_posts, settings

supabase/migrations
*.sql.txt → .sql

→

app · lib
validation, 2 API routes

→

tests
RLS + unit

→

npx supabase db reset
then npm test

Status: written to the architecture guide. Not yet run, typechecked or tested.

AI Workflow Router & Benchmark Lab

Prototype in ui_kits/workflow-lab/ → production source in handoff/workflow-router-app/

Prototype · ui_kits/workflow-lab/

Router
6 inputs → tool, confidence, reasons, limits, escalation, prompts

Matrix
16-column feature matrix, 12 canonical workflows

Benchmark
Persistence tests, 0–5 grading, weighted /5 and /100, leaderboard, CSV/MD export

Saved on this device only; labelled prototype mode.

Production · handoff/workflow-router-app/

Stack
Next.js App Router, TypeScript strict, Tailwind, Supabase (@supabase/ssr, magic link), Zod, Server Actions

Data
001_core_workflow_router, 002_benchmark_grading; org-scoped RLS (owner · admin · member · viewer); seed with 12 workflows

Tests
vitest suites, RLS test plan

Run install.sh. Not yet typechecked, linted, tested or built.

Claude Code toolkit

Token pipeline, files and commands — handoff/claude-code-toolkit/

lib/look.ts
tokensFor — every value
→
styles/tokens.css
semantic names
→
lib/tokenexport.ts
buildTokenExport
→
semester.tokens.json
generated · never hand-edit
→
FIGMA-MAPPING.md
Figma variables · checked

ADDS TO THE REPO

.claude/skills/audit-semester-design-sync/ read-only audit

.claude/skills/build-semester-ui/ build rules

.mcp.json Figma remote MCP

docs/FIGMA-MAPPING.md parity table

app/scripts/design-system-audit.mjs 4 checks

app/scripts/design-system-report.mjs CI markdown

app/src/lib/designsystem.test.ts proves each check

RUN

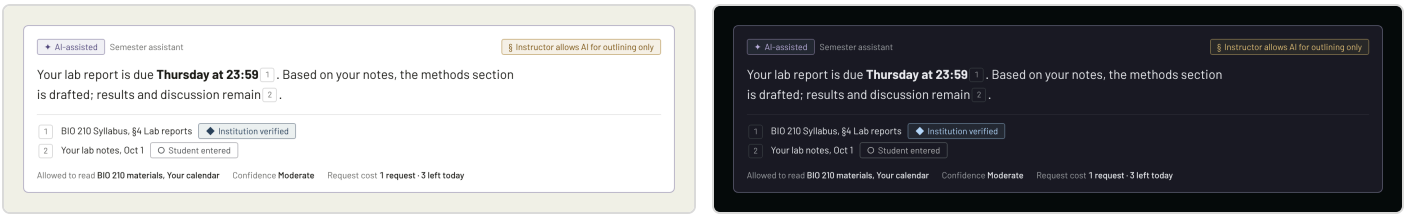
```
claude mcp add --scope project \
  --transport http figma https://mcp.figma.com/mcp
/audit-semester-design-sync app/src
cd app && npm run design-system:check
```

No new dependency, token source, Tailwind, Storybook or package.

Every component, light and dark

Each component in Parchment (light, left) and Ink (dark, right).

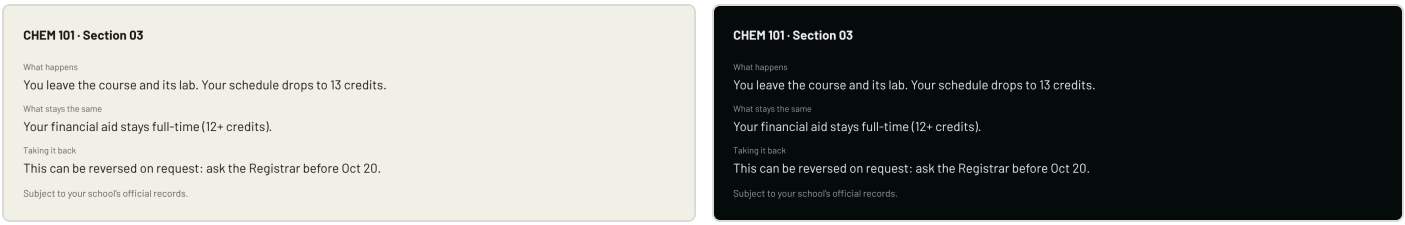
AIResponse AI



ProvenanceChips AI



ActionPreview ACTIONS



Button ACTIONS



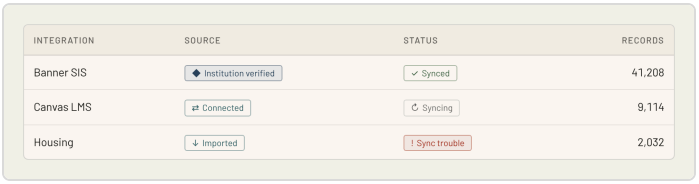
IconButton ACTIONS

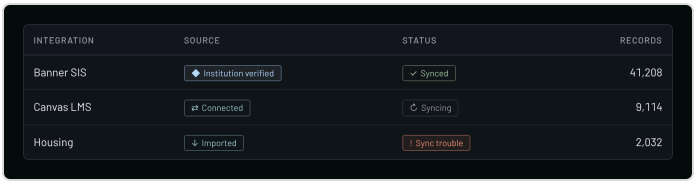


QuickActions ACTIONS



DataTable DATA





DecisionTrail DATA



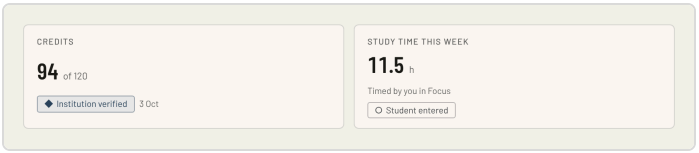


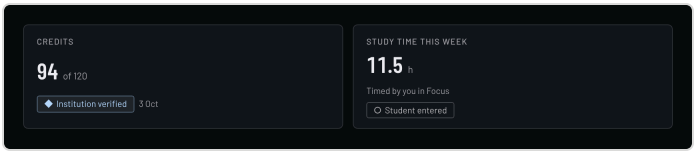
Fields DATA



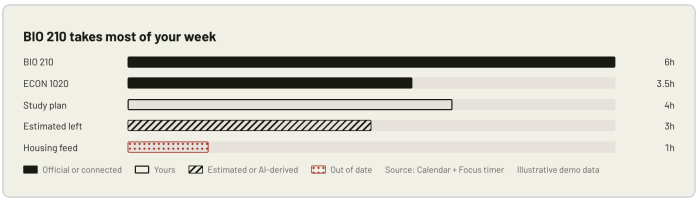


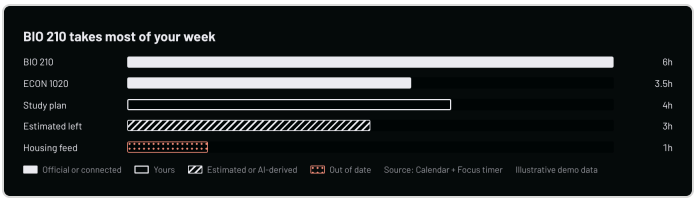
MetricTile DATA





ProvenanceChart DATA



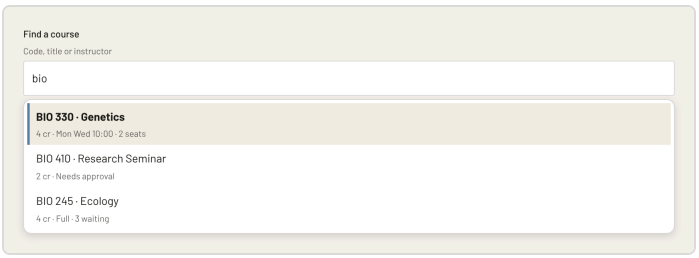


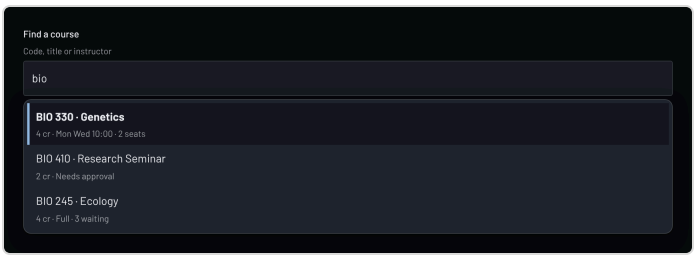
Choice FORMS



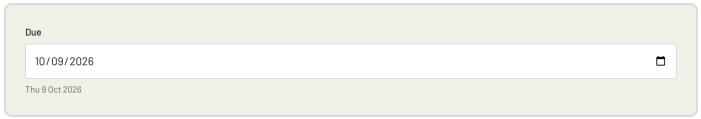


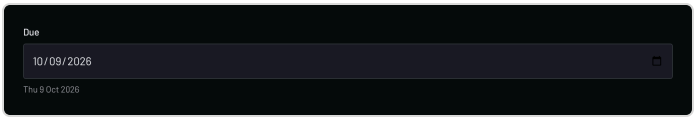
Combobox FORMS





DateField FORMS





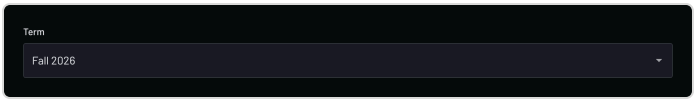
Segmented FORMS





Select FORMS





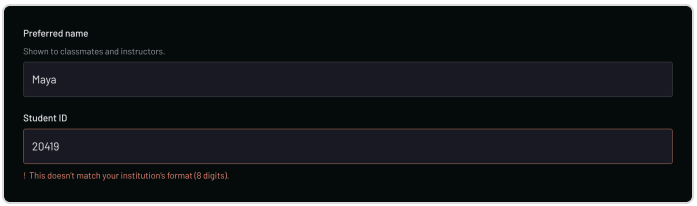
Switch FORMS



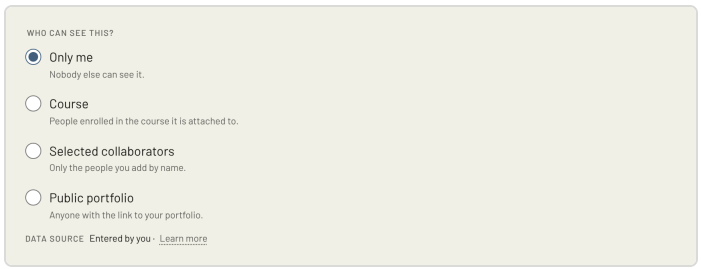


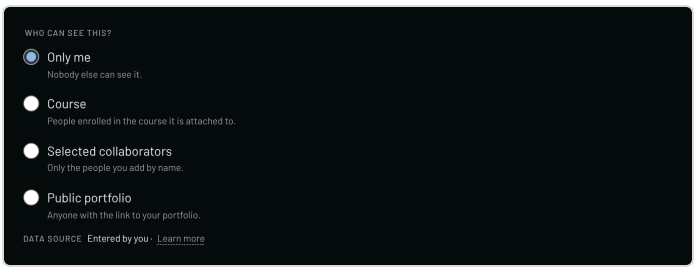
TextField FORMS





VisibilityPicker FORMS





Avatar GLYPH





Icon GLYPH

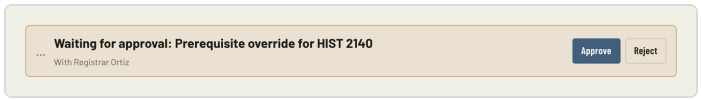


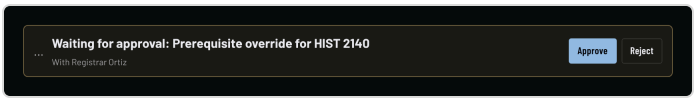
Wordmark GLYPH



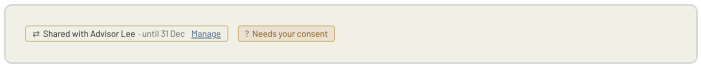


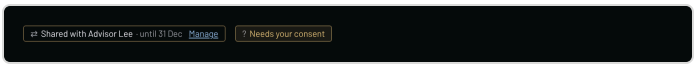
ApprovalBanner GOVERNANCE





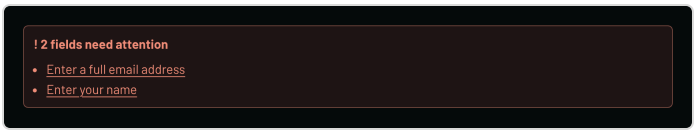
ConsentBadge GOVERNANCE





ErrorSummary GOVERNANCE



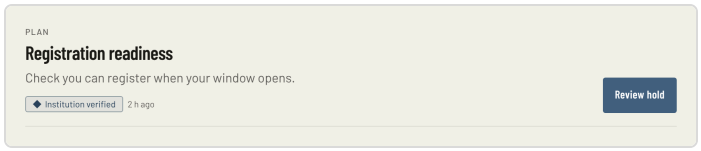


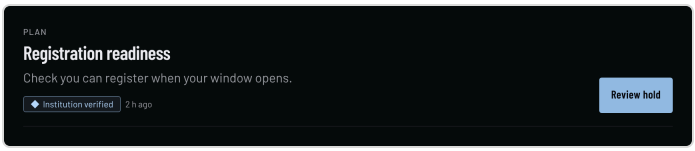
HealthBadge GOVERNANCE





PageHeader GOVERNANCE

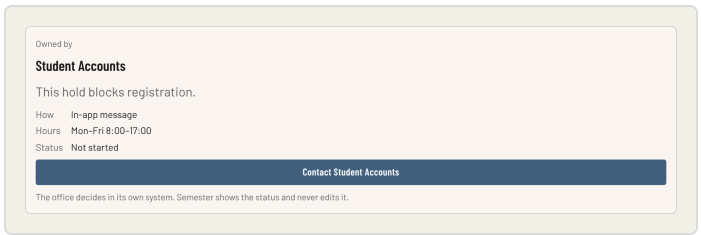


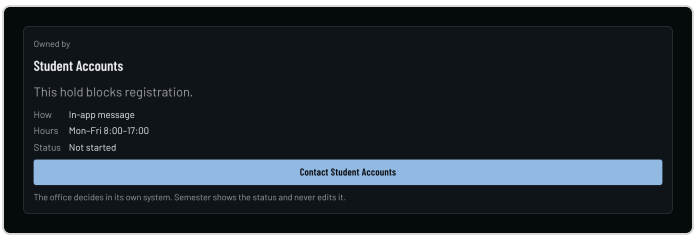


PolicyBadge GOVERNANCE

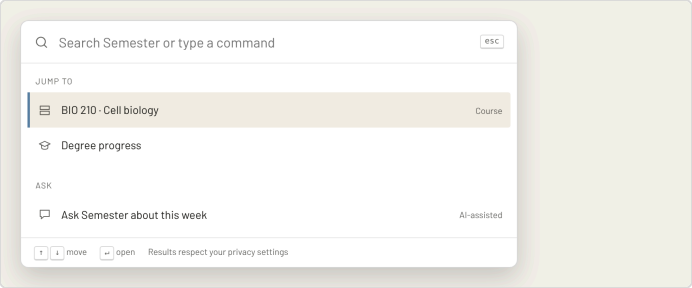


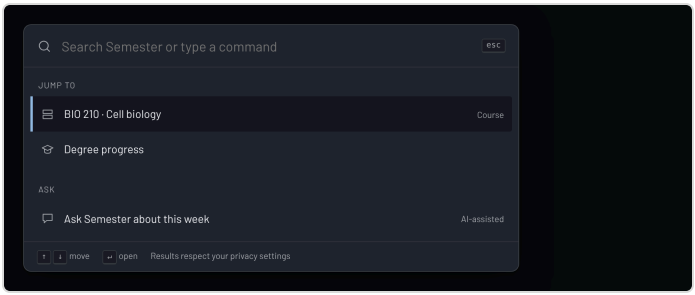
SupportHandoff GOVERNANCE



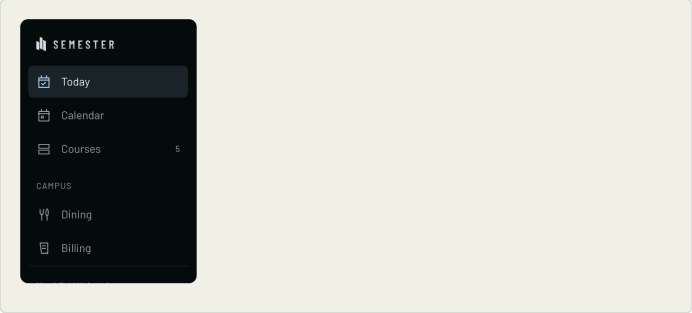


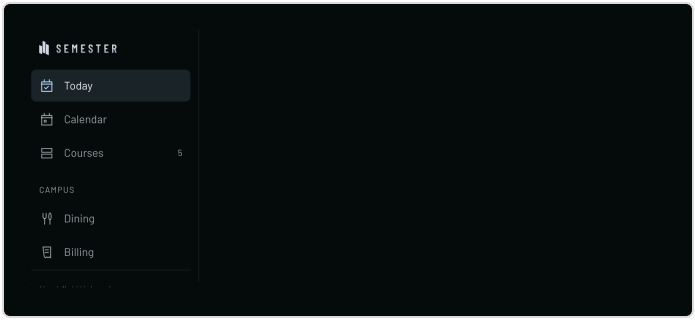
CommandPalette NAVIGATION



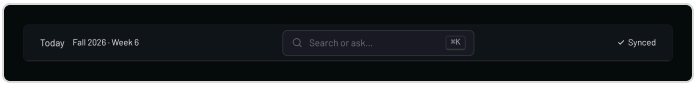


SideRail NAVIGATION

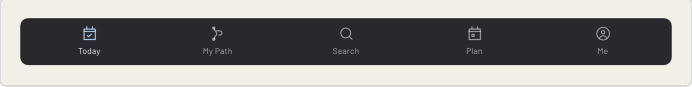


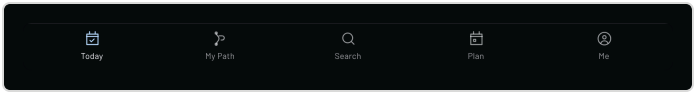


SystemContextBar NAVIGATION

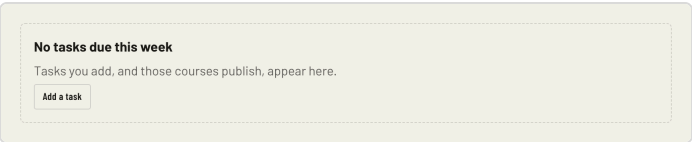


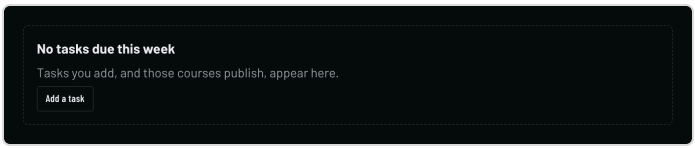
TabBar NAVIGATION



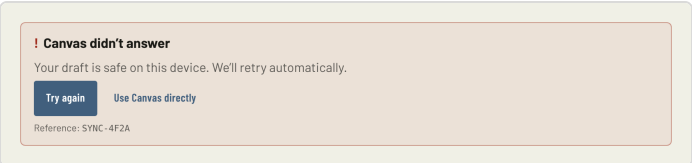


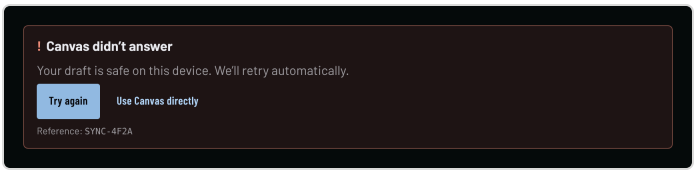
EmptyState STATES





ErrorState STATES





LoadingState STATES



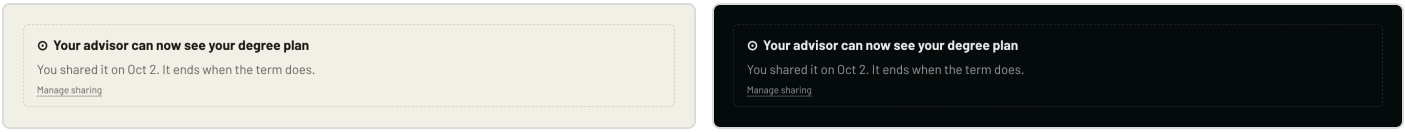
Notice STATES



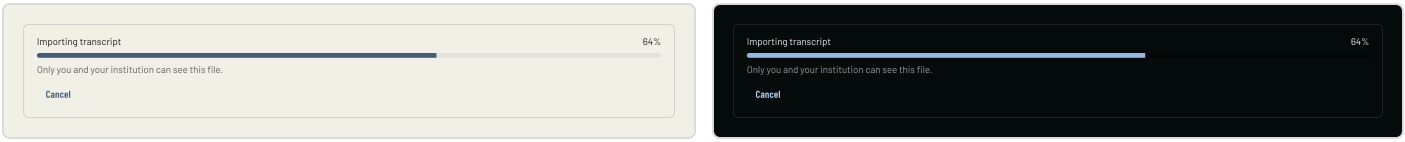
OfflineStrip STATES



PermissionNotice STATES



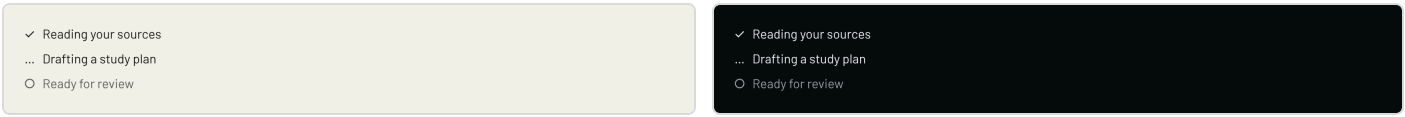
ProgressState STATES



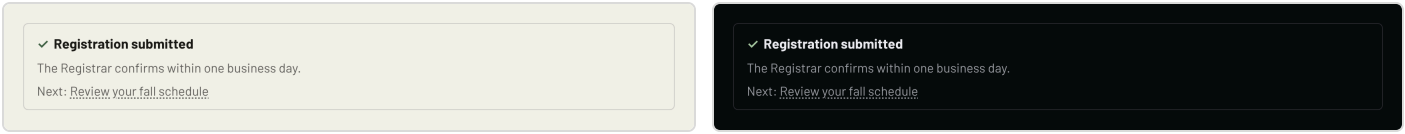
SaveState STATES



StepStatus STATES



SuccessState STATES

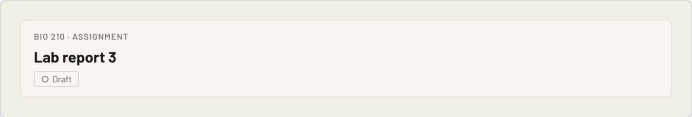


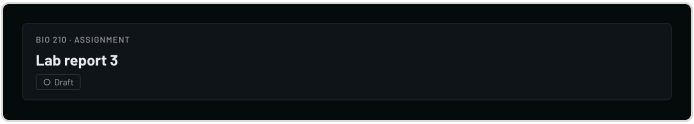
UndoToast STATES



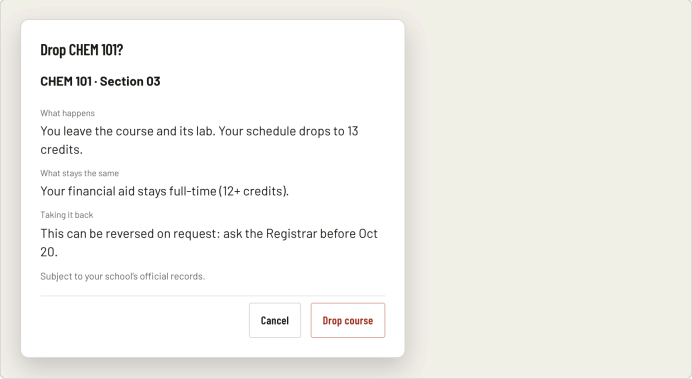


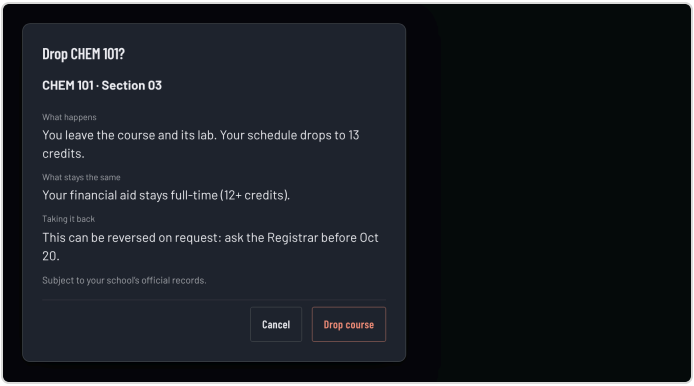
ContextBar SURFACES





Dialog SURFACES

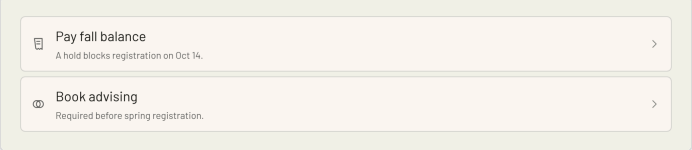


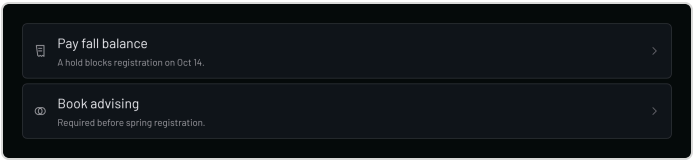


FocusBar SURFACES

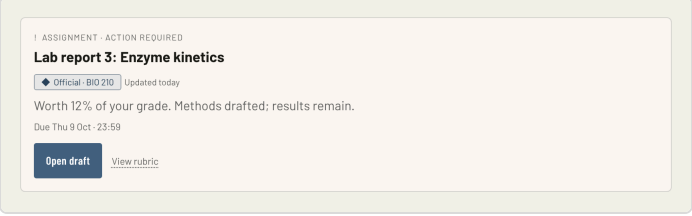


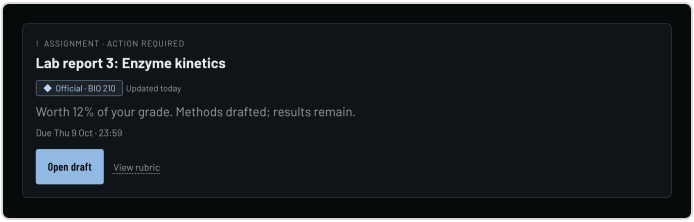
NextSteps SURFACES





ObjectCard SURFACES

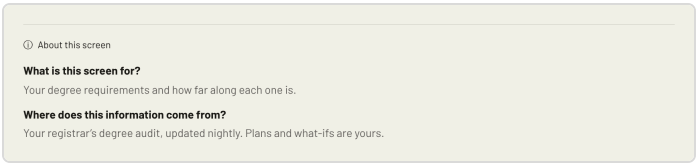


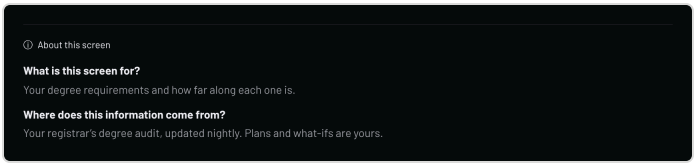


OpenIn SURFACES



ScreenGuide SURFACES

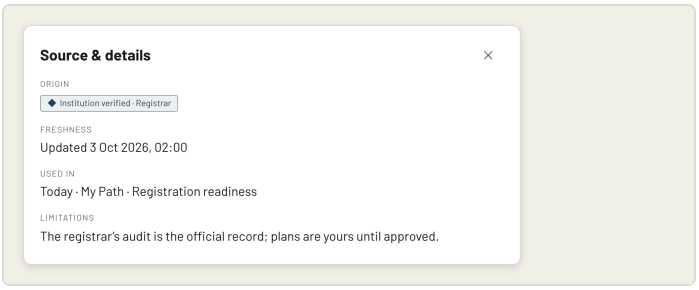


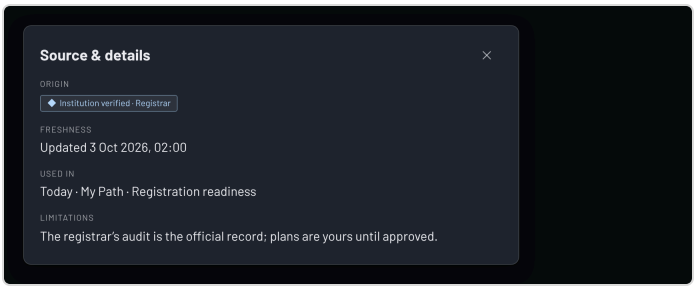


SectionLabel SURFACES



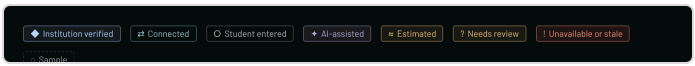
Sheet SURFACES





SourceBadge TRUST & PROVENANCE



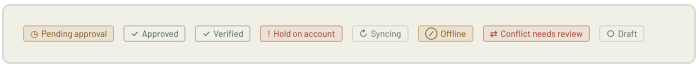


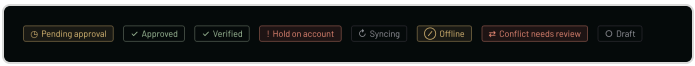
SourceLine TRUST & PROVENANCE





StatusChip TRUST & PROVENANCE





Design language

docs/system/README.md

Design, intelligence, workflow and data system

One brand language · one source-authority language · one action language · one policy language · one privacy language · one error-recovery language · one accessibility standard · one workflow grammar · one education graph = one Semester OS.

- design/SEMESTER_BRAND_IDENTITY.md
- design/SEMESTER_VISUAL_LANGUAGE.md
- design/SEMESTER_UI_GRAMMAR.md
- design/SEMESTER_COMPONENT_LIBRARY.md
- design/SEMESTER_INTERACTION_LANGUAGE.md
- design/SEMESTER_MOTION_AND_FEEDBACK.md
- design/SEMESTER_DATA_TRUST_UI.md
- design/SEMESTER_RESPONSIVE_STRATEGY.md
- algorithms/SEMESTER_ALGORITHM_CATALOG.md
- algorithms/SEMESTER_ALGORITHM_GOVERNANCE.md
- algorithms/SEMESTER_NEXT_ACTION_RANKING.md
- algorithms/SEMESTER_WORKLOAD_MODEL.md
- algorithms/SEMESTER_REGISTRATION_READINESS.md
- algorithms/SEMESTER_SEARCH_RANKING.md
- algorithms/SEMESTER_AI_ROUTING.md
- workflows/SEMESTER_WORKFLOW_ARCHITECTURE.md
- workflows/SEMESTER_WORKFLOW_CATALOG.md
- data/SEMESTER_DATA_FLOW_ARCHITECTURE.md
- data/SEMESTER_DATA_CLASSIFICATION.md
- data/SEMESTER_SOURCE_AUTHORITY.md
- data/SEMESTER_PROVENANCE_MODEL.md

Next 25 implementation actions

1. Decide brand hue: keep Semester blue or move to indigo
2. Build PolicyBadge
3. Build ConsentBadge
4. Build HealthBadge
5. Build SupportHandoff
6. Build ApprovalBanner

7. Build ErrorSummary
8. Promote PageHeader to a component
9. Add data-trust panel to every critical screen
10. Wire freshness.ts to SourceBadge
11. Ship next-action ranking with explanation UI
12. Ship registration readiness states
13. Write algorithm governance records for the 6 pilot algorithms
14. Bias/fairness review for caseload and risk signals
15. Evaluation dataset for next-action ranking
16. Search permission filter + authority ranking
17. AI router transparency panel
18. Implement the SemesterWorkflow type in the repo
19. Universal workflow state machine
20. Data classification tags on every table
21. Provenance columns on canonical tables
22. Analytics event taxonomy (no PII)
23. axe + keyboard tests per component
24. Screen consistency audit against the UI grammar
25. Regenerate docs/master from the repo

`docs/system/design/SEMESTER_UI_GRAMMAR.md`

UI grammar

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Universal page shell

Context bar (logo, location, search ⌘K, notifications, profile) → primary navigation (Today · Path · Plan · Courses · Workspace · Support · Career; rail on desktop, tab bar on mobile) → page title + what it helps you do + source/freshness → Layer 1 → Layer 2 → Layer 3. Built from SystemContextBar, SideRail, TabBar and the student-screen template.

Three-layer disclosure

Layer	Purpose	Example
1	Immediate decision	“Your registration opens Tuesday at 8:00 AM. You have one hold.”
2	Explanation and action	“The hold is from Student Accounts. Here’s why it matters and who can help.”

Layer	Purpose	Example
3	Advanced detail	Source system, policy text, history, export, integration state, audit trail

A student never needs Layer 3 for a normal daily action. Staff use Layer 3 often but start collapsed.

Every screen answers

- Where am I?
- What is this screen for?
- What matters now?
- What can I do?
- What is official?
- What is stale or uncertain?
- What happens if I take this action?
- Who can help?
- Where is deeper detail?

Standard panels

Data trust: Source · Updated · Authority · Status · Policy (with version) · Action (view official details). Built from SourceBadge + SourceLine + Fields.

AI answer: Answer · Sources · What this means · Suggested next step · Limits and uncertainty · Course/institution policy · Give feedback · Ask a person / official office. Built from AIResponse + ProvenanceChips.

Error: What happened · What data may be affected · What you can do now · Whether it saved · Whether it will retry · Who can help · Reference ID. Built from ErrorState.

`docs/system/design/SEMESTER_COMPONENT_LIBRARY.md`

Component library

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Required shared components mapped to this design system (`components/`, namespace `SemesterDesignSystem_e9f3ba`).

Required	In this system	Status
AppShell	Shell templates (student-screen, console-screen, platform-shell)	Have
ContextBar	ContextBar, SystemContextBar	Have

Required	In this system	Status
GlobalNavigation	SideRail	Have
MobileNavigation	TabBar	Have
GlobalSearch / CommandPalette	CommandPalette	Have
PageHeader	PageHeader	Have
SourceBadge	SourceBadge	Have
FreshnessBadge	SourceBadge updated prop + freshness.ts	Partial
AuthorityBadge	SourceLine by/authority	Partial
PolicyBadge	PolicyBadge	Have
ConsentBadge	ConsentBadge (+ VisibilityPicker input)	Have
StatusBadge	StatusChip	Have
HealthBadge	HealthBadge	Have
RiskBanner	Notice	Partial
ActionCard	ObjectCard	Have
ActionCenter	NextSteps	Have
WorkflowStepper	StepStatus	Have
DecisionCard	ActionPreview + Dialog	Have
EvidenceDrawer	Sheet + SourceDetails	Have
AuditTimeline	DecisionTrail	Have
DataTable	DataTable	Have
EmptyState / LoadingState / ErrorState	EmptyState, LoadingState, ErrorState	Have
OfflineState	OfflineStrip	Have
ForbiddenState	PermissionNotice	Have
SupportHandoff	SupportHandoff	Have
AIAnswerPanel	AIResponse	Have
CitationPanel	ProvenanceChips	Have
ActionPreview	ActionPreview	Have
ApprovalBanner	ApprovalBanner	Have
FormField	TextField, Select, Combobox, DateField, Choice, Switch	Have
ErrorSummary	ErrorSummary	Have
Modal	Dialog, Sheet	Have
Toast	UndoToast	Have
NotificationCenter	Student Notifications screen	Pattern only

Gaps closed (Oct 2026): PolicyBadge, ConsentBadge, HealthBadge, SupportHandoff, ApprovalBanner, ErrorSummary, PageHeader in components/governance/.

Every component defines

Semantic token usage · accessibility requirement · responsive behaviour · loading/empty/error/offline/forbidden states · analytics event · source/freshness display · test coverage. Recorded in each component's `.prompt.md` and `.d.ts`.

Modules may not invent their own colours, spacing, status vocabulary, permission behaviour or source labels.

`docs/system/design/SEMESTER_INTERACTION_LANGUAGE.md`

Interaction language

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Situation	Rule
Alerts	Only actionable; one primary per view; glyph + words; never colour alone
Official data	Source + authority + freshness shown; official items outrank derived ones
AI uncertainty	Confidence and limits stated; sources numbered; human or office offered
Approvals	Preview first, consequence in words, second approver when consequential, audit before apply
Policy restrictions	Say which policy, who owns it, what you can do instead
Recovery after errors	Say whether it saved, whether it retries, who can help, reference ID
Consent	Recipient, category, purpose, duration, items; revoke always visible
Destructive actions	Undo where possible; otherwise a request path; never silent

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`docs/system/design/SEMESTER_MOTION_AND_FEEDBACK.md`

Motion and feedback

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Event	Behaviour	Token
Hover / press	Wash, no scale	130 ms
State change	Direct	180 ms
Panel	Subtle	240 ms
Sheet / modal	Rise 8 px + fade, direct	280 ms
Save state	Always visible (SaveState)	—
Completion	Calm check, no confetti	—

Event	Behaviour	Token
Loading	Non-distracting skeleton (LoadingState), no spinners	—
Reduced motion / calm mode	All motion zeroed	prefers-reduced-motion, data-calm

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docs/system/design/SEMESTER_RESPONSIVE_STRATEGY.md

Responsive strategy

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Surface	Strategy
Student	Mobile-first; five-destination tab bar; offline-safe for student-owned data
Staff / institution / ops	Desktop-first; dense tables; rail + context bar
Public site	Fluid; single column under 860 px
All	Reflow at 320 px; 200% zoom; 44 px targets (24 px floor in dense composites)

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docs/system/design/SEMESTER_DATA_TRUST_UI.md

Data trust UI

Semester design, intelligence, workflow and data system · 2026-10-05 · Source: uploads/expand further on brand identy algorthims workflow.pdf, reconciled with this design system (readme.md, tokens/, components/). Where they differ, this design system wins and the difference is listed.

> Semester should make an institutionally complex world feel personally clear.

Every critical screen shows: Source · Updated · Authority · Status · Policy (versioned) · Action. Example: "Source: official institution record · Updated 2 hours ago · Authority: Registrar · Status: current · Policy: registration policy v4.2 · View official details."

Freshness states come from semester-core freshness.ts (live ≤ 5 s, recent ≤ 60 s, delayed, failed, rebuilding); delayed data shows a badge and blocks destructive actions until confirmed at the source.

Figma mapping and design-system baseline

handoff/claude-code-toolkit/docs/design-system/README.md

Semester design-system workflow

Order of authority: `lib/look.ts` → `styles/tokens.css` → `lib/tokenexport.ts` → `design-tokens/semester.tokens.json` (generated) → `docs/design-system/FIGMA-MAPPING.md` → Figma.

Never hand-edit: `app/design-tokens/semester.tokens.json`, `app/reports/`.

Skills

- `/build-semester-ui <screen> [figma-url]` — build or change UI from existing primitives.
- `/audit-semester-design-sync [figma-url] [path]` — read-only report.
- `/create-semester-component <name> <screens>` — spec-first shared component.

Figma MCP (each developer, once)

```
 claude mcp add --scope project --transport http figma https://mcp.figma.com/mcp
claude    # then run /mcp and authenticate in the browser
```

`.mcp.json` holds only the endpoint; authentication is interactive and per developer. CI never authenticates to Figma.

Checks (from `app/`)

Command	What it does
`npm run tokens:check`	Fails if the committed JSON differs from `buildTokenExport`
`npm run design-system:audit`	Raw values vs the hex LEDGER, undefined variables, tokens.css ↔ export, Figma mapping — file:line
`npm run design-system:report`	Writes `app/reports/design-system/report.md` + `.json`, runs 8 existing contract tests
`npm run design-system:check`	All of the above that gate a PR

Adding a mapping

Add a row to `FIGMA-MAPPING.md` whose first path cell is `semantic.<name>` or `primitive.<name>` as it appears in the JSON. Run `npm run design-system:audit`; a path that does not resolve fails as *missing*.

Reading the report

Violations (blocker/major) fail CI. **Warnings** (minor) are raw inline values with the token to use instead; they do not fail until a baseline is agreed. **Unmapped candidates** are exported semantic tokens no Figma variable uses yet — informational.

---

handoff/claude-code-toolkit/docs/design-system/SEMESTER-DESIGN-SYSTEM-BASELINE.md

Semester design-system baseline

Read-only audit of harrisonjrubin7-cmyk/semester @ main (5eba494), 2026-10-05. Evidence is quoted from files read through the GitHub API; nothing was executed.

1 · Current architecture

Concern	Evidence
Stack	Vite + React + TypeScript under `app/` (`app/package.json`: `"build": "tsc -b && vite build"`); npm workspaces at root, one lockfile (`CLAUDE.md`)
Styling	Plain CSS: `styles/app.css` (~321 KB), `industry.css`, `unity.css`, `features.css`, `tokens.css`, `typefaces.css` + inline style objects using `var(--*)`
Value authority	`lib/look.ts` `tokensFor` — 13 grounds × 11 accents, density, corners, text size, text spacing, written as CSS custom properties on the root
Semantic authority	`styles/tokens.css` — "No colour is written here as a hex"; held by `styles/tokens.test.ts`
Export	`lib/tokenexport.ts` `buildTokenExport(tokensCss, appCss)` → `design-tokens/semester.tokens.json` (`$schema: semester.tokens/1`), regenerated by `npm run tokens:export` (`TOKENS=write vitest run src/lib/tokenexport.test.ts`)
Components	`components/` (~273 files), shared primitives in `components/unity/` (ObjectCard, ContextBar, Status, ProvenanceChips, ActionPreview, States, Visibility, Table, Combobox, DateField, OpenIn, NextSteps, ScreenGuide…), `components/ui.tsx`
Screens	`app/src/screens.tsx` — exhaustive `Record<Screen, ComponentType>` of ~95 lazy screens; areas in `lib/navareas.ts`
Enforcement	19 `styles/*.test.ts` contracts (breakpoints, deadcss, density, fields, glass, gutter, hex, inset, mapcredit, motion, print, reach, scale, stacking, sweepdensity, taps, textbuttons, textscale, tokens); `lib/contrast.test.ts` (143 pairings); `lib/tokenexport.test.ts`; lint = `oxlint` + `scripts/styles.mjs` + `labels.mjs` + `terms.mjs`; `census:design`; `sweep:contrast/targets/spacing/walls`
CI	12 workflows incl. `ci.yml` (build job runs `npm run lint`, `npm test`, `npm run build` in `app/`), `contrast.yml`, `drift.yml`
Claude	`CLAUDE.md` (check main first; decisions take PR numbers; gate list; proof standards); skills `add-course`, `debugging-and-error-recovery`, `run`, `security-and-hardening`, `test-driven-development`

2 · Source-of-truth map

Layer	File	Guard
Values	`app/src/lib/look.ts`	`lib/contrast.test.ts`, `look.test.ts`

Layer	File	Guard
Fixed primitives	`app/src/styles/app.css` first `:root` (`--type-*` , `--sp-*` , `--leading-*` , `--lift-*` , `--ease` , `--fast`)	`scale.test.ts` , `density.test.ts`
Semantic	`app/src/styles/tokens.css`	`tokens.test.ts`
Generated	`app/design-tokens/semester.tokens.json`	`tokenexport.test.ts` snapshot
Docs	`DESIGN-SYSTEM-GUIDE.md` , `docs/DESIGN-TOKENS.md` , `docs/COLOR-AND-DARK-MODE-SPEC.md` , `docs/TRUST-CUES-AND-SOURCE-PRESENTATION.md`	—
Figma	Intended consumer of the JSON (`tokenexport.ts` header)	**None** — no `.mcp.json` , no mapping file

3 · Strengths to preserve

The value/semantic split with no colours in `tokens.css` ; the generated export and its snapshot test; the shrinking hex LEDGER in `hex.test.ts` (10 files, each with a reason); per-ground contrast measured on every surface; motion roles zeroed by `prefers-reduced-motion` and `data-calm` ; one trust vocabulary (`lib/status.ts` , `lib/source.ts`); the exhaustive screen table; CLAUDE.md's "revert the fix and watch the guard go red" standard.

4 · Gaps

Area	Gap	Evidence
Persistent instructions	CLAUDE.md has no design-system section (source-of-truth order, raw-value rule, Figma protocol)	CLAUDE.md headings: main check, decisions, gates, proving, contrast
UI-generation procedure	No skill for building UI from unity primitives	`.claude/skills/` list
Audit procedure	No design-sync audit skill	same
Component creation	No rule against one-off wrappers	same
Figma governance	No MCP config, no mapping	tree search: no `.mcp.json` , no `docs/design-system/`
Raw-value enforcement	Hex in .tsx is enforced; numeric zIndex / ms / cubic-bezier / box-shadow / fontSize / borderRadius in inline styles are not reported with file:line	`hex.test.ts` covers quoted hex only; `DESIGN-TOKENS.md` : layer literals "were not replaced"
Token drift	Enforced (snapshot) but no named check script	`tokens:export` writes; no read-only `tokens:check`
CI reporting	No design-system report artifact	`ci.yml` build job

5 · Risk classification

Severity	Finding	Evidence
Major	Figma parity is unenforceable: no mapping exists, so a Figma variable rename or a new Figma-only token cannot be caught	no mapping file; `tokenexport.ts` names Figma as a consumer
Major	Agent sessions can introduce raw inline z-index/duration/shadow values that no test reports by line	`DESIGN-TOKENS.md` Layers note; only `stacking.test.ts` covers stylesheet literals
Minor	No read-only token drift command; `tokens:export` both checks and (with `TOKENS=write`) writes	`app/package.json:16`
Minor	No design-system section in CLAUDE.md	CLAUDE.md

6 · Minimal plan

Priority	Path	Change	Acceptance
P0	`CLAUDE.md`	Append the section in `CLAUDE.md.design-system-section.md`	Section present; existing sections unchanged
P0	`.claude/skills/{build-semester-ui,audit-semester-design-sync,create-semester-component}/SKILL.md`	Add three skills	`/build-semester-ui` etc. listed in Claude Code
P0	`app/scripts/design-system-audit.mjs`, `design-system-report.mjs`	Add (zero deps)	`npm run design-system:audit` exits 0 on main
P0	`app/src/lib/designsystem.test.ts`	Add	Each check fails on its fixture; real tree clean
P1	`app/package.json`	`tokens:check`, `design-system:audit`, `:report`, `:check`	Scripts run from `app/`
P1	`docs/design-system/FIGMA-MAPPING.md`	Template with verified code paths	Audit: 0 missing mappings
P1	`.mcp.json`	Remote Figma endpoint only	No credentials in file
P2	`.github/workflows/ci.yml`	Two steps in the build job after lint + artifact upload	Report artifact on PRs
P2	`.gitignore`	`app/reports/`	Report not committed

Deferred: replacing existing inline z-index/duration literals (warnings only until a baseline is agreed); fetched Figma variable names; visual regression; Figma writes (never).

`handoff/claude-code-toolkit/docs/design-system/FIGMA-MAPPING.md`

Figma ↔ code mapping

> **Status: template.** No Figma file has been inspected yet, so every *Figma* name below is a **proposed** name, not a fetched one. The *Exported path* column is real: each path was checked against `app/design-tokens/semester.tokens.json` at main@5eba494. When a Figma file is connected, replace proposed names

with fetched ones and set **Status** per row.

Ownership

Concern	Owner
Behaviour, accessibility, responsive layout, public component APIs, token values	Code
Approved visual intent, exploration	Figma
Token pipeline	`lib/look.ts` (values) → `styles/tokens.css` (semantic names) → `lib/tokenexport.ts` → `design-tokens/semester.tokens.json` (generated)

Mapping rules

1. No Figma variable becomes a production token without a code PR to `look.ts` / `tokens.css` and a regenerated export.
2. No Figma component becomes a React component automatically — map it to an existing pattern first (`build-semester-ui`, `create-semester-component`).
3. Missing states in Figma never remove code requirements (loading, empty, error, disabled, permission, keyboard, focus, responsive).
4. Anything unresolved goes in **Unresolved** below with an owner.
5. Status vocabulary: `Match` · `Partial` · `Missing in Figma` · `Missing in code` · `Obsolete` · `Proposed (unverified)`.

Unresolved

Item	Side	Owner	Note
Figma file URL	Figma	Design	Not provided; all Figma names are proposals

Figma variables are *imports* of `app/design-tokens/semester.tokens.json` (generated by `npm run tokens:export`), never the other way round. `npm run design-system:check` fails when a path in the **Exported path** column does not exist in that file, and the report lists semantic tokens with no mapping as non-required candidates.

Rows are machine-read: the first column that starts with `primitive.` or `semantic.` is checked. Keep one variable per row.

Collections

Figma collection	Modes	Exported collection	Notes
Semester / Ground	ink, graphite, midnight, basalt, oxide, forest, wine, industry-dark, parchment, paper, bone, industry, fog	ground	default **ink**
Semester / Accent	sterling, brass, copper, jade, slate, pewter, industry, oxblood, moss, ink, gold	accent	values resolved on the default ground

Figma collection	Modes	Exported collection	Notes
Semester / Density	comfortable, snug, tight	density	multiplies `--sp-*`
Semester / Corners	drawn, square, soft, round	corners	drives `--r-*`
Semester / Text size	compact, normal, large, largest	textSize	`text-scale` 0.94 / 1 / 1.09 / 1.18
Semester / Text spacing	normal, open, wide	textSpacing	
Semester / Base	default	stylesheet + constant	type, spacing, leading, lift, easing

Semantic variables

Exported path	Figma variable (proposed)	CSS	Use
semantic.surface-sunken	surface/sunken	—surface-sunken	Input wells, tracks
semantic.surface-canvas	surface/canvas	—surface-canvas	The page
semantic.surface-base	surface/base	—surface-base	Standard content surface
semantic.surface-raised	surface/raised	—surface-raised	Cards above base
semantic.surface-overlay	surface/overlay	—surface-overlay	Sheets, menus, Focus bar
semantic.surface-critical	surface/critical	—surface-critical	Error wash
semantic.text-primary	text/primary	—text-primary	Anything read to act
semantic.text-secondary	text/secondary	—text-secondary	Held to 4.5:1
semantic.text-tertiary	text/tertiary	—text-tertiary	Metadata only, 3:1
semantic.text-on-action	text/on-action	—text-on-action	On the primary button
semantic.border-subtle	border/subtle	—border-subtle	
semantic.border-default	border/default	—border-default	
semantic.border-strong	border/strong	—border-strong	
semantic.action-primary	action/primary	—action-primary	One CTA colour: the ground's metal
semantic.action-primary-strong	action/primary-strong	—action-primary-strong	Hover/emphasis
semantic.action-secondary	action/secondary	—action-secondary	Wash
semantic.status-attention	status/attention	—status-attention	The one colour the app spends
semantic.status-attention-line	status/attention-line	—status-attention-line	
semantic.status-attention-wash	status/attention-wash	—status-attention-wash	
semantic.status-danger	status/danger	—status-danger	6:1
semantic.status-danger-line	status/danger-line	—status-danger-line	
semantic.status-danger-wash	status/danger-wash	—status-danger-wash	
semantic.status-success	status/success	—status-success	= accent
semantic.status-info	status/info	—status-info	= accent
semantic.status-neutral	status/neutral	—status-neutral	

Exported path	Figma variable (proposed)	CSS	Use
semantic.chart-verified	chart/verified	—chart-verified	
semantic.chart-estimated	chart/estimated	—chart-estimated	
semantic.chart-stale	chart/stale	—chart-stale	
semantic.chart-student-entered	chart/student-entered	—chart-student-entered	
semantic.focus-color	focus/color	—focus-color	4.5:1 on panel
semantic.focus-ring-width	focus/ring-width	—focus-ring-width	2px
semantic.target-min	target/min	—target-min	24
semantic.target-primary	target/primary	—target-primary	44
semantic.shape-control	shape/control	—shape-control	
semantic.shape-card	shape/card	—shape-card	
semantic.shape-surface	shape/surface	—shape-surface	
semantic.shape-sheet	shape/sheet	—shape-sheet	
semantic.elevation-raised	elevation/raised	—elevation-raised	
semantic.elevation-floating	elevation/floating	—elevation-floating	
semantic.elevation-modal	elevation/modal	—elevation-modal	
semantic.type-role-display	type/display	—type-role-display	
semantic.type-role-page	type/page	—type-role-page	
semantic.type-role-section	type/section	—type-role-section	
semantic.type-role-body	type/body	—type-role-body	
semantic.type-role-label	type/label	—type-role-label	
semantic.duration-fast	motion/fast	—duration-fast	130ms
semantic.duration-standard	motion/standard	—duration-standard	180ms
semantic.layer-overlay	layer/overlay	—layer-overlay	80

Primitive variables Figma needs directly

Exported path	Figma variable (proposed)	Varies with
primitive.app-bg	base/bg	ground
primitive.app-panel	base/panel	ground
primitive.app-fg	base/fg	ground
primitive.app-accent	base/accent	accent, ground
primitive.app-warn	base/warn	ground
primitive.r-sm	radius/sm	corners
primitive.r-md	radius/md	corners
primitive.r-lg	radius/lg	corners
primitive.sp-4	space/4	stylesheet × density

Exported path	Figma variable (proposed)	Varies with
primitive.sp-7	space/7	stylesheet × density
primitive.type-base	type/base	stylesheet × textSize
primitive.text-scale	type/scale	textSize

Components

Figma component	React implementation	Required variants / states
Button	`.btn` + `ActionButton` (`components/ui.tsx`)	primary · secondary · ghost · icon · disabled · busy
Status chip	`unity/Status.tsx` → `StatusChip`	tone neutral/info/success/attention/danger · glyph + word
Source badge	`components/SourceBadge.tsx`	8 trust kinds · with age · with Report
Provenance chips	`unity/ProvenanceChips.tsx`	fill solid/outline/dashed · ≤2 cues
Object card	`unity/ObjectCard.tsx`	with/without primary · Detailed mode source line
Context bar	`unity/ContextBar.tsx`	inside a panel (frameless)
Action preview	`unity/ActionPreview.tsx`	recovery undo/request/none
States	`unity/States.tsx`	loading · error · success · progress · steps · permission · offline
Visibility	`unity/Visibility.tsx`	4 audiences · locked
Table	`unity/Table.tsx`	empty · caption hidden
Combobox	`unity/Combobox.tsx`	open · selected row (edge + weight)

Design-system project additions (proposals, not in tokens.css)

The Semester design-system project (Claude Design) adds brief-driven tokens that **do not exist in code**. They are listed so a Figma file built from that project can be reconciled; they must not be imported as if exported.

Design-system token	Status	Closest code token	Resolution
—source-official / -line / -wash	Proposal	—status-success (= accent)	Decide in a DD whether provenance gets its own hue
—source-connected / -line / -wash	Proposal	—	Same
—source-personal	Proposal	—status-neutral	Same
—source-ai / -line / -wash	Proposal	—status-attention (AI is styled with estimates)	Same
—status-pending	Proposal	—status-attention	Alias if accepted
—blue-100…900 (Semester blue)	Proposal	Industry accent ramp in look.ts	Already an accent; no new ramp needed
—accent-editorial (Source Line)	Proposal	—app-accent-deep / Brass shade	Map to accent when Source Line ships

PART 2

Building

How Semester is built: architecture, engineering practice, testing, the build plan and what comes next.

SEMESTER

Architecture guide

For architects and senior engineers. v0.1, October 2026. A design-level description: verify everything against the repo (harrisonjrubin7-cmyk/semester) before relying on it.

1. Principles

1. **The institution is the system of record.** Semester reads, displays its provenance, and writes back only through approved paths.
2. **Student records never reach an AI model.** This is enforced in the gateway, not by convention.
3. **Company roles never see student data.** This is enforced by RLS and separate schemas.
4. **Every consequential action is previewed, attributed and logged.**
5. **Status over claims.** Freshness and source are shown on every record.

2. System context

```
Institution: SIS · LMS · IdP (SAML) · Calendar
        | read sync (pull, webhook)        ▲ approved write-backs
        ▼                                  |
Semester ingest workers → canonical store (Postgres) → read models (per role)
        |                                  |
        ▼                                  ▼
Web app (Next.js on Vercel)  ←→  API (route handlers / server actions)
        |                                  |
        ▼                                  ▼
AI gateway (redact → policy → model → cite) Audit log · Ops telemetry
```

3. Components

Component	Tech	Responsibility
Web app	Next.js App Router, TypeScript, design-system tokens	Student, staff, console, community, ops UIs
Auth	Supabase Auth + SAML SSO per institution	Identity; role claims mapped from the IdP
Database	Supabase Postgres + RLS	Schemas: `core`, `edu` (student data), `community`, `market`, `ops` (company, no PII), `audit`
Ingest	Scheduled workers + webhooks	SIS/LMS pulls, normalisation, reconciliation reports
Read models	Materialised views / tables	Registration readiness, caseload, today plan; each row stamped with source and synced_at

Component	Tech	Responsibility
AI gateway	Server-side module (`semester-core/aigateway.ts`)	Field allow-list, redaction, course AI policy, citations, cost metering
Payments (marketplace)	A licensed processor (e.g. Stripe Connect)	Semester never holds funds
Observability	Platform logs + error tracking + uptime checks	SLOs, alerts, the ops console

---

4. Data model (summary)

Tenancy: every row carries `institution_id`; RLS policies check it against the JWT claim. The `edu` schema holds student records (enrollments, holds, grades as shown, advising notes). `community` and `market` hold user-generated content, linked to a profile and never to edu tables. `ops` holds company data with no student identifiers. See `handoff/semester-platform/supabase/migrations/`.

---

5. Security boundaries

- RLS on every table; service-role keys only in workers, never in the browser.
- AI gateway allow-list: course content, the student's own typed text, public catalog data. Blocked: anything from `edu.*`.
- Marketplace and messages: rate limits, a report pipeline to the moderation queue, an integrity classifier on listings.

---

6. Key flows

SIS sync

1. Worker pulls a delta.
2. Normalise.
3. Upsert with `source` and `synced_at`.
4. Reconcile against the previous snapshot.
5. Mismatches go to the registrar queue.
6. Rebuild the affected read models.

Ask Semester

1. Request.
2. Policy check (course AI policy).
3. Build the context from allowed sources only.
4. Redact.
5. Model call.
6. Cite sources.
7. Label the answer AI-assisted.

8. A person confirms any action.

Marketplace sale

1. Listing.
2. Integrity and safety check.
3. Live.
4. Offer.
5. Processor checkout.
6. Pickup confirmation releases the payout.
7. Any dispute goes to T&S.

---

7. Environments

Env	Data	Access
Local	Seed and synthetic data	Developers
Preview	Synthetic data	Per pull request
Staging	Synthetic data, mirroring the institution's configuration	Team + institution IT
Production	Real data under the DPA	Just-in-time and logged

---

8. Decisions (ADRs)

- ADR-001: Postgres RLS as the primary authorisation layer.
- ADR-002: read models over live SIS queries, for speed and to show freshness.
- ADR-003: the AI gateway as the single egress point to models.
- ADR-004: marketplace payments through a processor; no money transmission.
- ADR-005: company ops data in a separate schema with no student identifiers.